

GENIFY

LECTURE 16

28-August-2025

TOPICS COVERED:

RIS

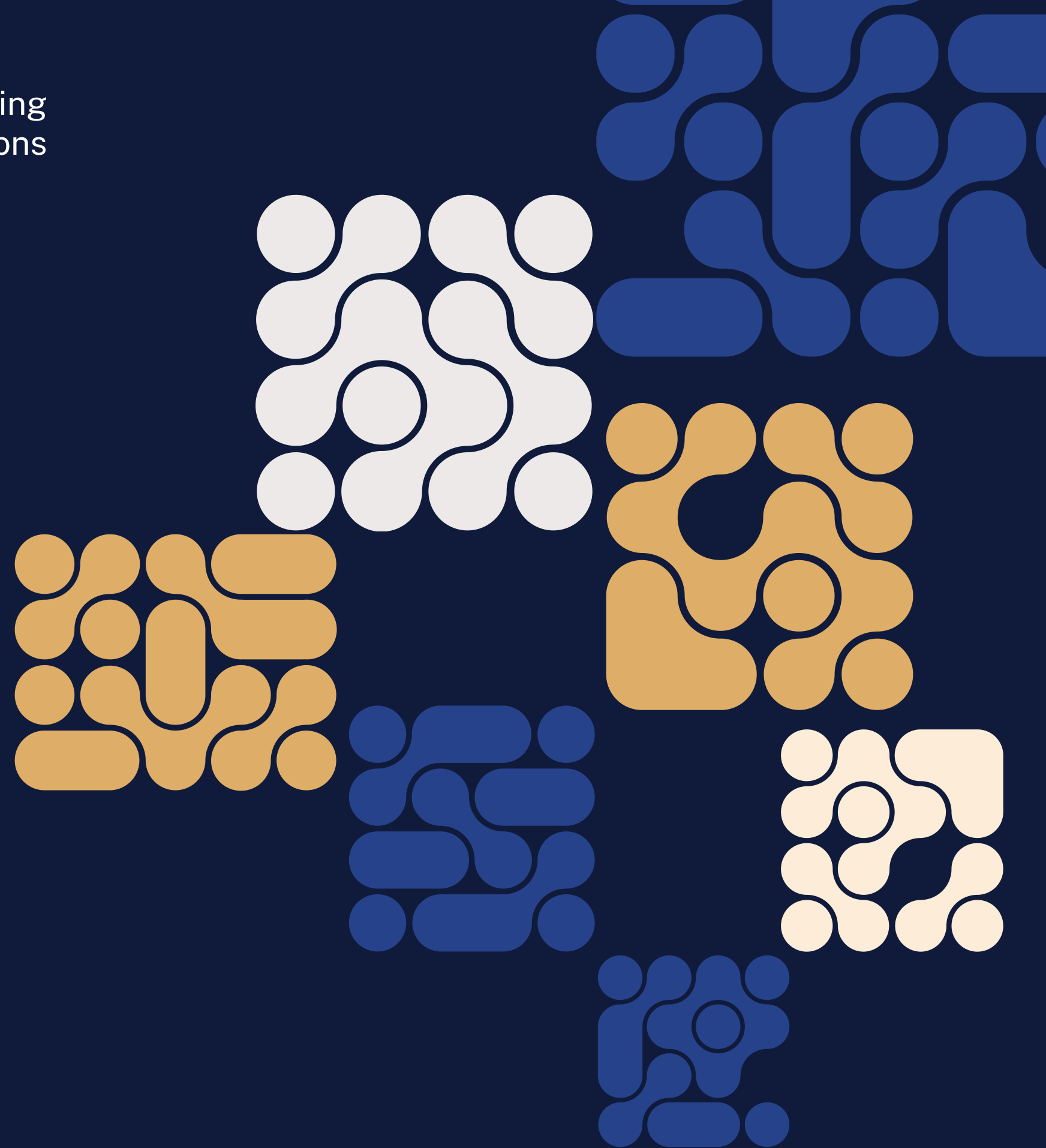
- TRANSFORMERS
 - ~~SELF-ATTENTION~~
 - MULTI-HEADED ATTENTION
 - POSITIONAL ENCODING
 - ENCODER

Presenter

Shahzaib Kashif

Co-Presenter

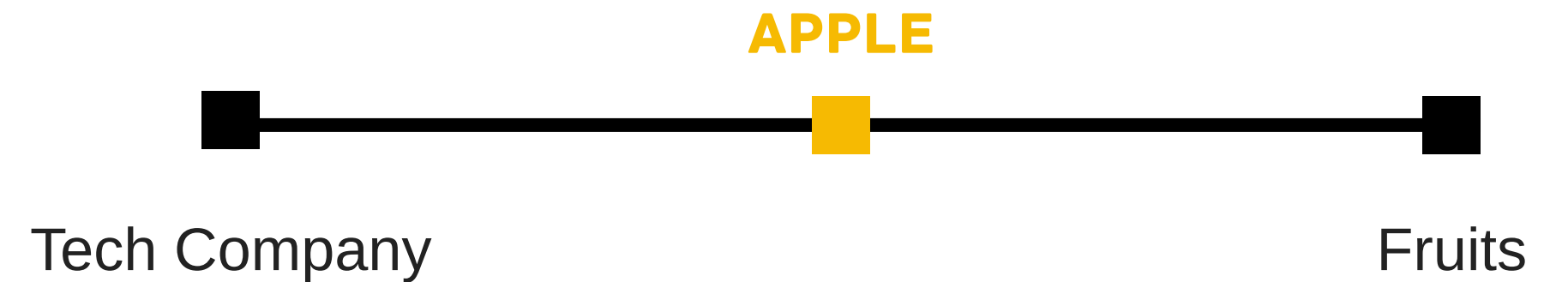
Shayan Hassan Baig



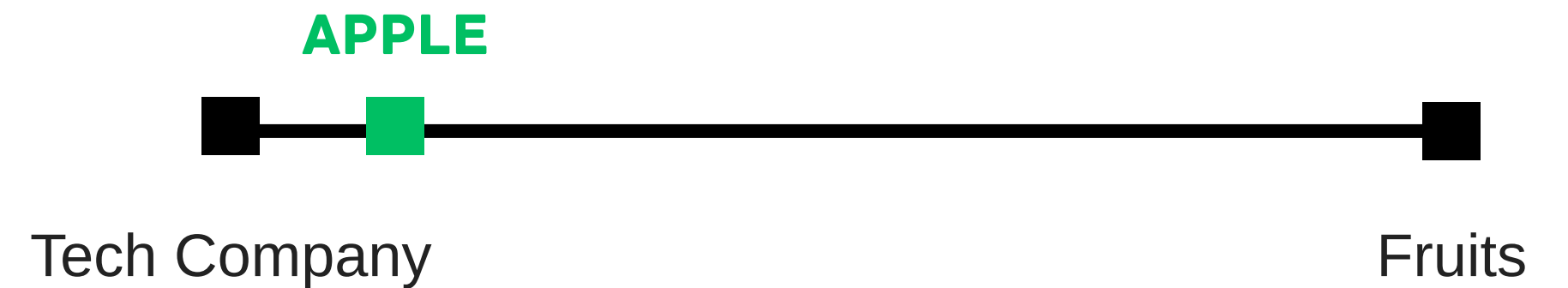
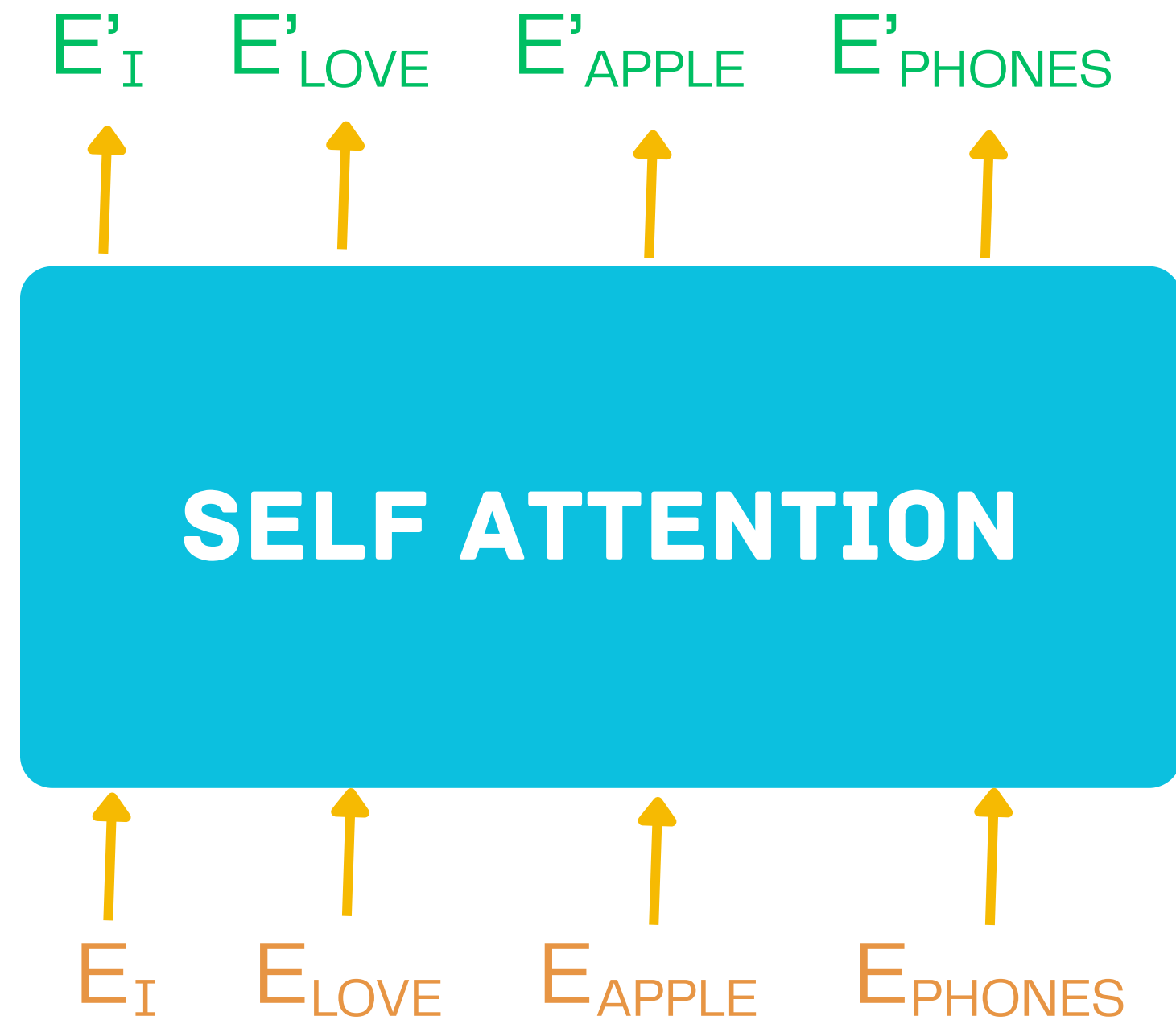


MULTI-HEADED ATTENTION

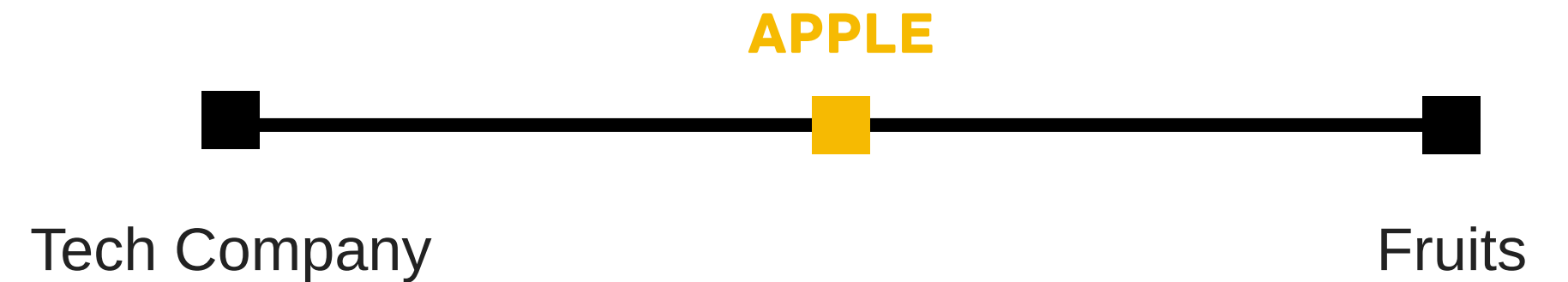
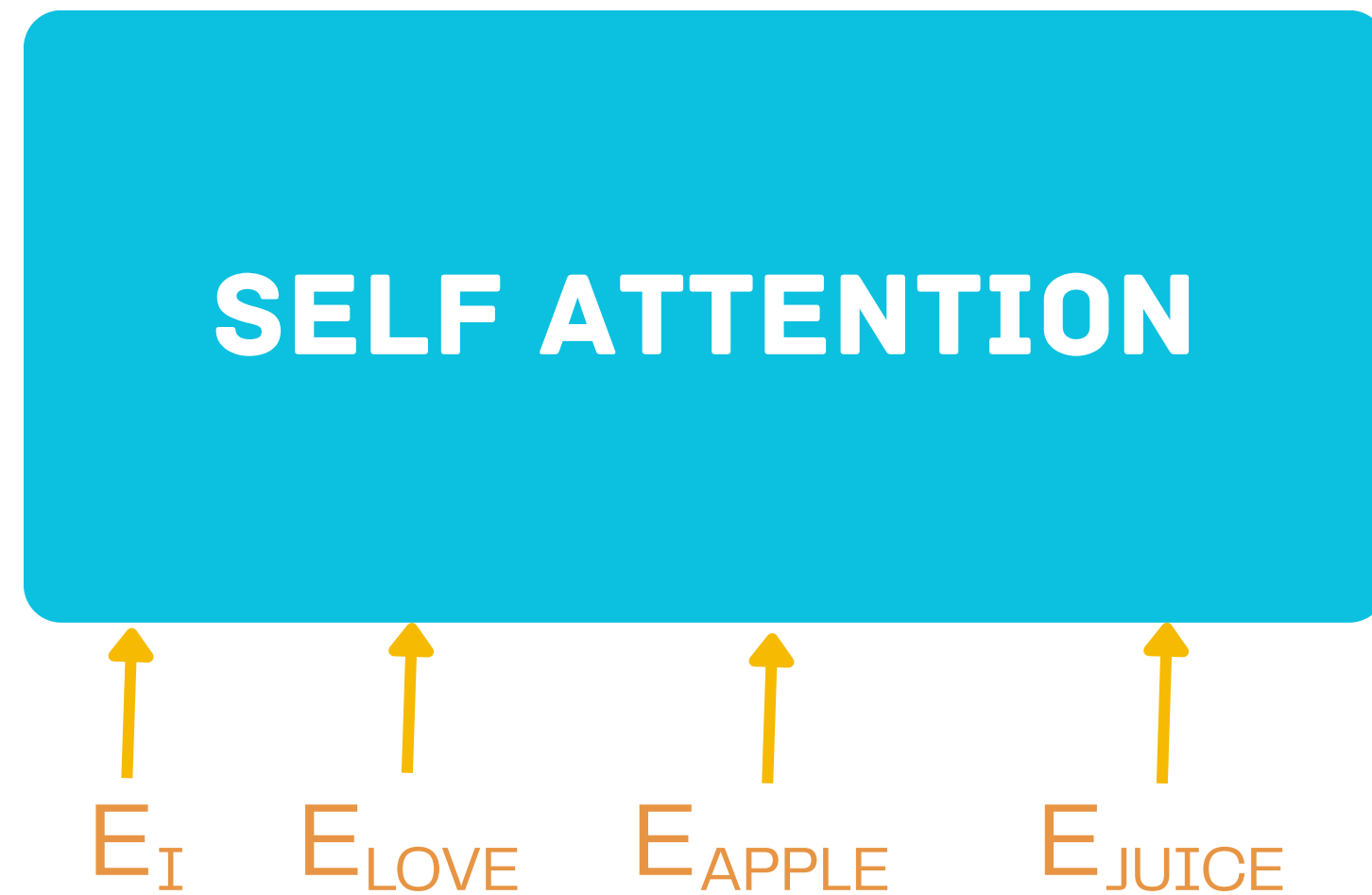
Self Attention - Brief Overview



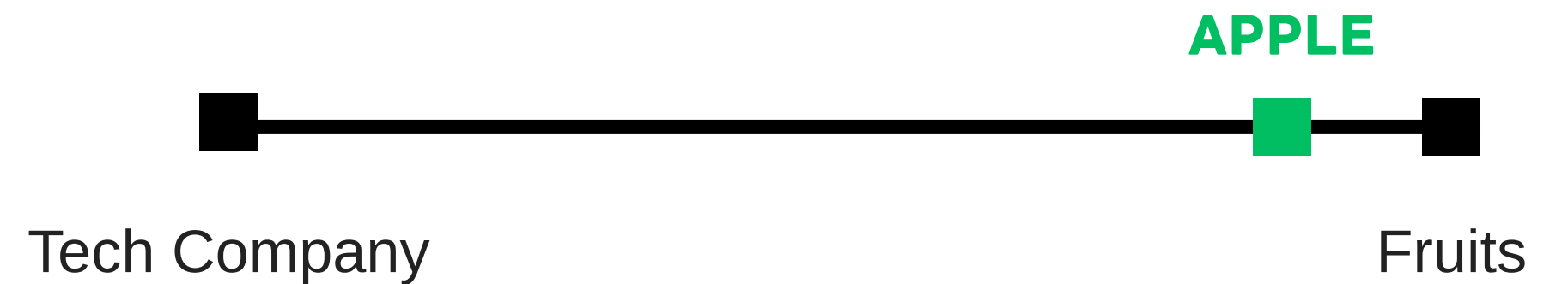
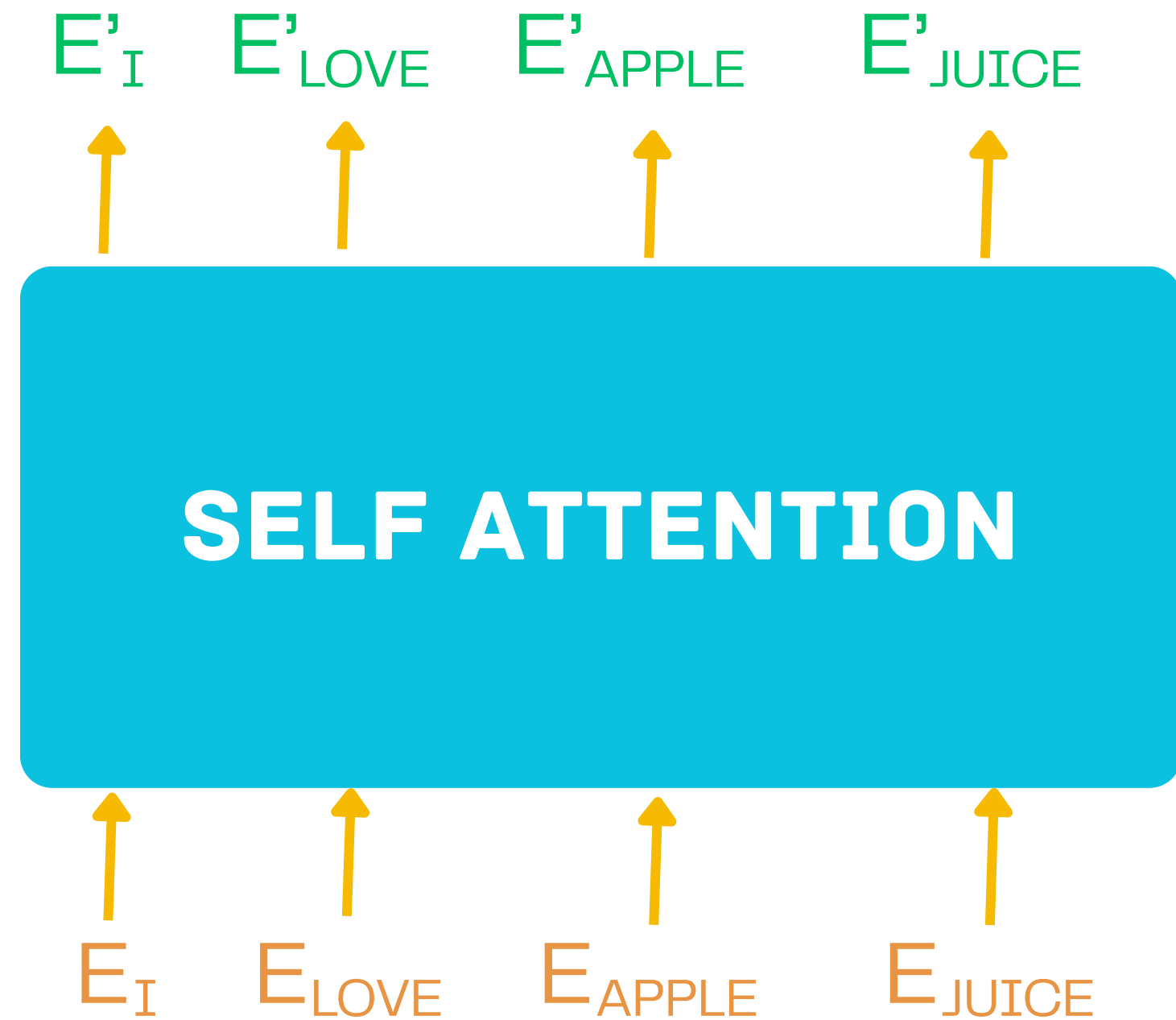
Self Attention - Brief Overview

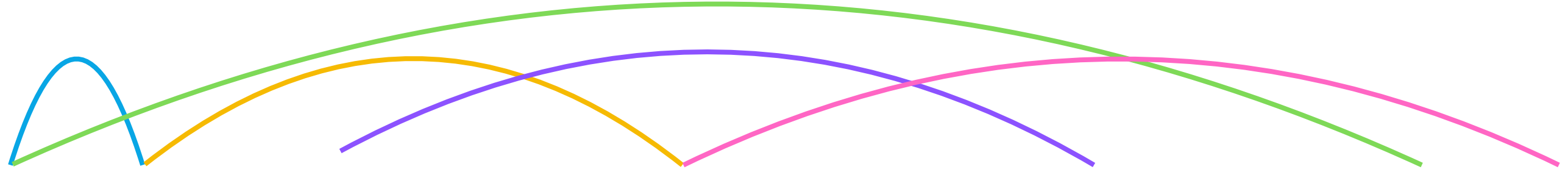


Self Attention - Brief Overview



Self Attention - Brief Overview





When Jake first laid eyes on the Apple iPhone, resting on the front shelf, he wasn't expecting much. It felt like holding the future. His old phone had served him well, and he wasn't one to chase the latest tech trends. But the moment he held the sleek, glassy device in his hand, there was something undeniably different about it.

She saw a man with the telescope

1- She saw a man who was holding a telescope



2- She used a telescope to see the man



She saw a man with the telescope

1- She saw a man who was holding a telescope



She saw a man with the telescope

2- She used a telescope to see the man



She saw a man with the telescope

Keys on the table belongs to Mano

Keys on the table belongs to Mano

Spatial Relationship



Keys on the table belongs to Mano

Spatial Relationship



Possession



- - - - -



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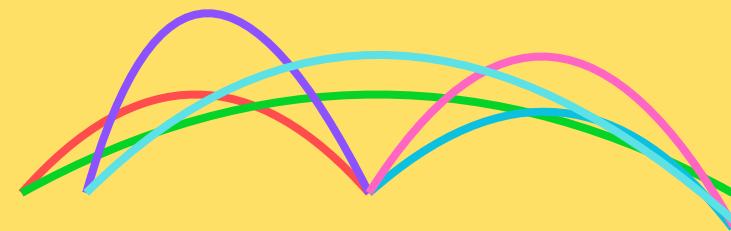
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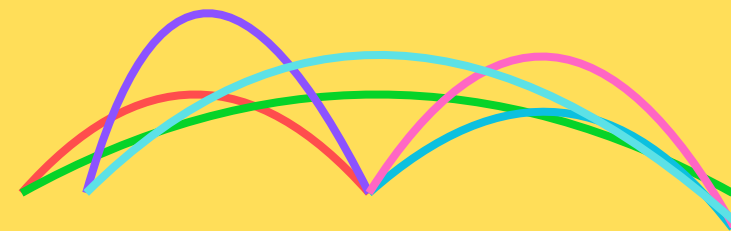
LOVE APPLE PHONES

Self Attention



LOVE APPLE PHONES

Self Attention



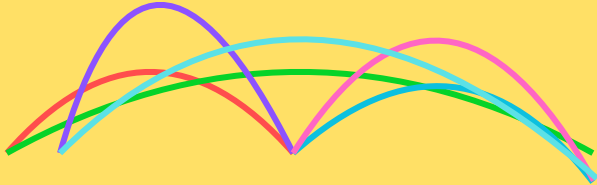
LOVE APPLE PHONES

Self Attention

Spatial Relationship

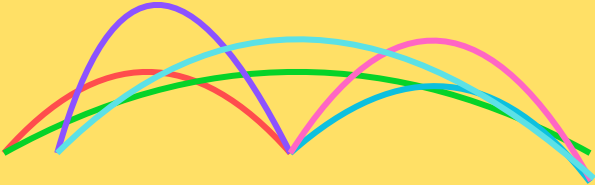
Subject verb object

Time



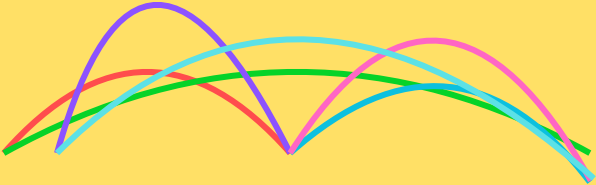
LOVE APPLE PHONES

Self Attention



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Self Attention

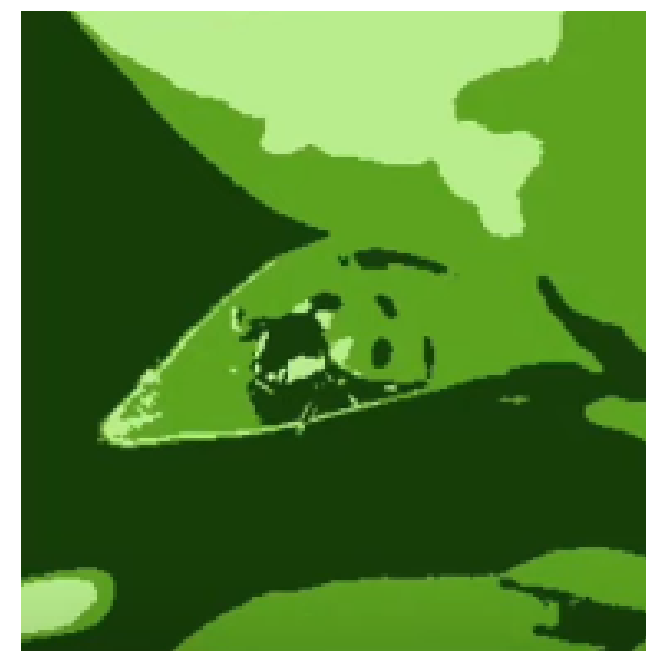
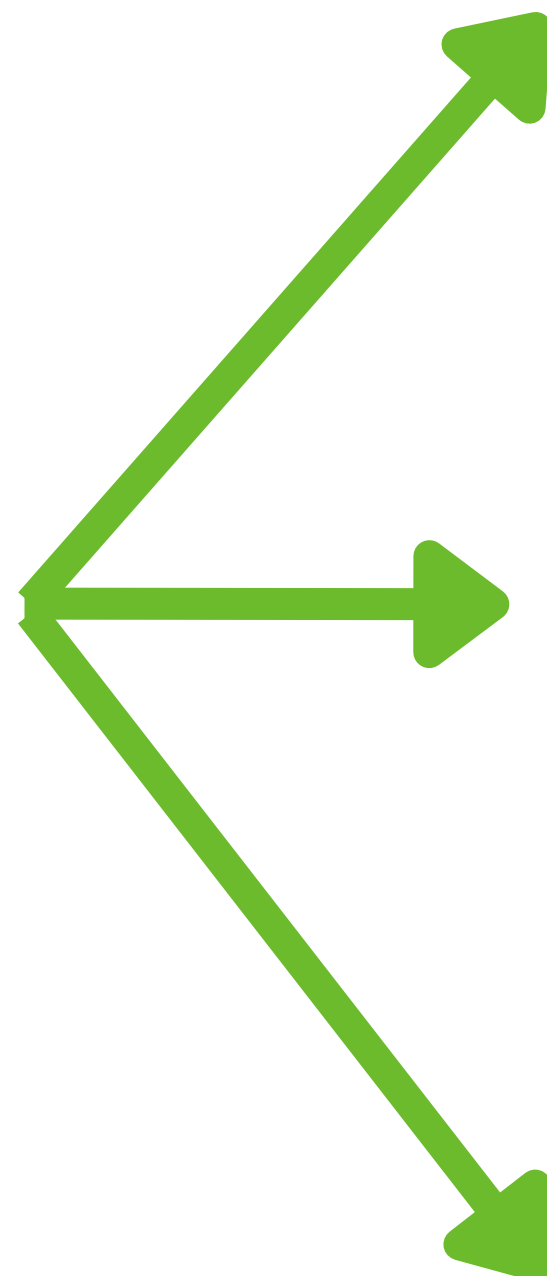
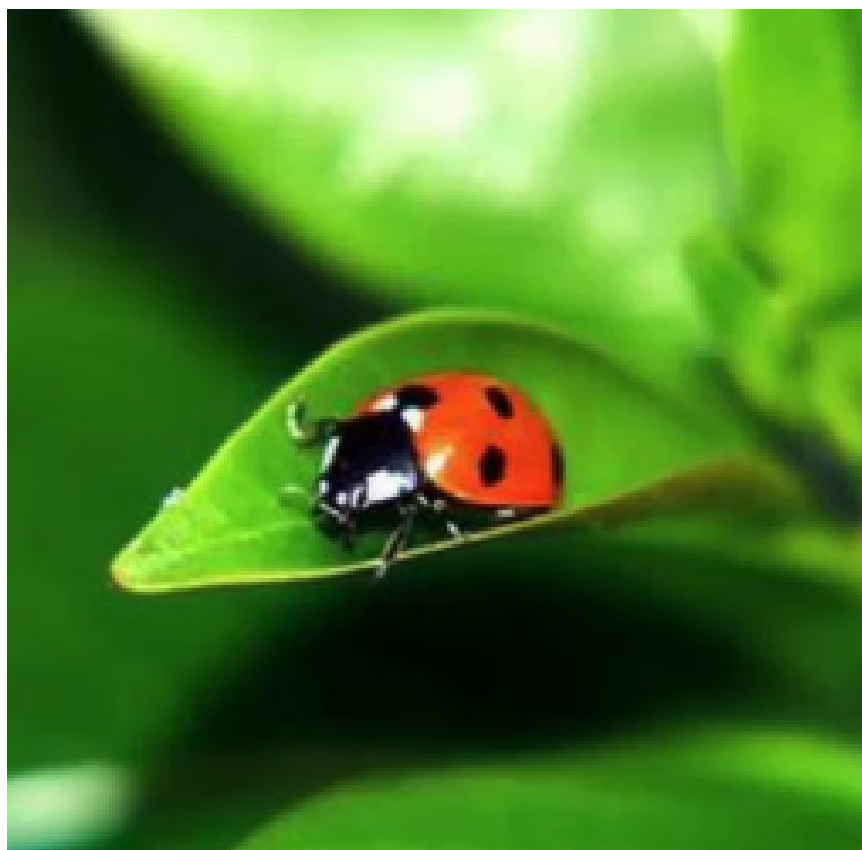


LOVE APPLE PHONES

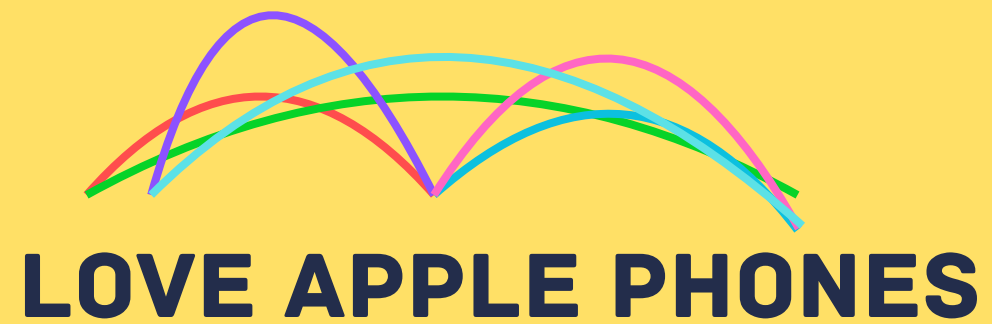
Self Attention





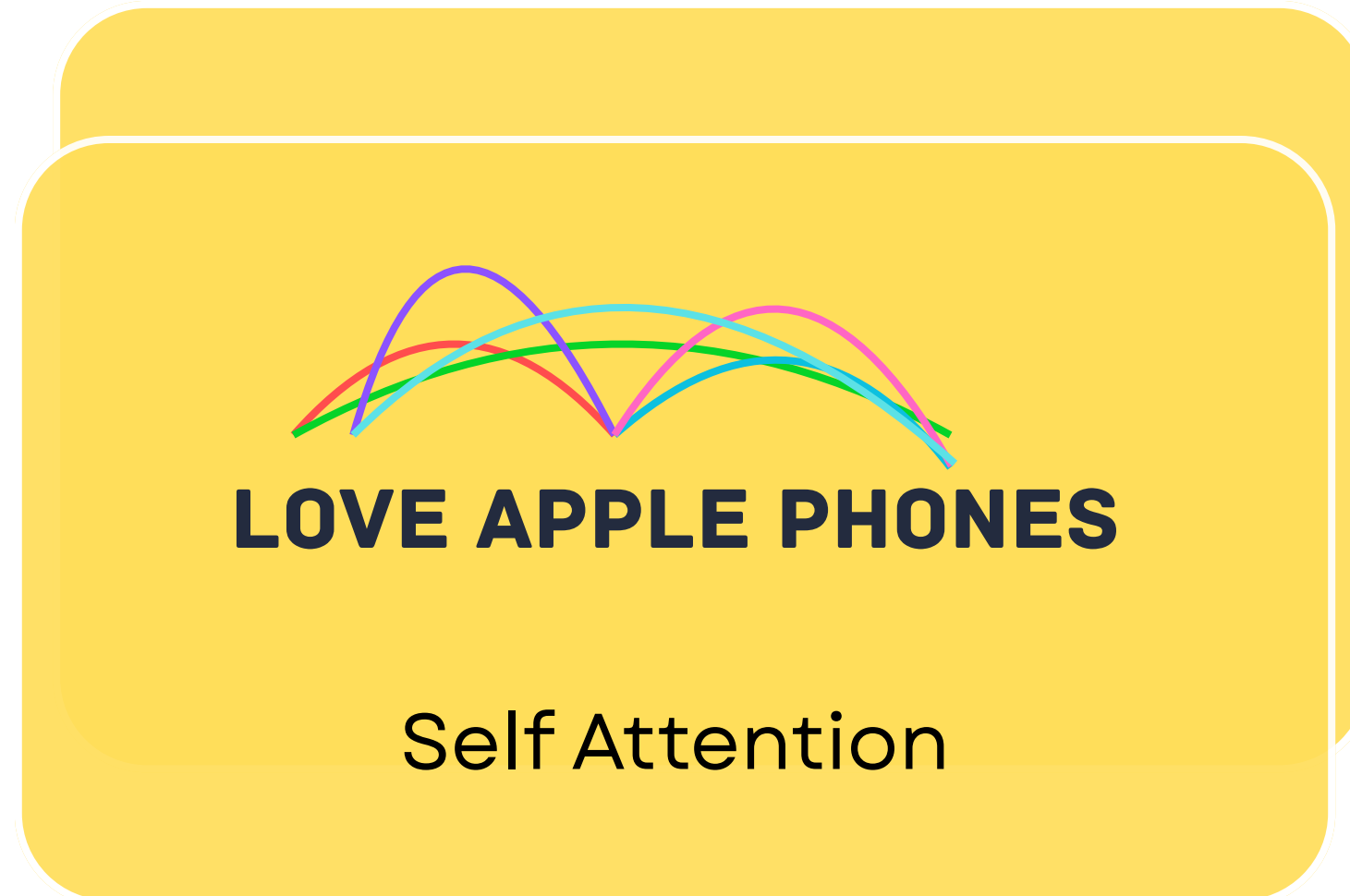


Keys on the table belong to Mano. She must have left them there before heading out for work. The table, cluttered with books and papers, made them easy to overlook. I decided to place them in the drawer for safekeeping. Hopefully, she'll remember to grab them when she gets back home.

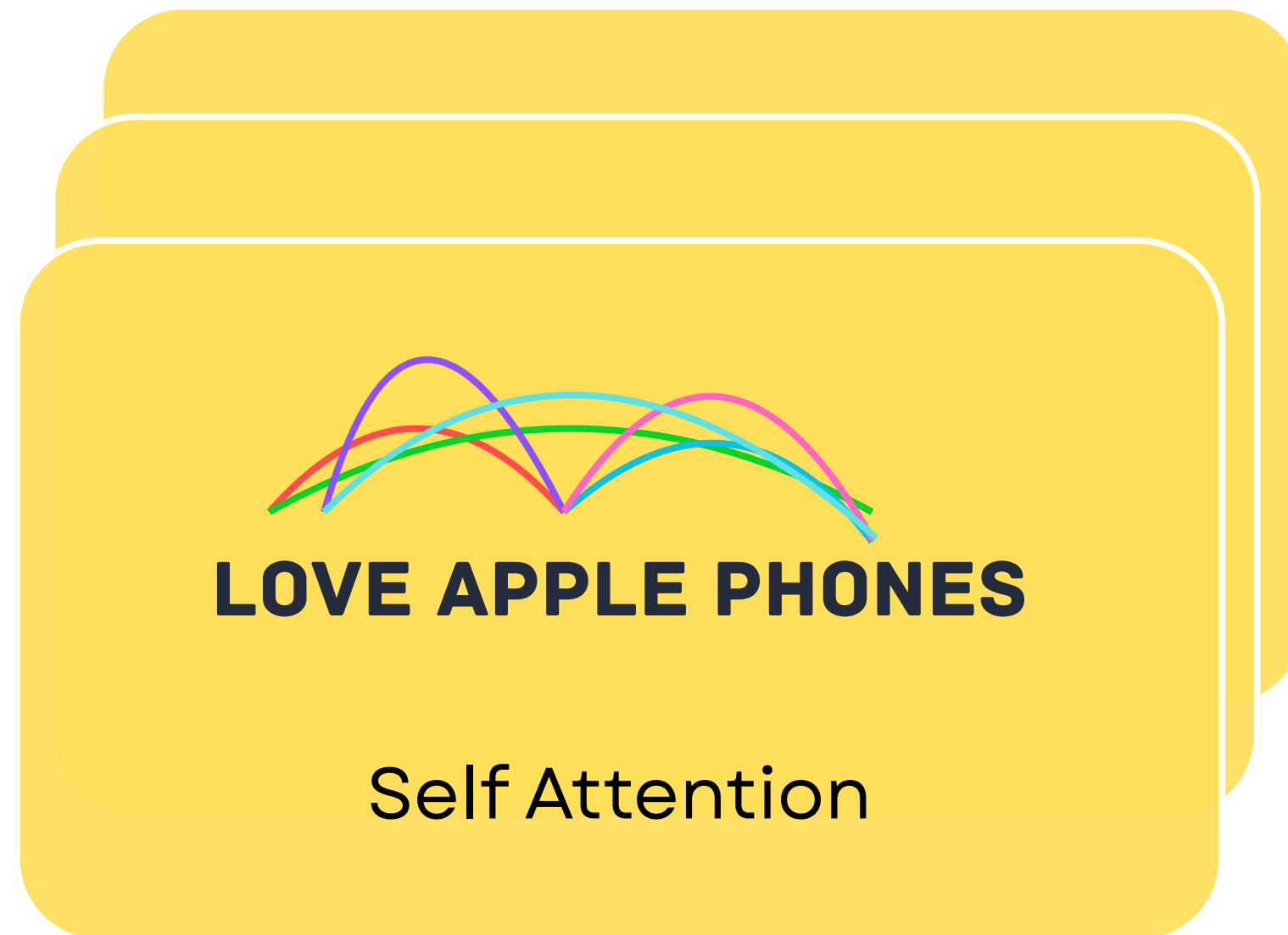


Self Attention

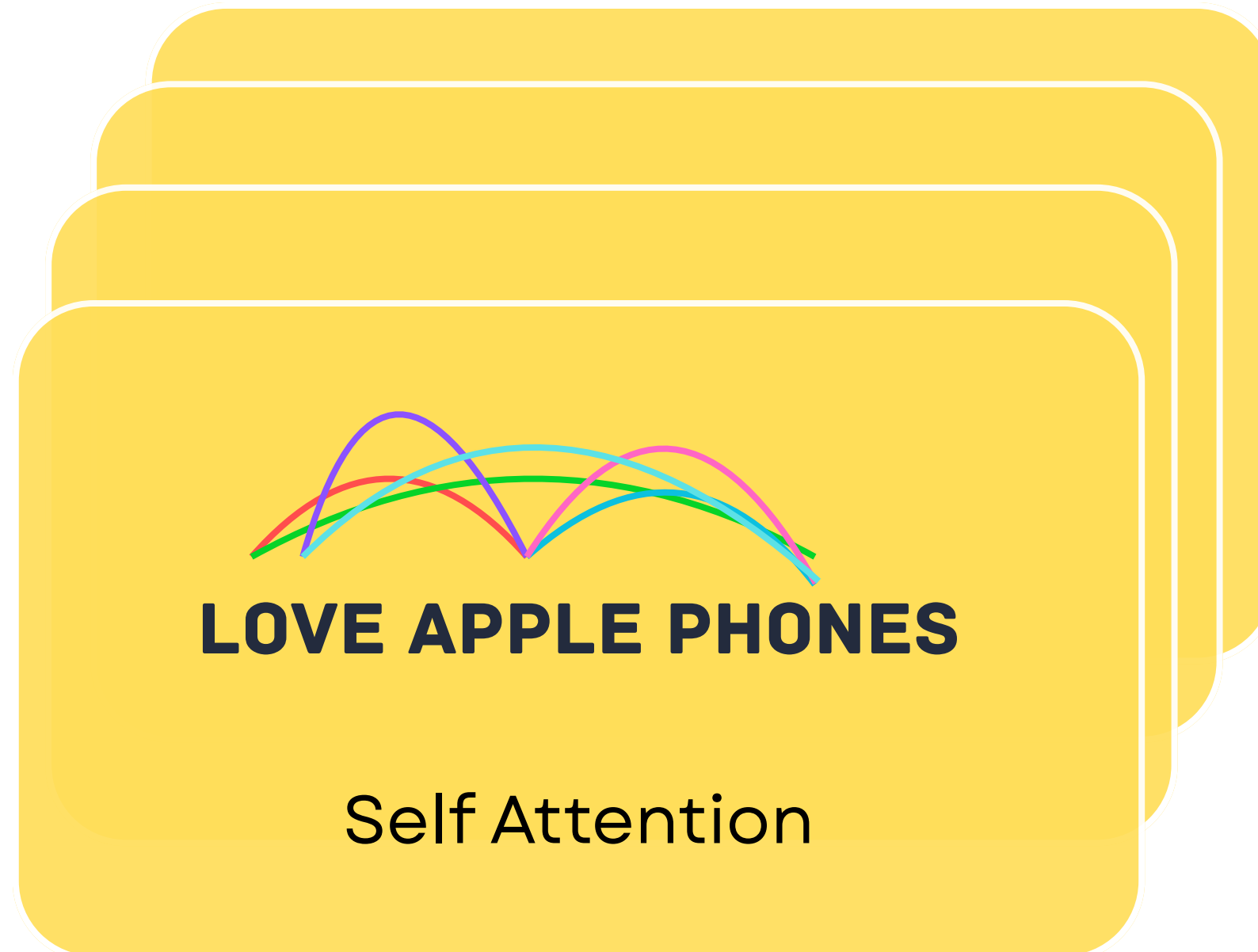
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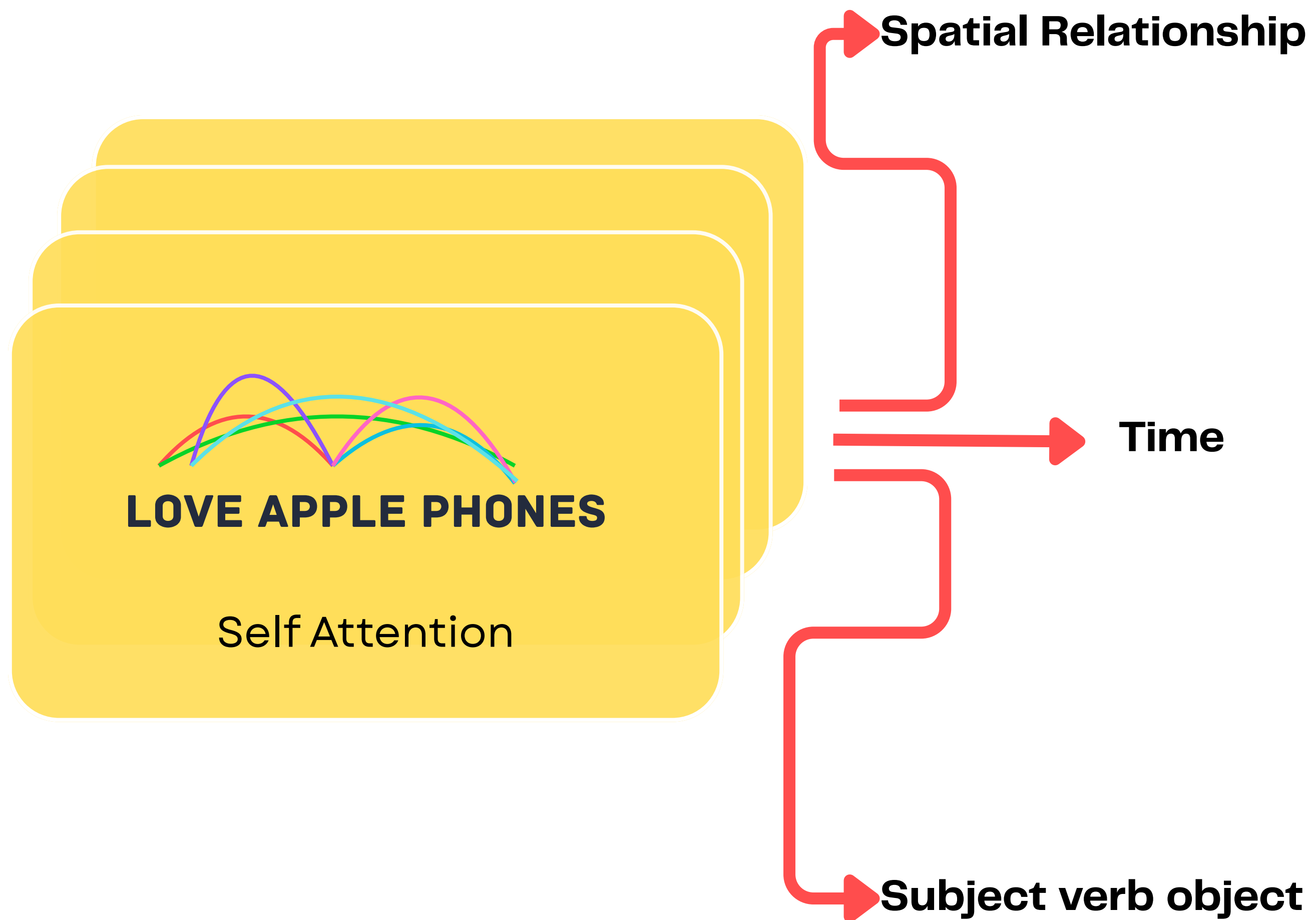
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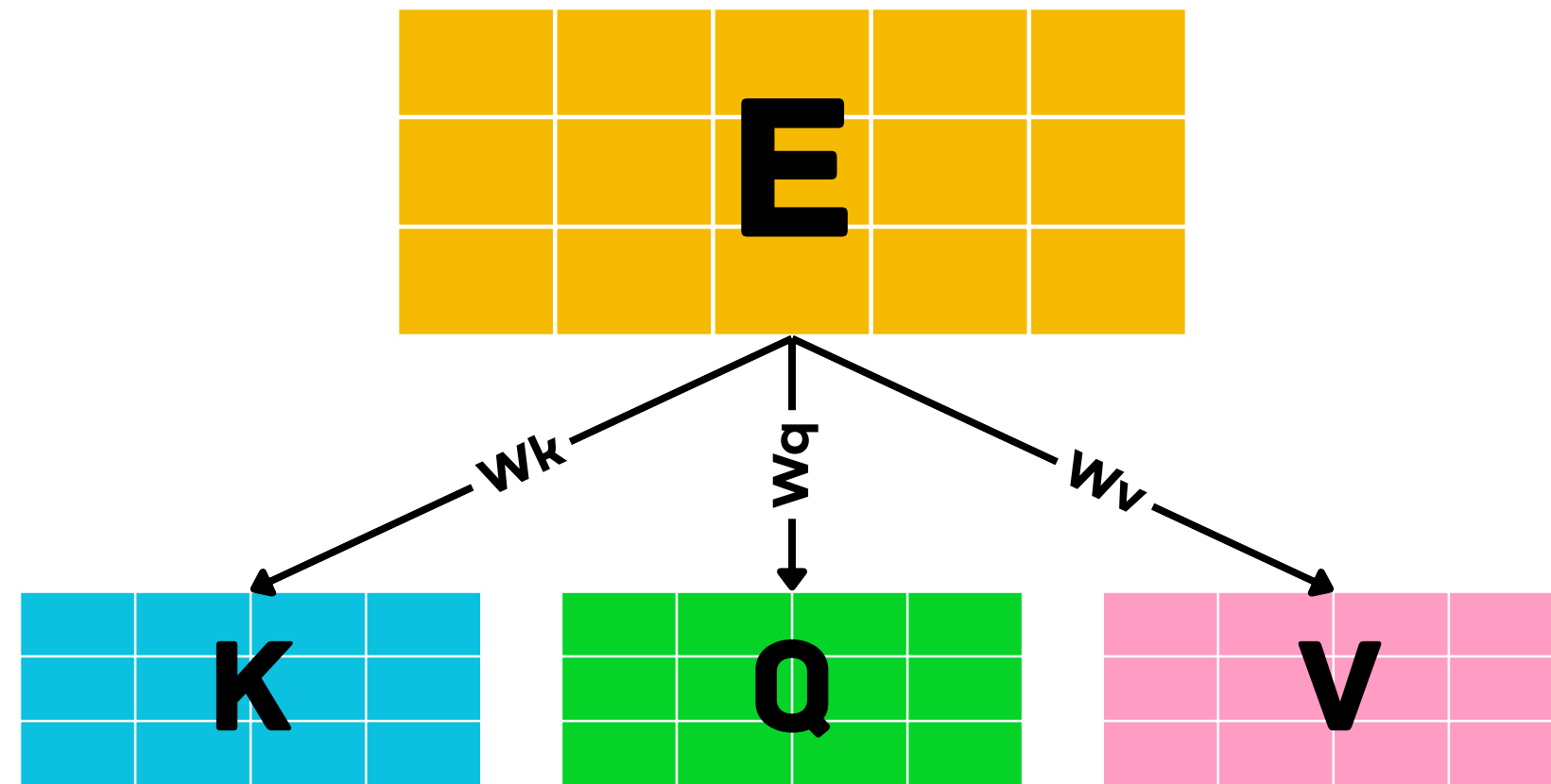
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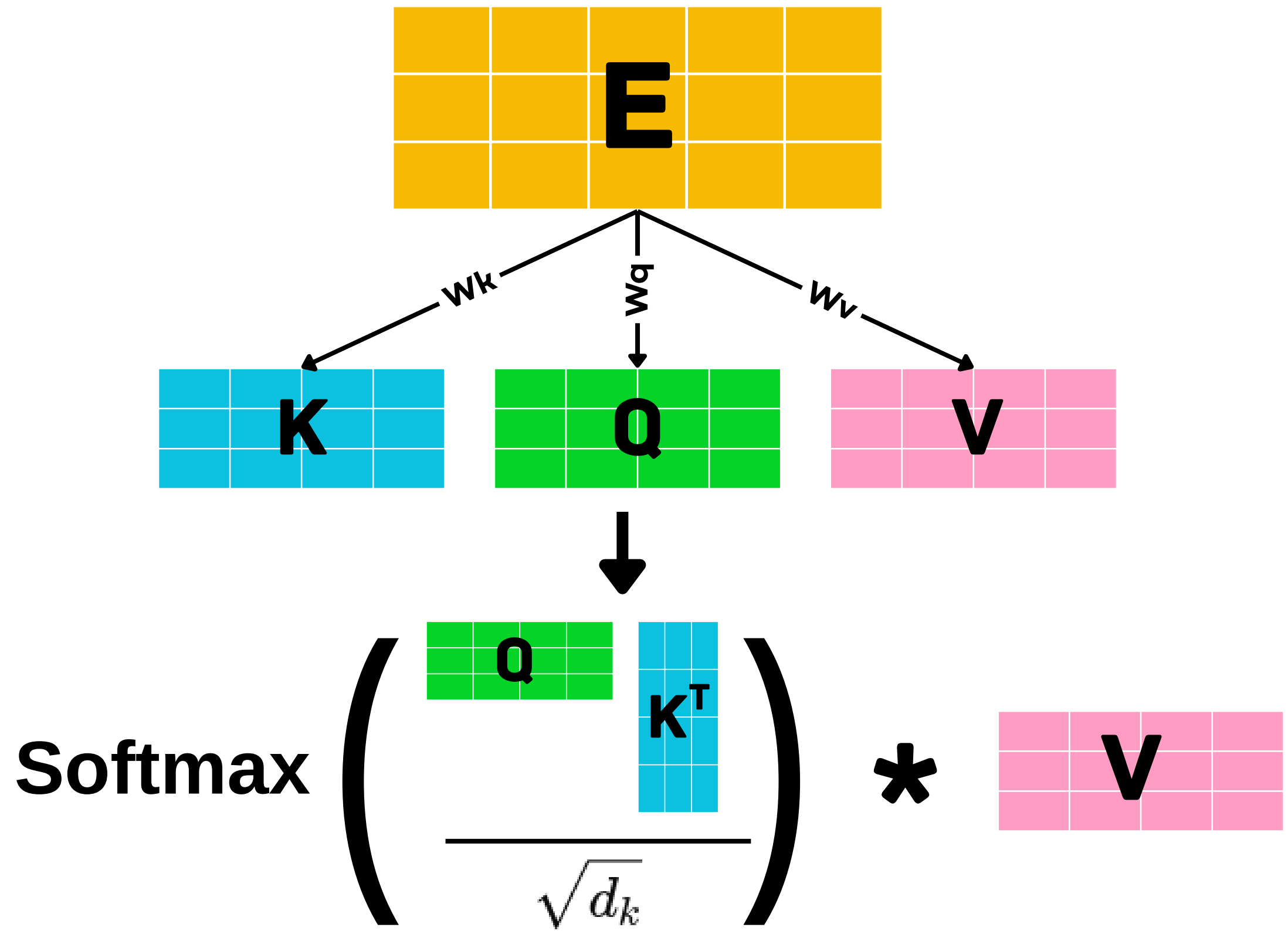


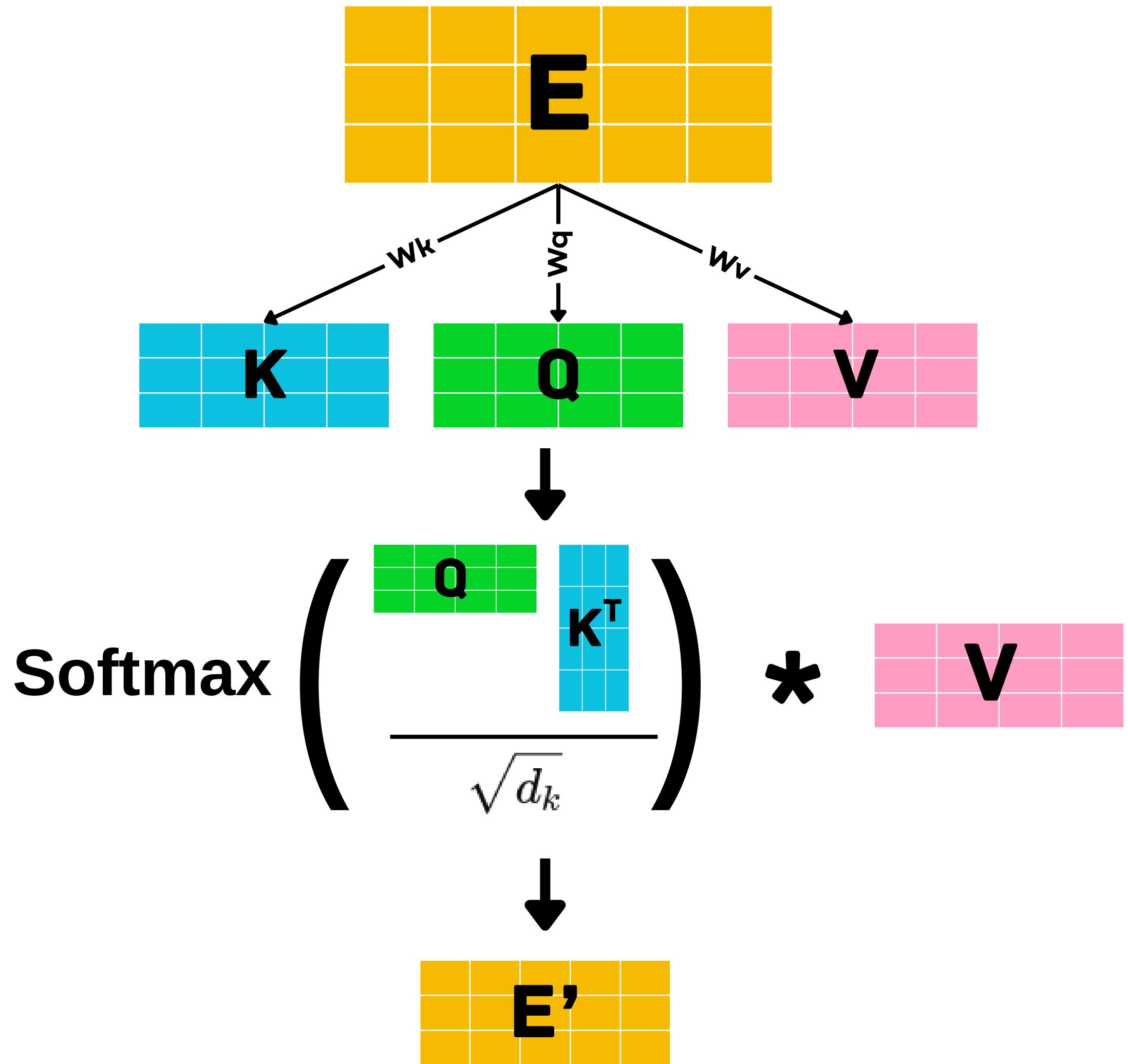
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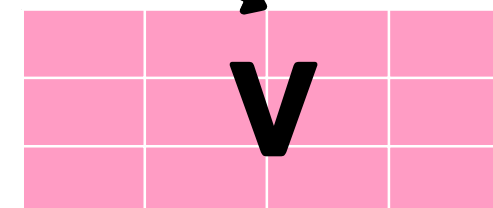
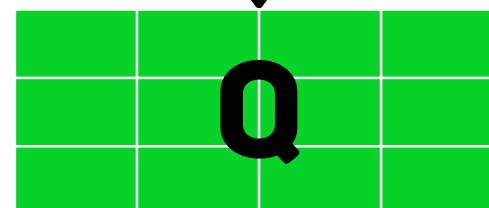
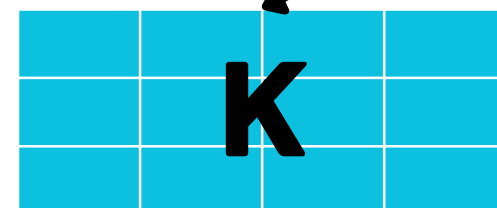
love
apple
phones



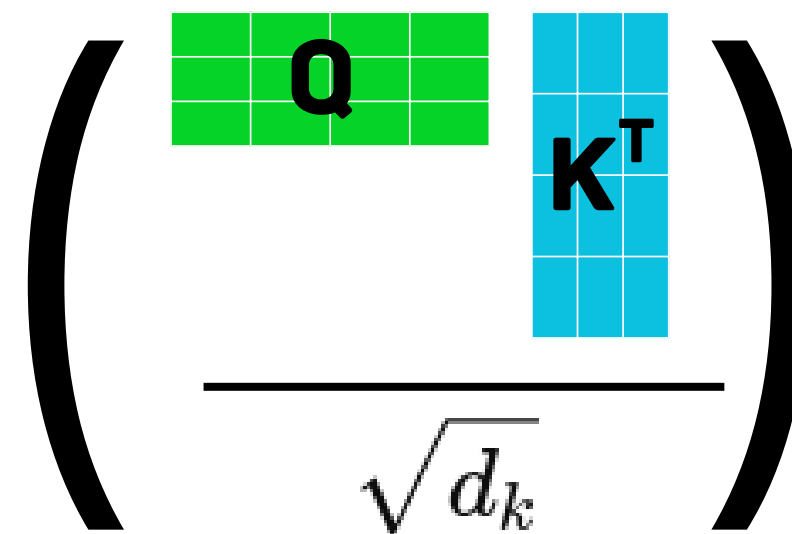
w_k

w_q

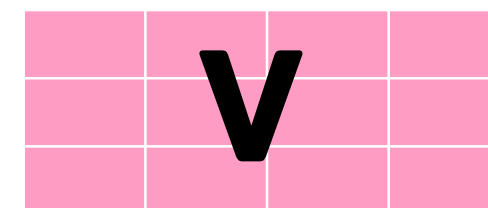
w_v



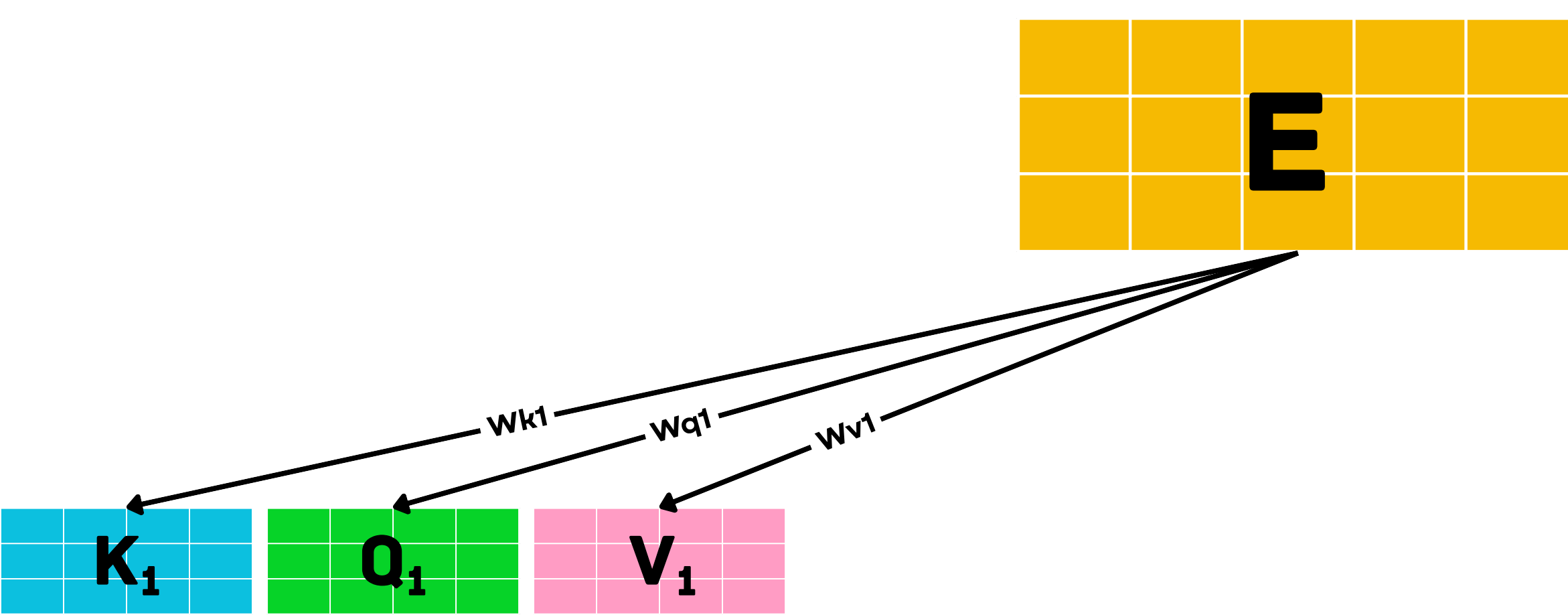
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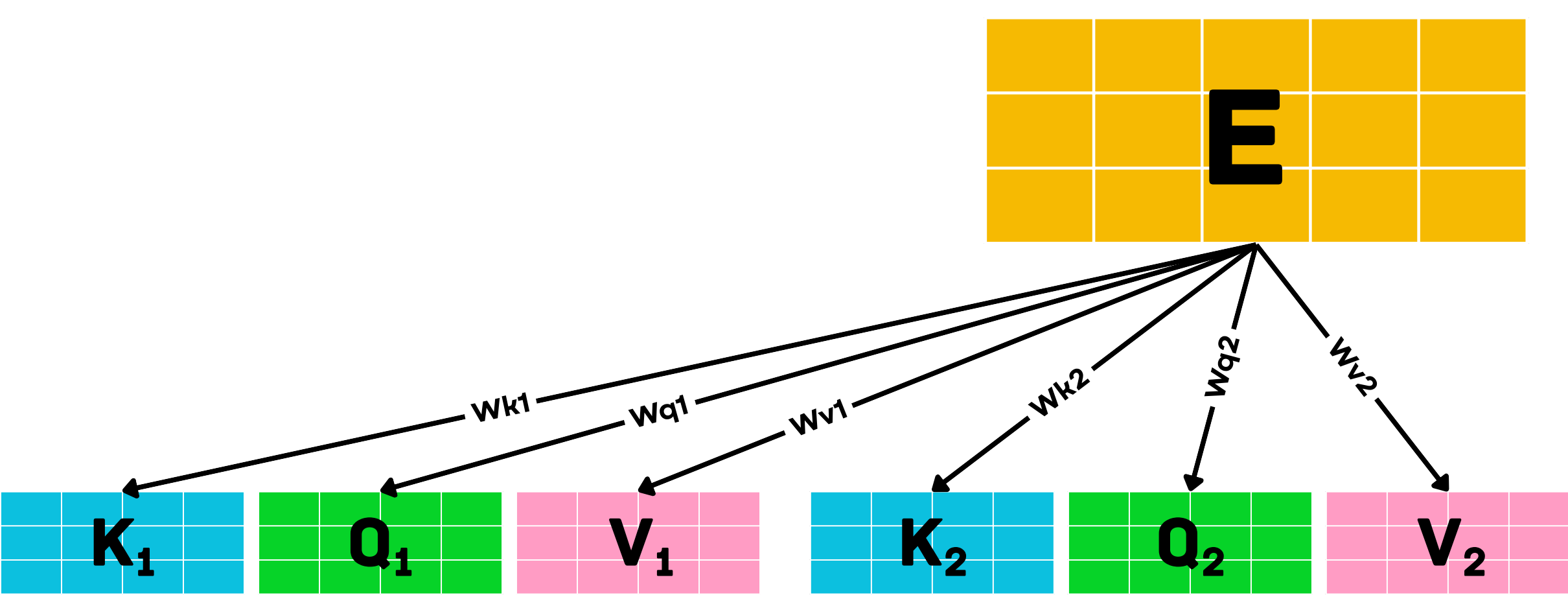


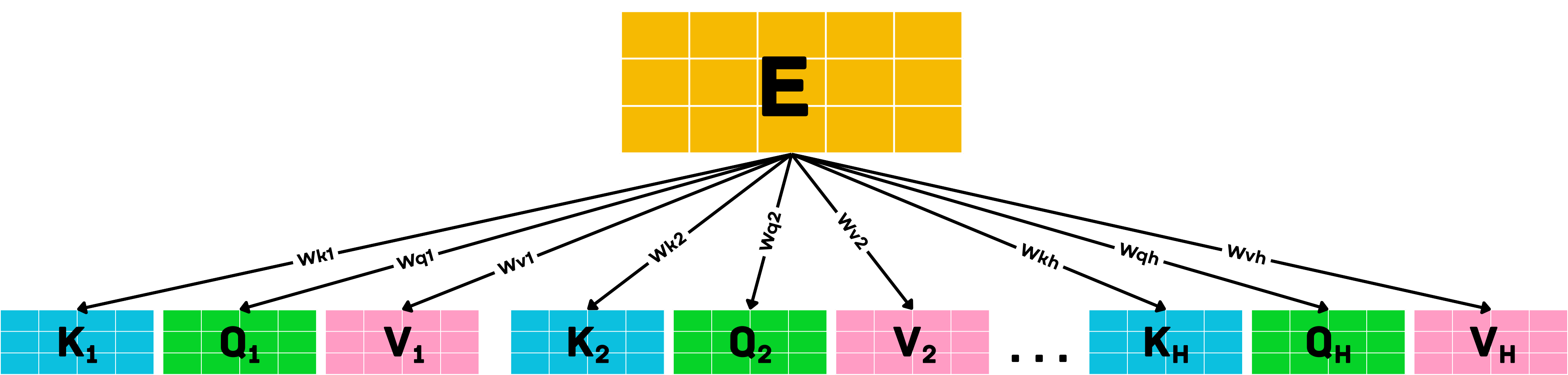
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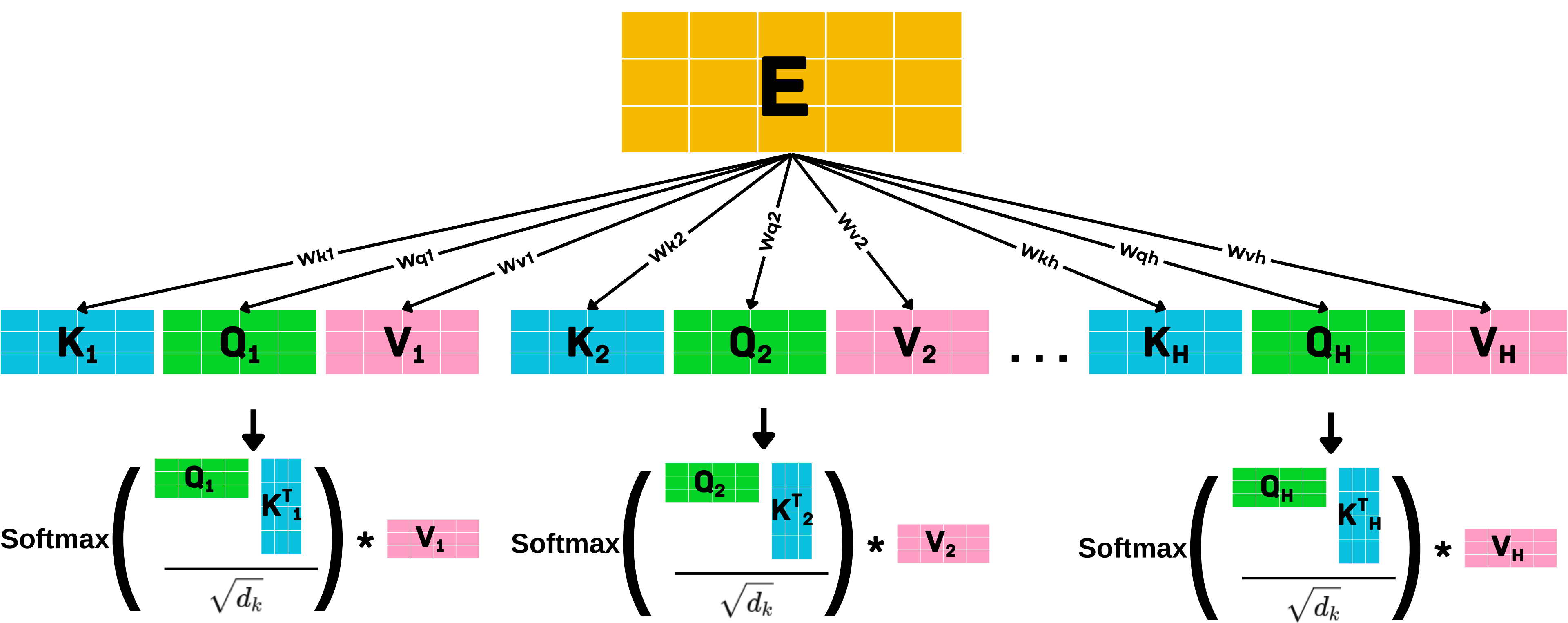


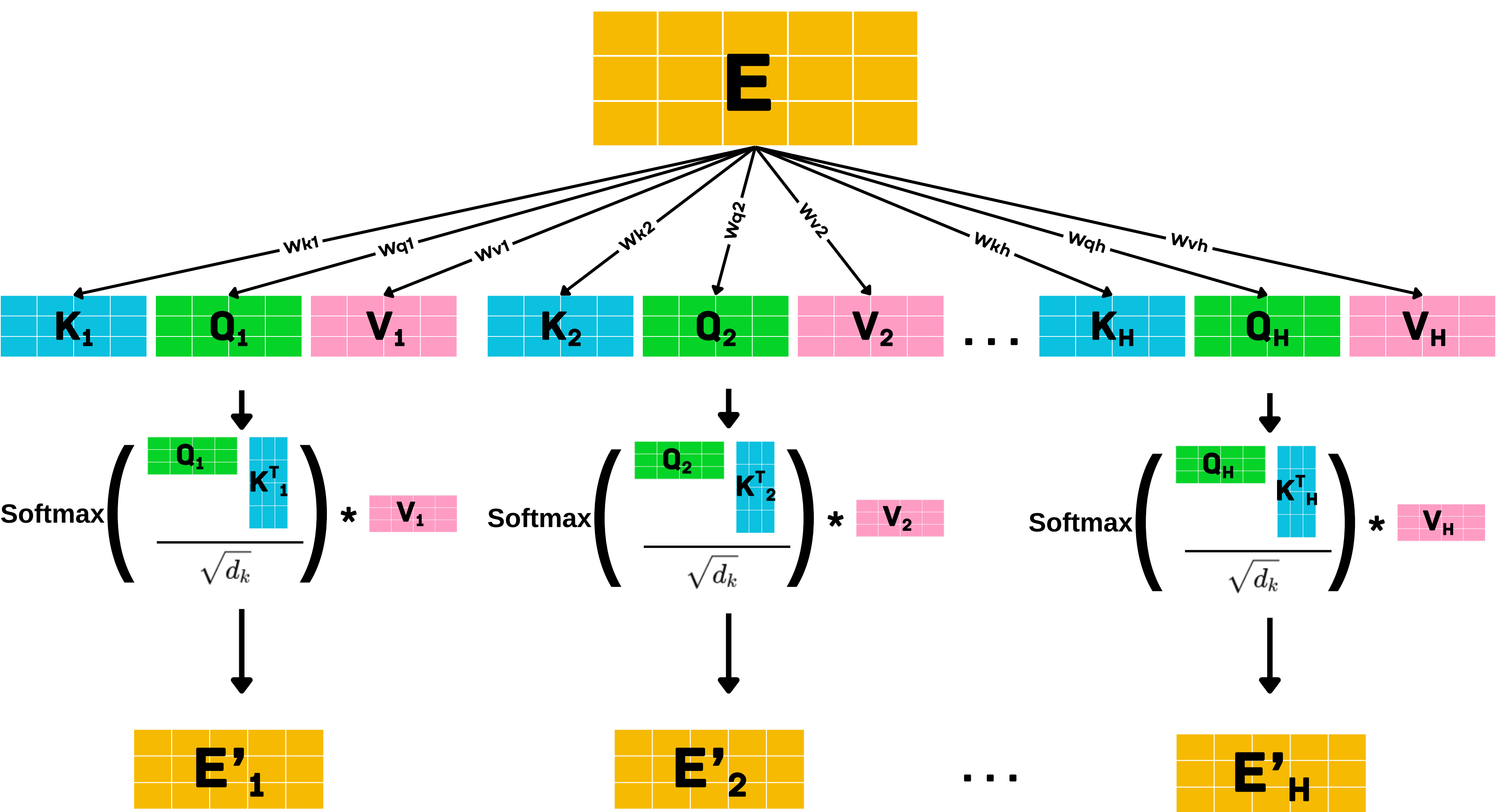


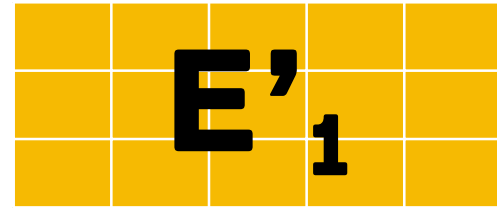








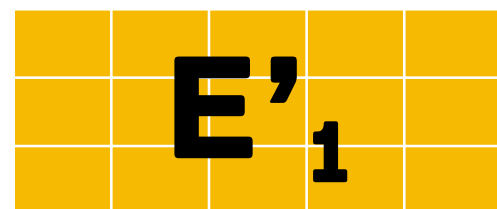




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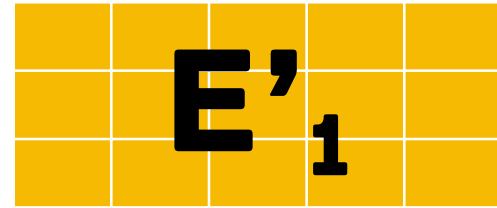
When Jake first laid eyes on the Apple iPhone resting on the front shelf he wasn't expecting much. It felt like holding the future. His old phone served him well, and he wasn't one to chase the latest tech trends. But the moment he held the sleek, glassy device in his hands, there was something different about it.



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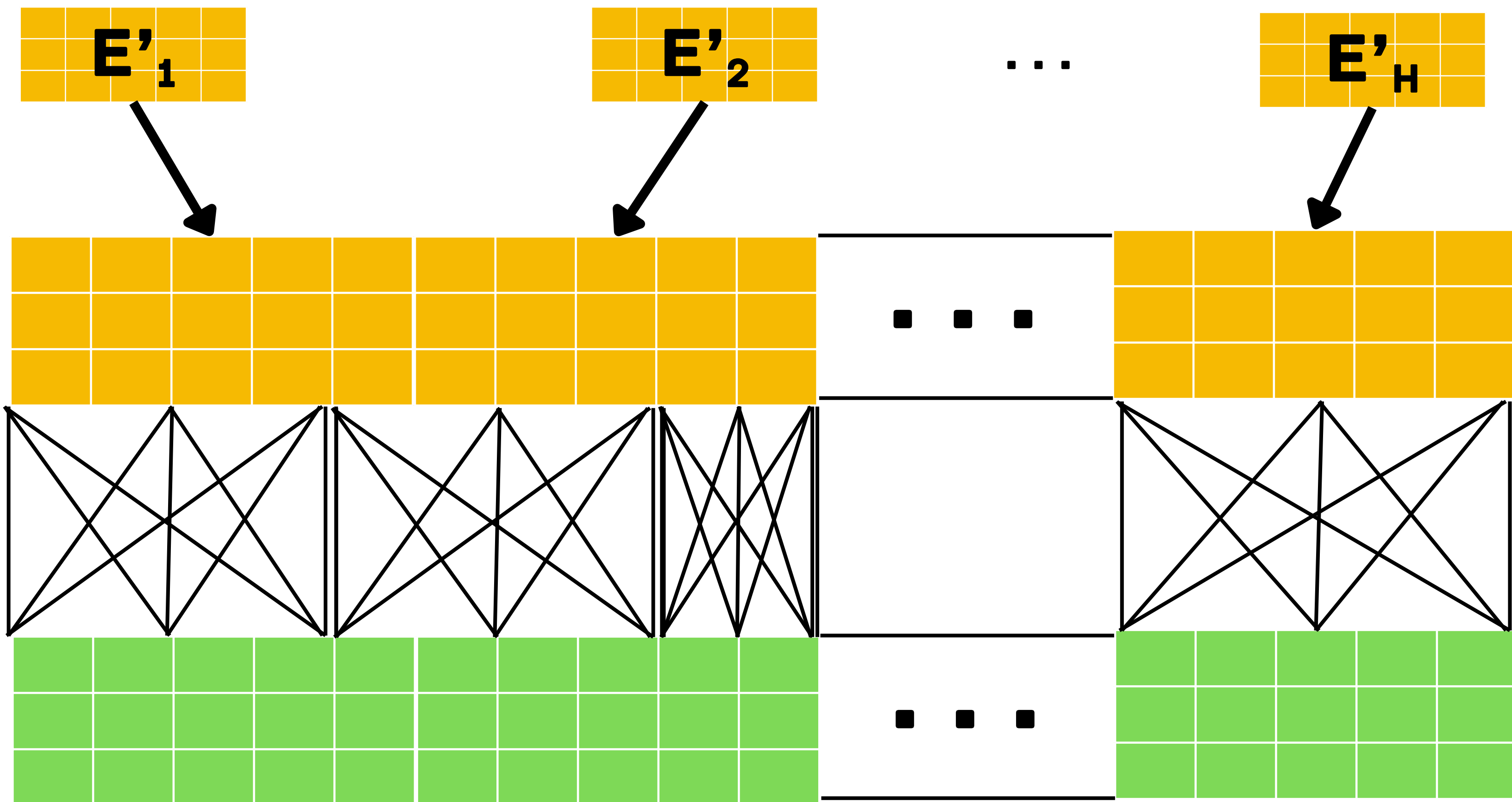
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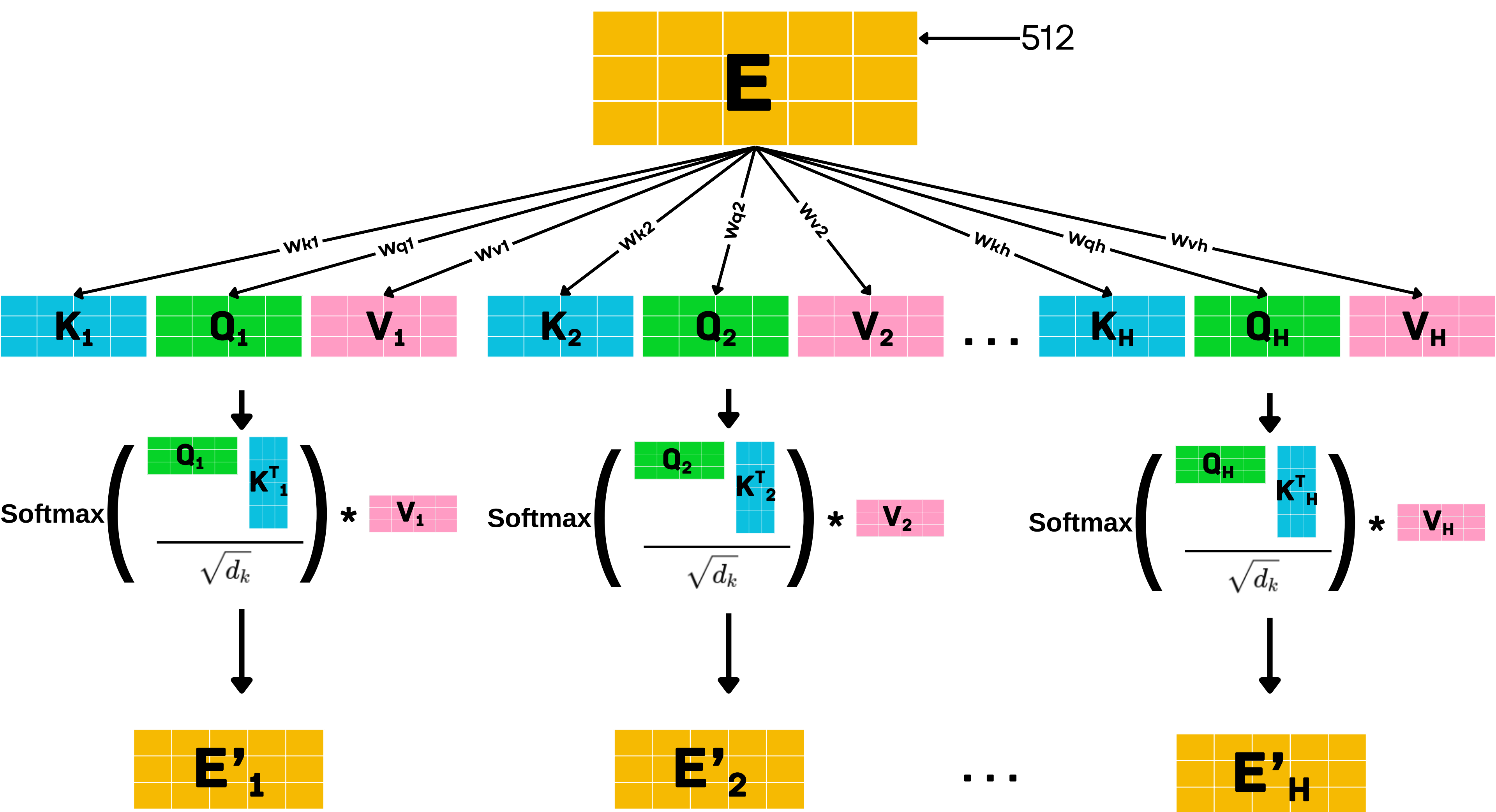


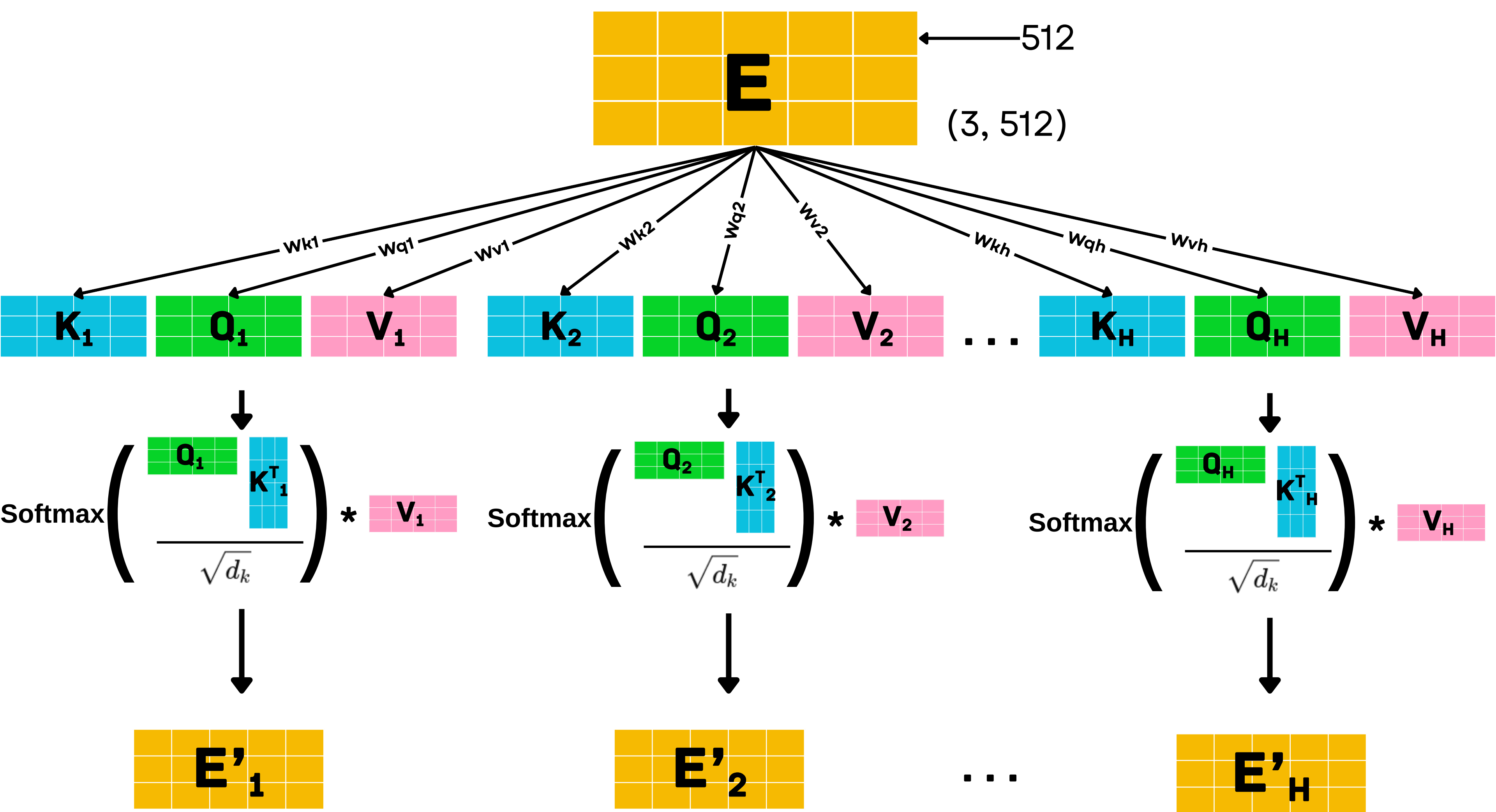
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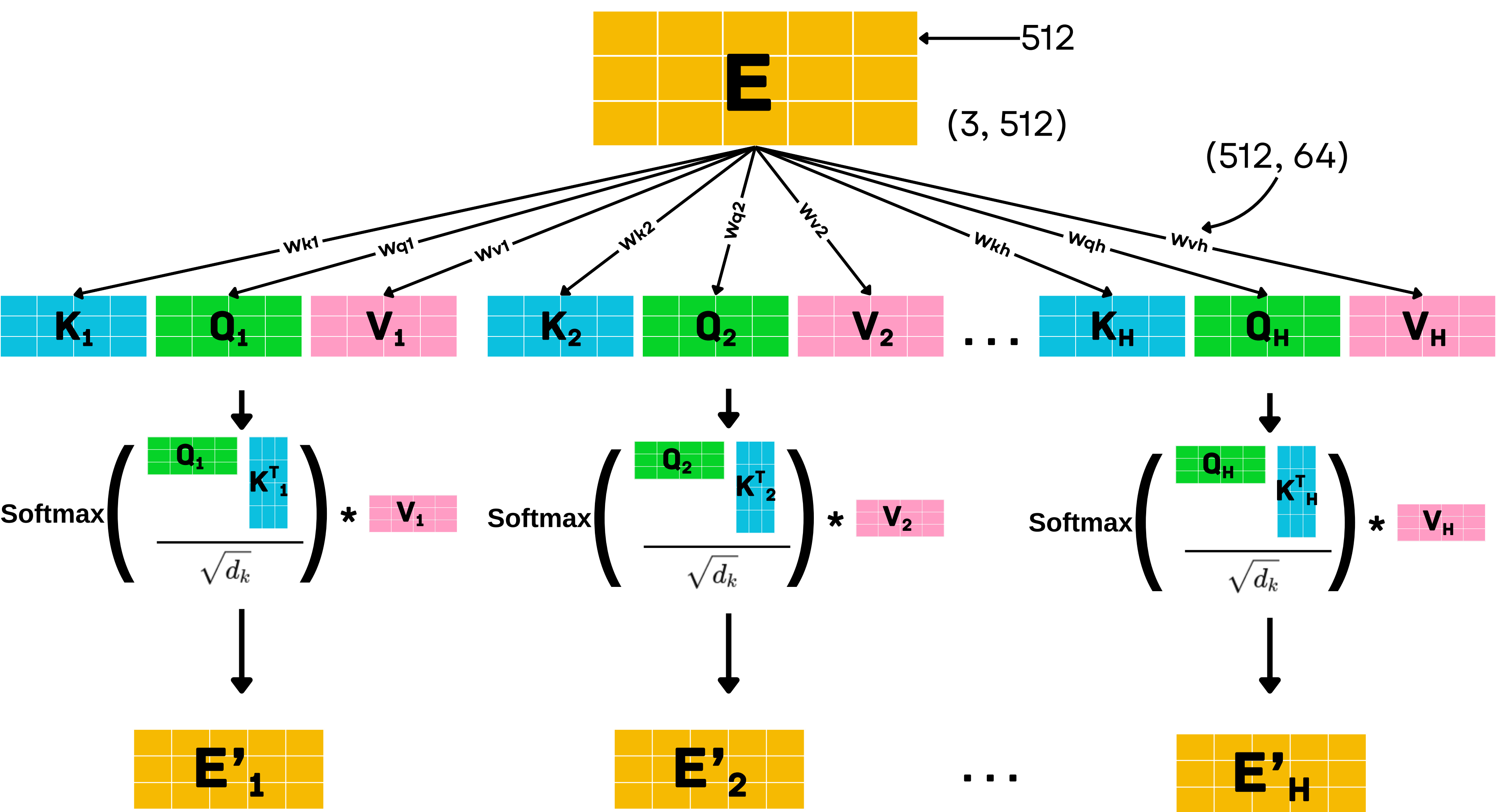


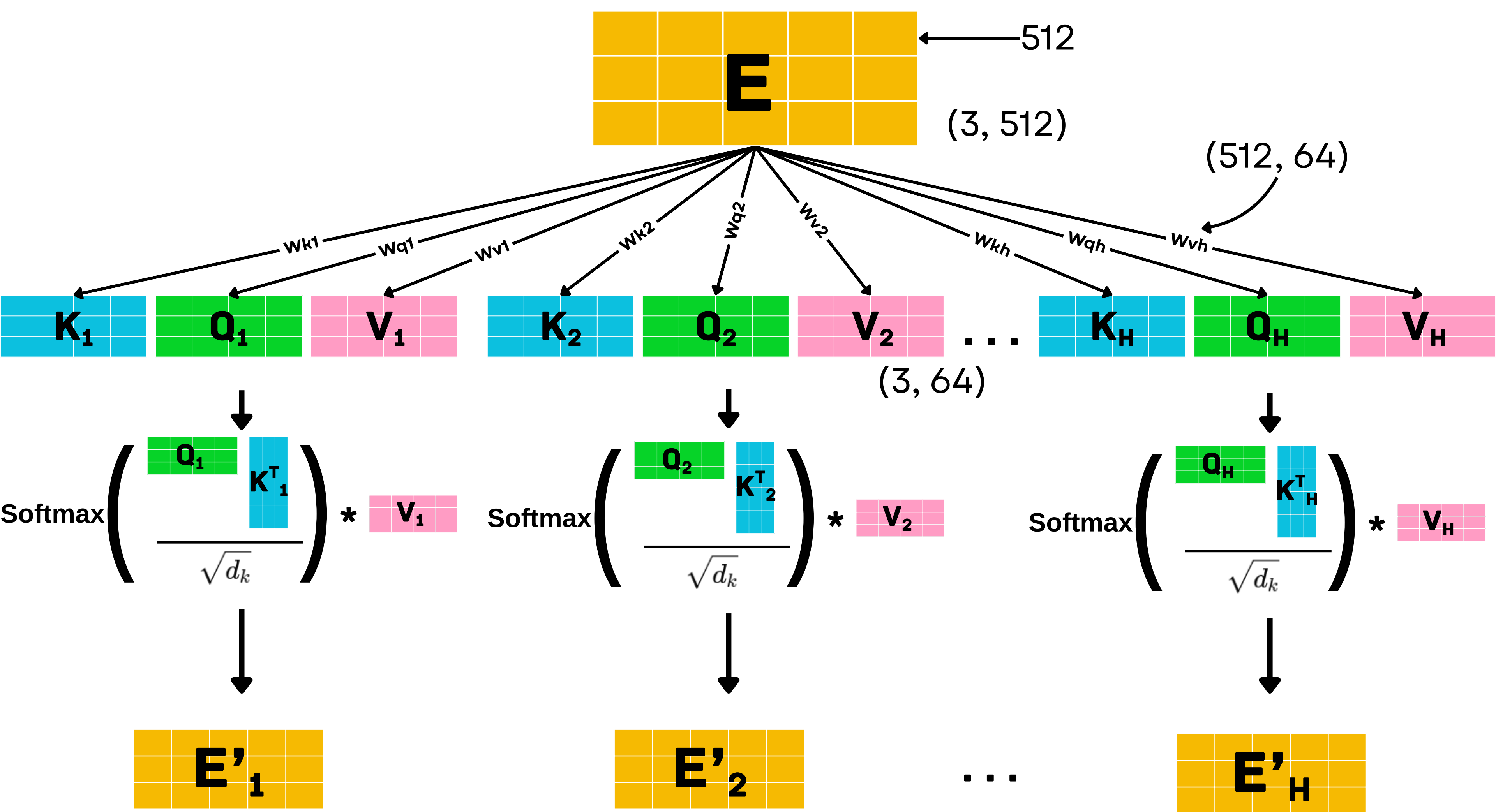
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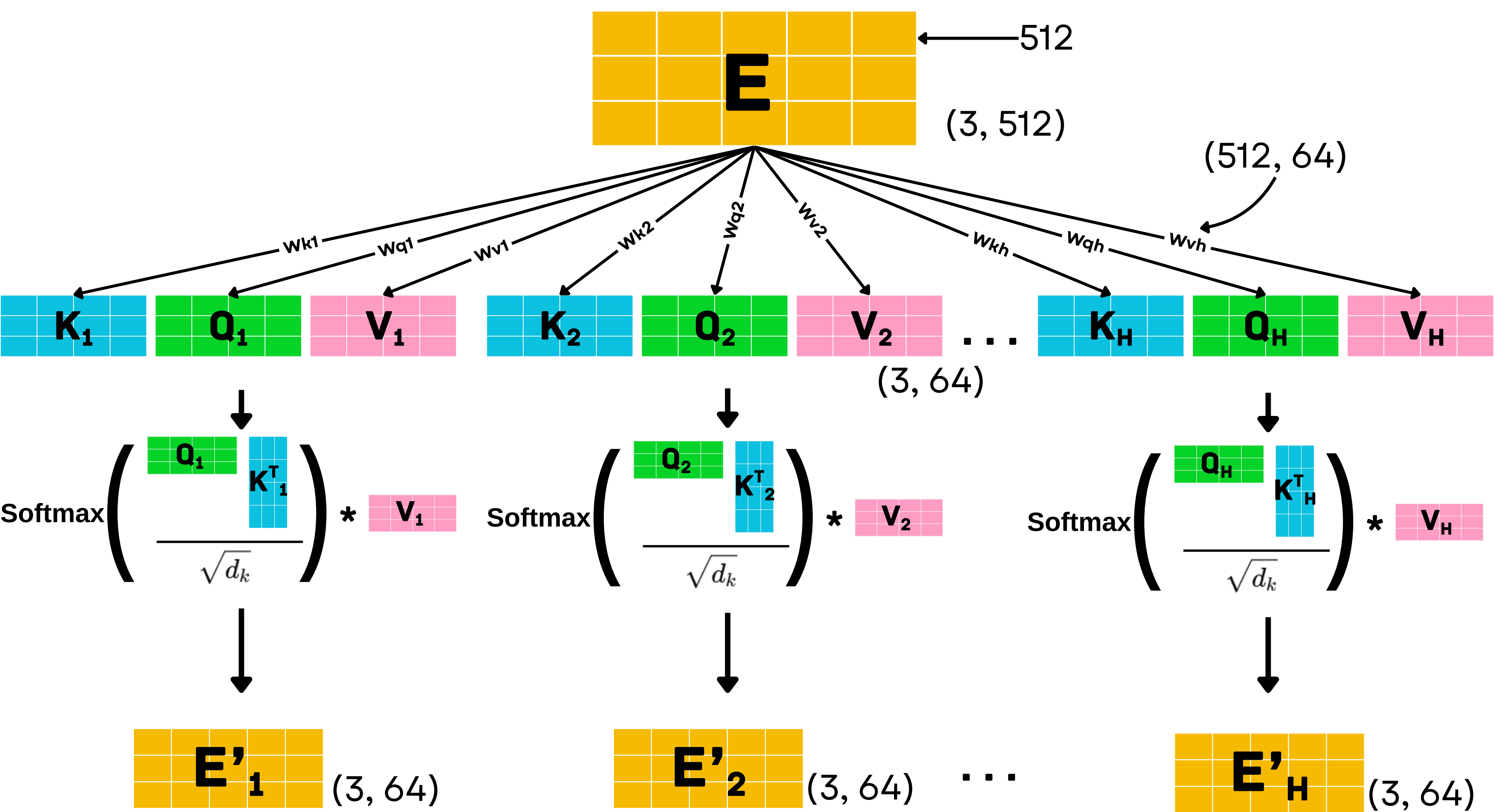


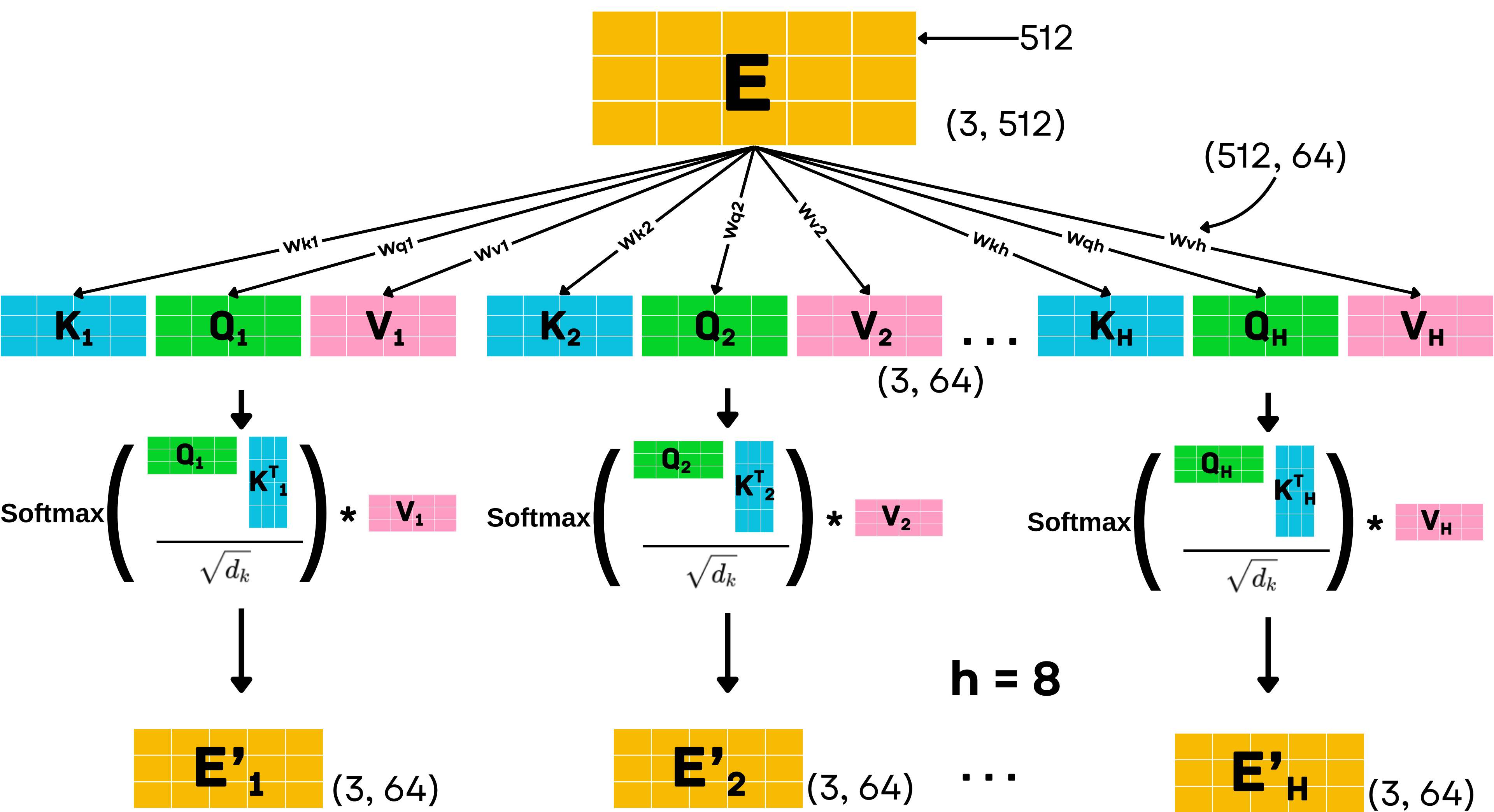


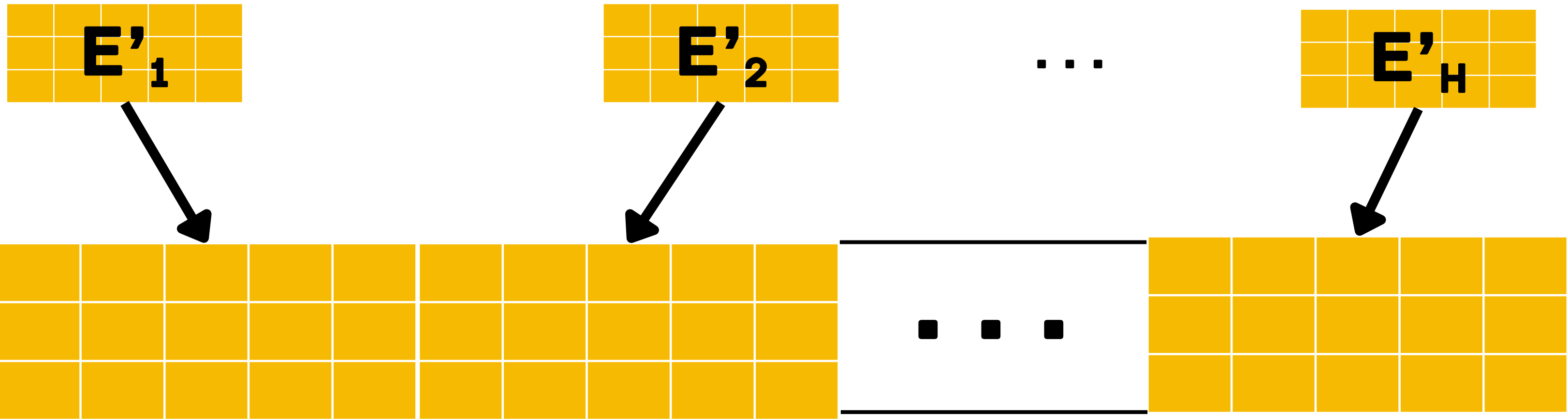


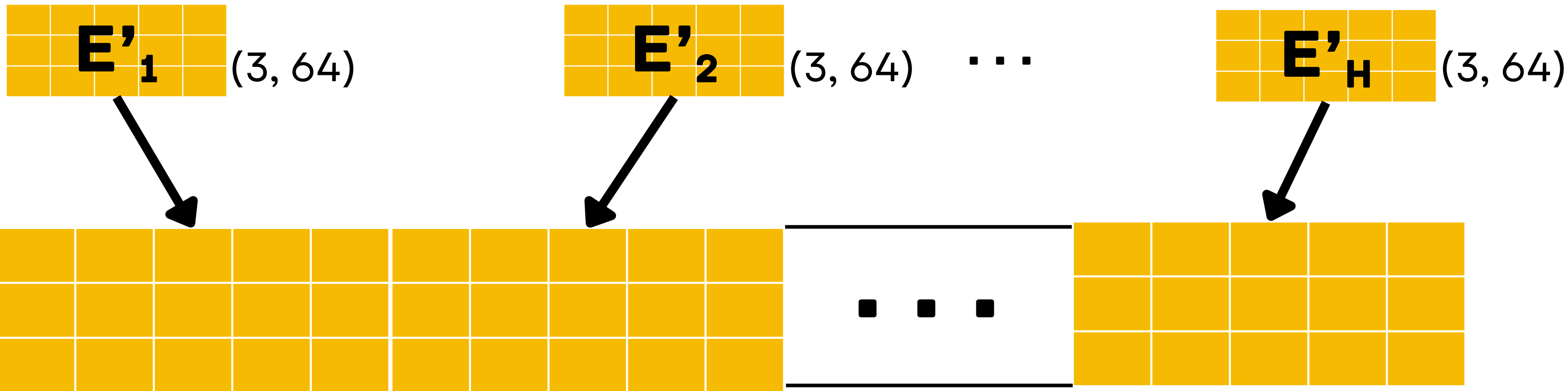


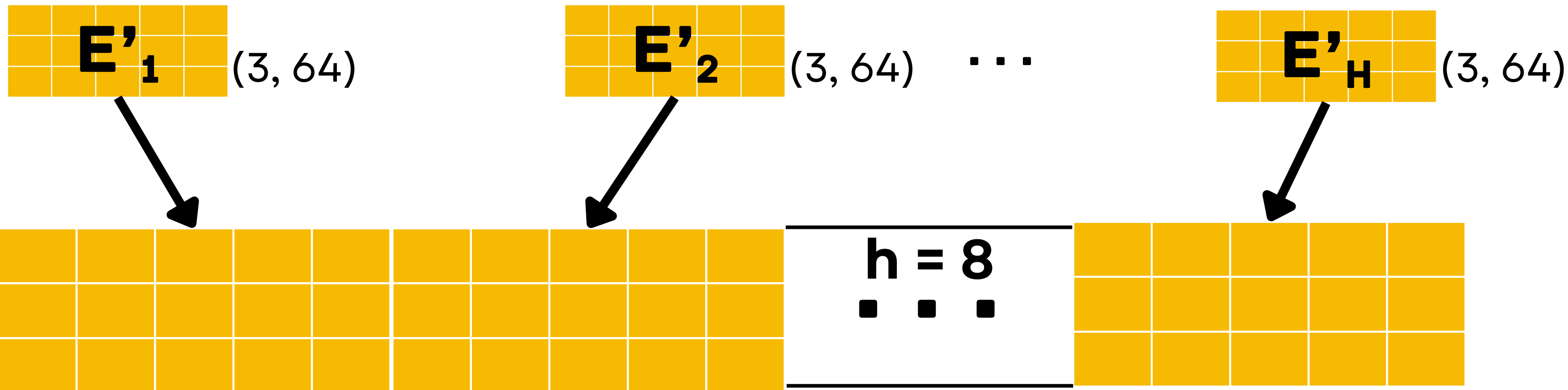


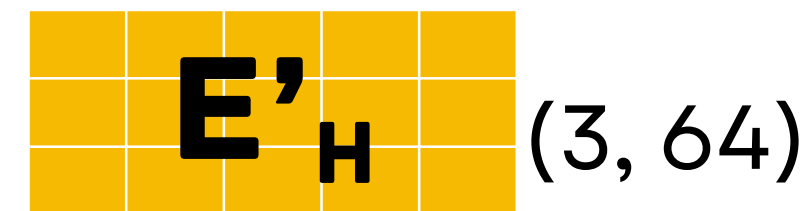
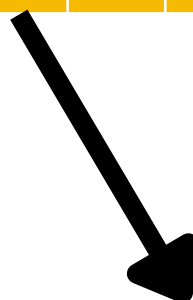
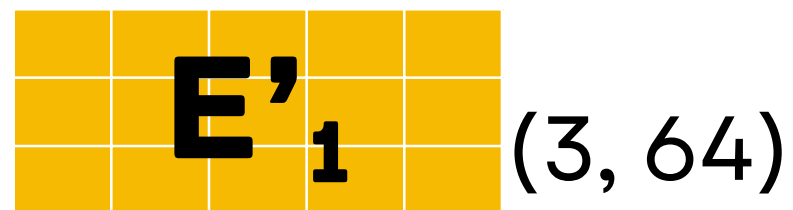


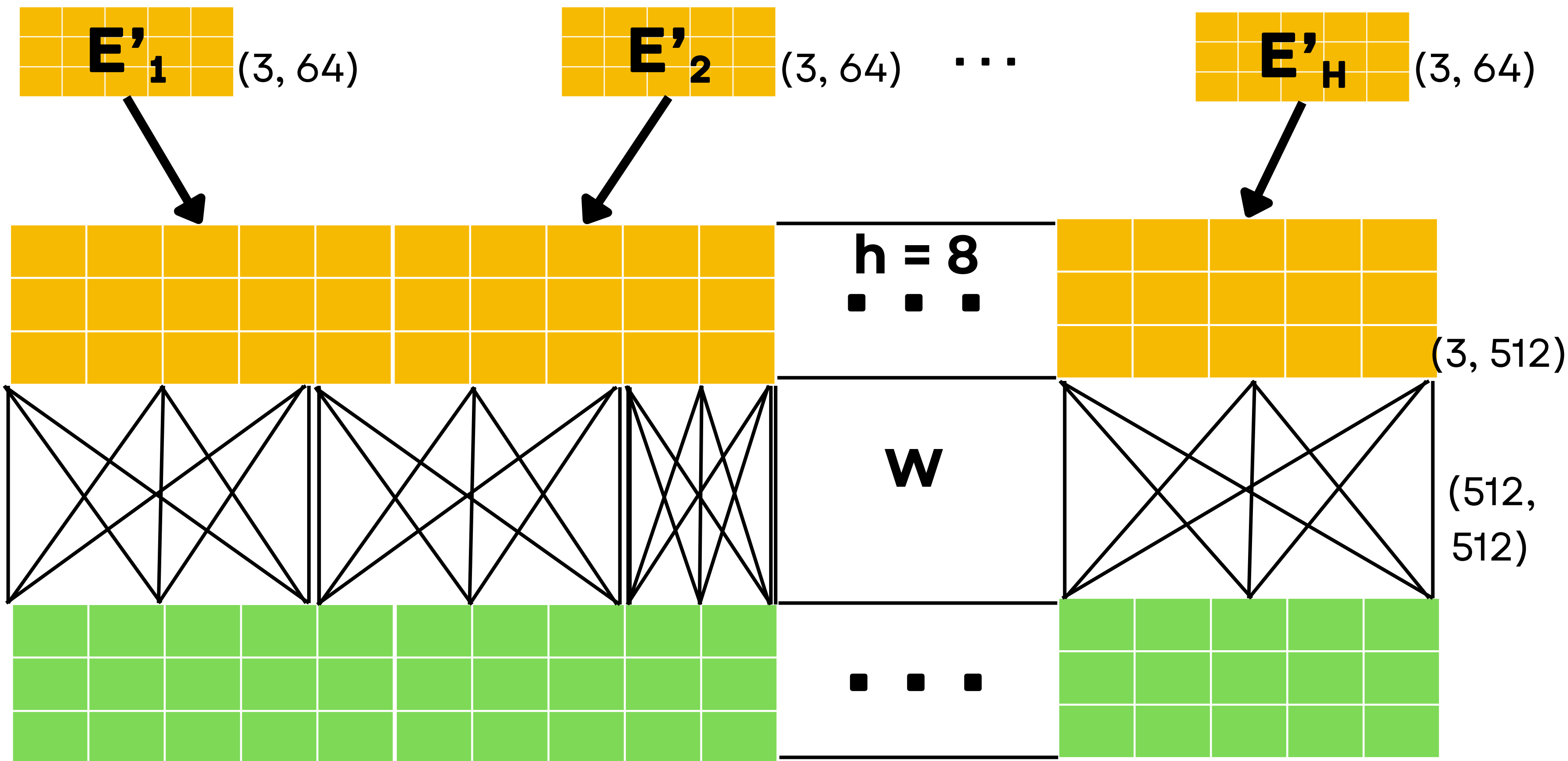


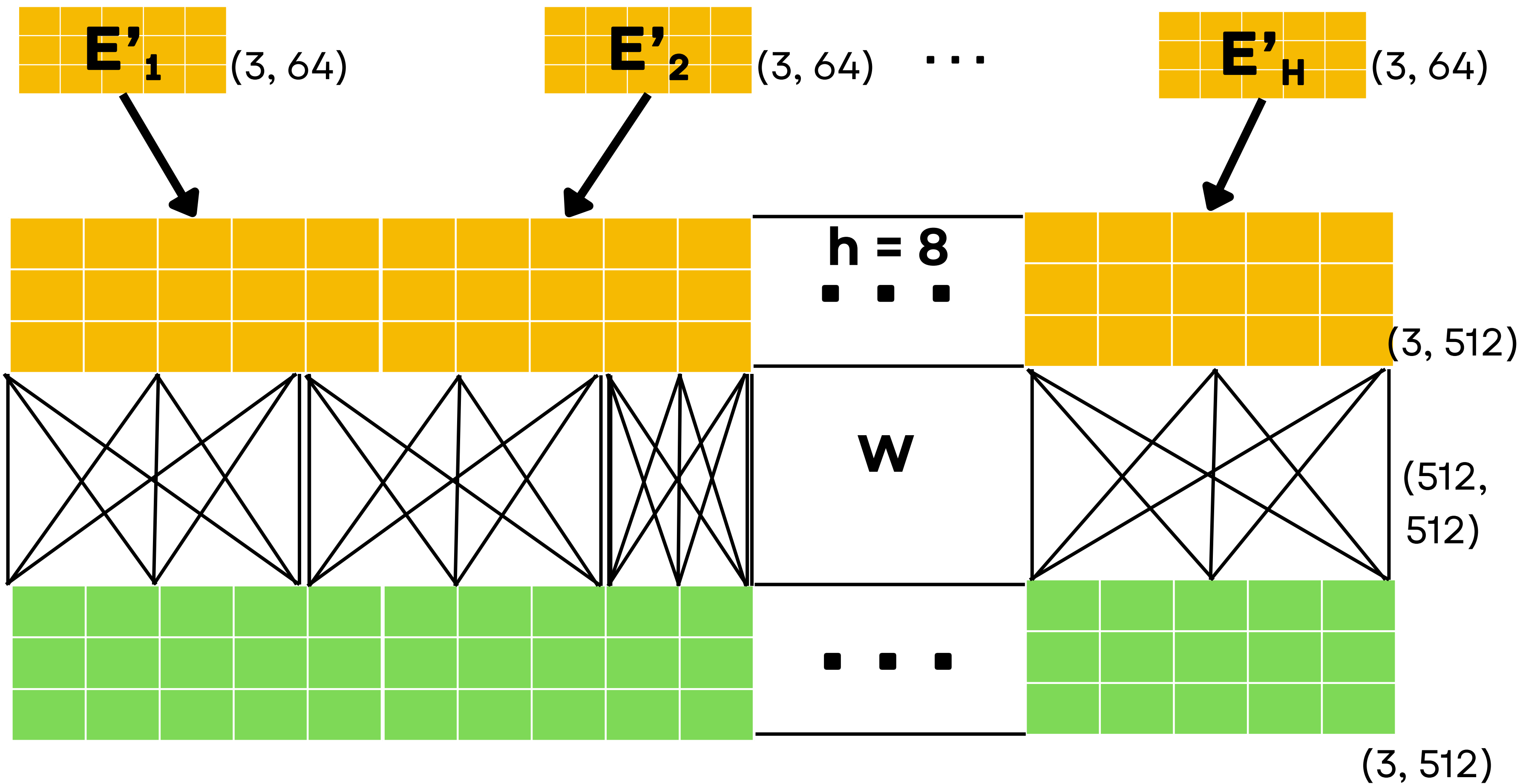












Q: How many heads in Multi-Head Attention does GPT-3 has?

Q: How many heads in Multi-Head Attention does GPT-3 has?

A: 96

Q: How many heads in Multi-Head Attention does GPT-3 has?

A: 96

Q: How many heads in Multi-Head Attention does GPT-4 has?

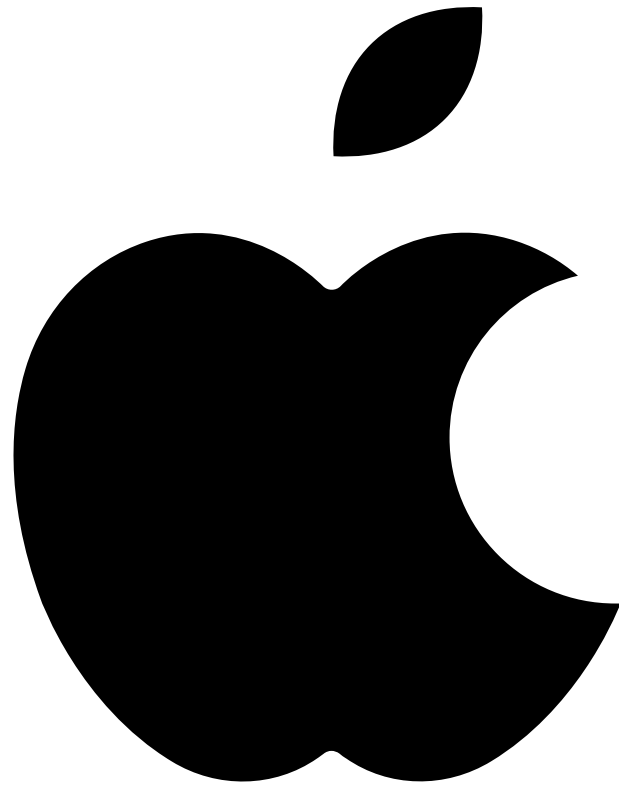
Home Task



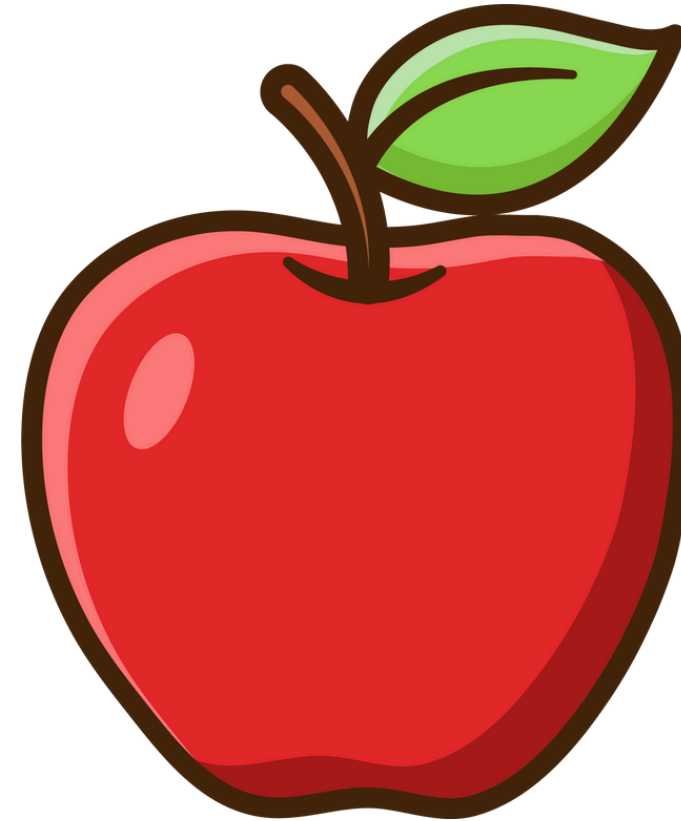
POSITIONAL ENCODINGS

Self Attention - Mathematics

LOVE APPLE PHONE



LOVE APPLE JUICE



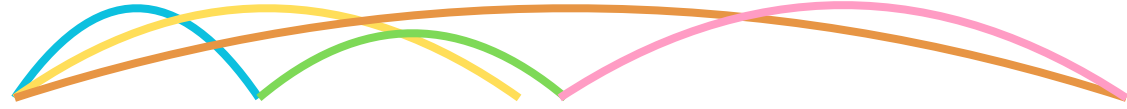
When Jake first set eyes on Apple iPhone



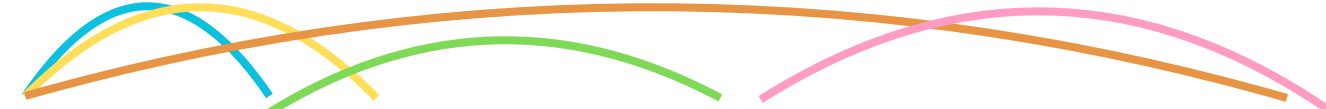
When Jake first set eyes on Apple iPhone

The **cat** chased the **mouse**

The **mouse** chased the **cat**

A series of five overlapping, colorful arcs (blue, yellow, orange, green, pink) positioned above the text.

The **cat** chased the **mouse**

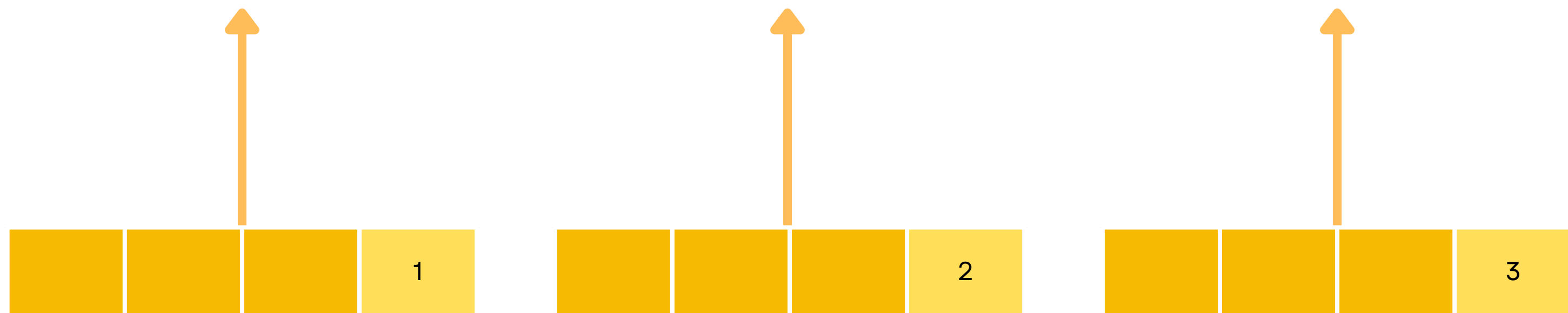
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The **mouse** chased the **cat**

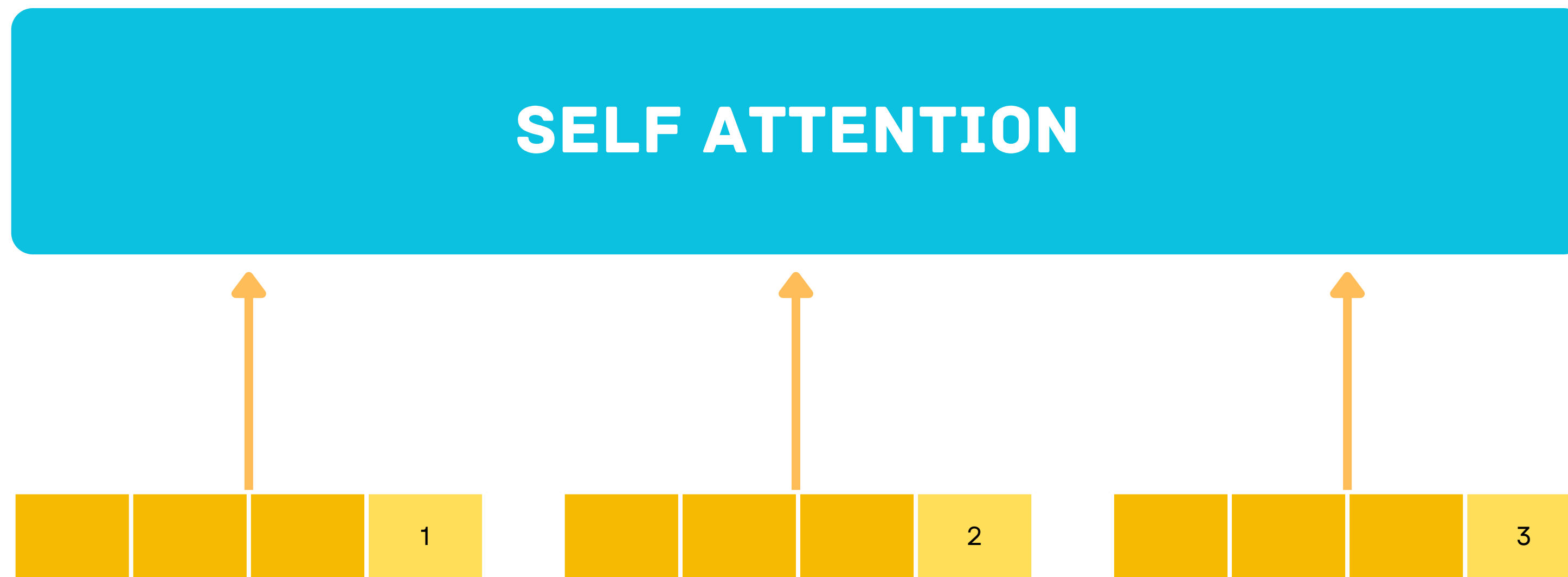
SELF ATTENTION



SELF ATTENTION

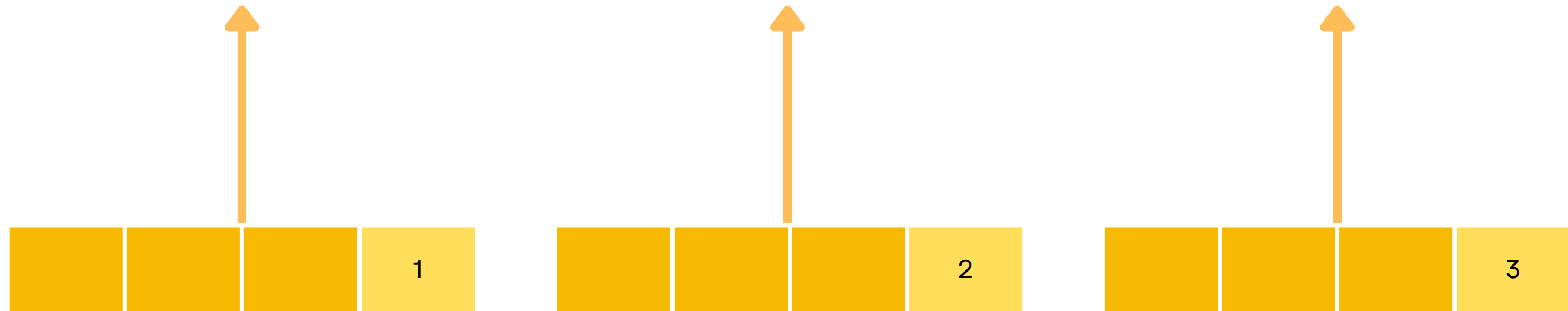


Q: What if we have 1000 words sentence



- ✗ • Discrete
- ✗ • Unbounded
- ✗ • Not Trainable

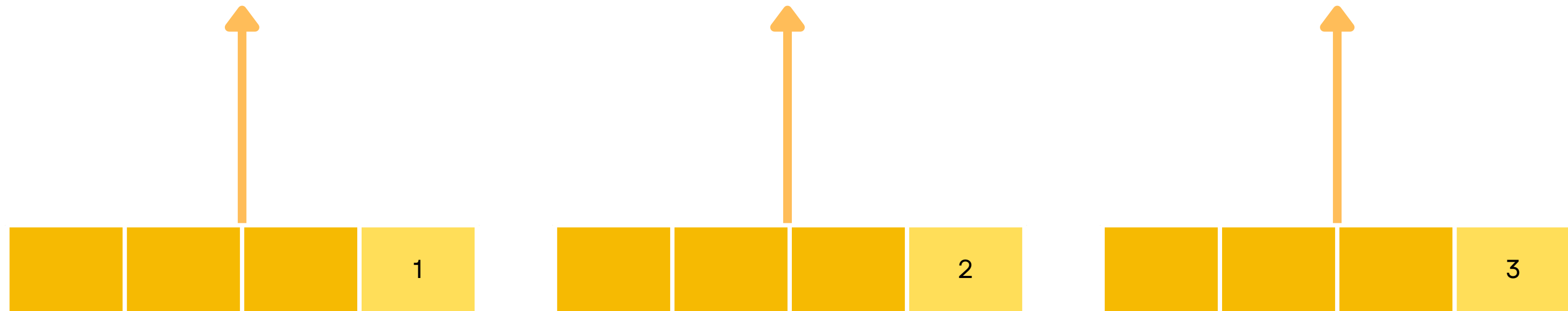
SELF ATTENTION



- ✓ • Continuous
- ✓ • Bounded
- ✓ • Trainable

- ✗ • Discrete
- ✗ • Unbounded
- ✗ • Not Trainable

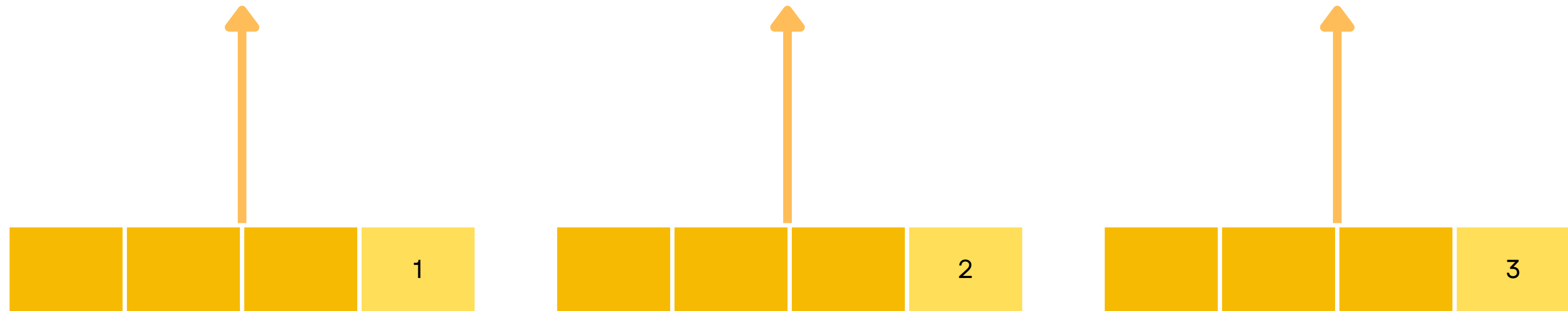
SELF ATTENTION

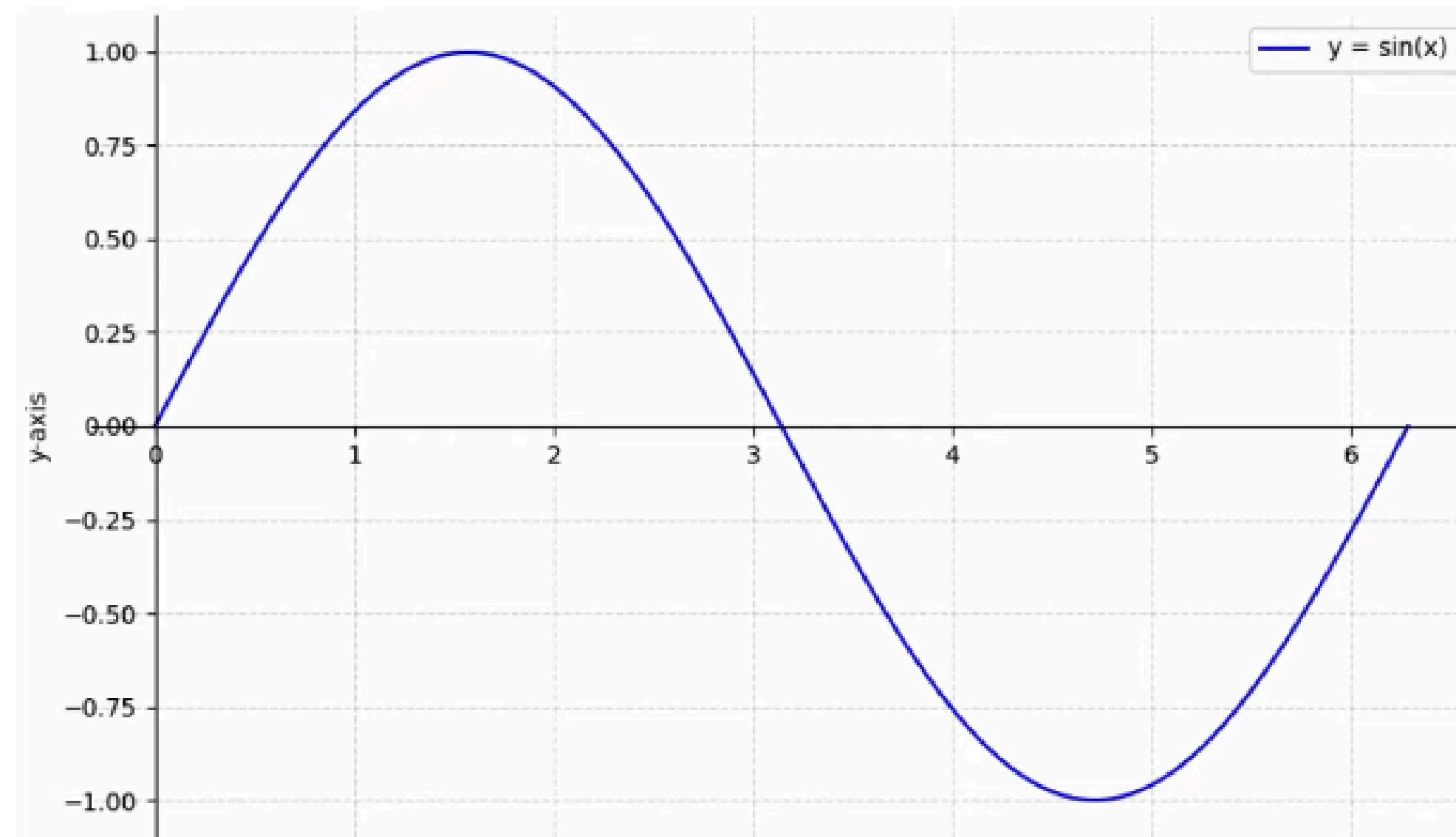


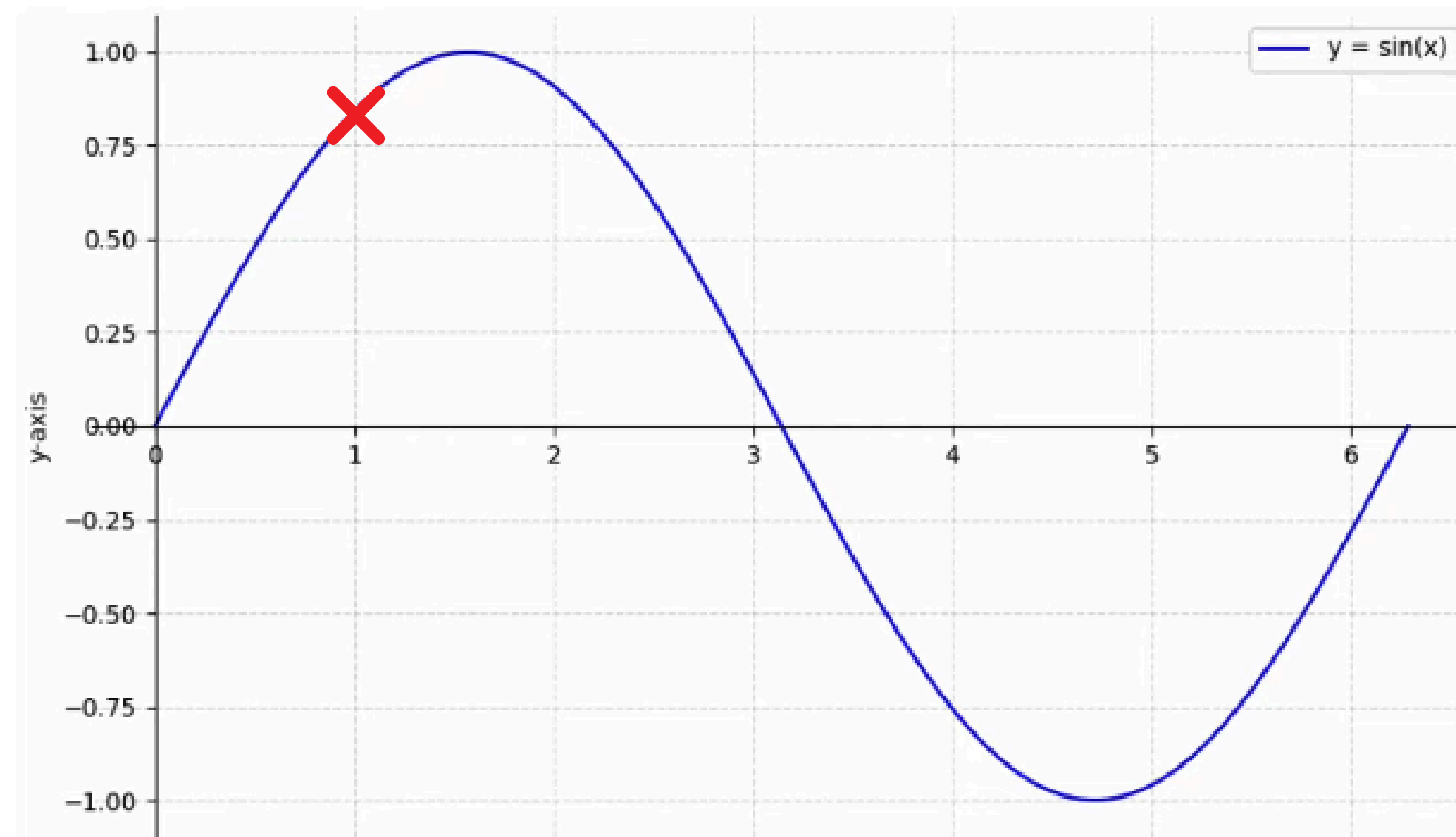
- ✓ • Continious
- ✓ • Bounded
- ✓ • Trainable

- ✗ • Discrete
- ✗ • Unbounded
- ✗ • Not Trainable

SELF ATTENTION



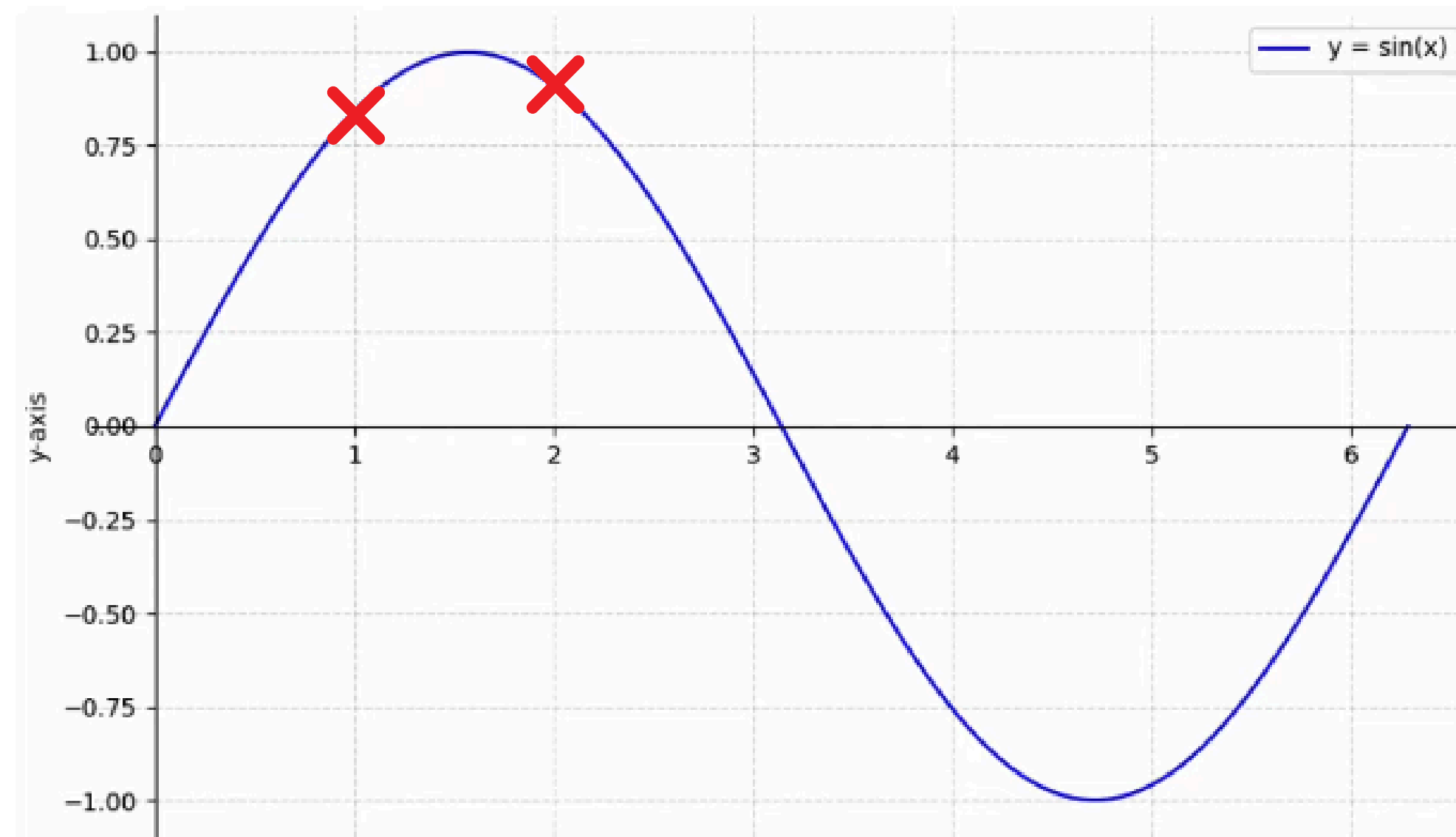




☐ ☐ ☐ ☒ Sin(1)

☐ ☐ ☐ ☒

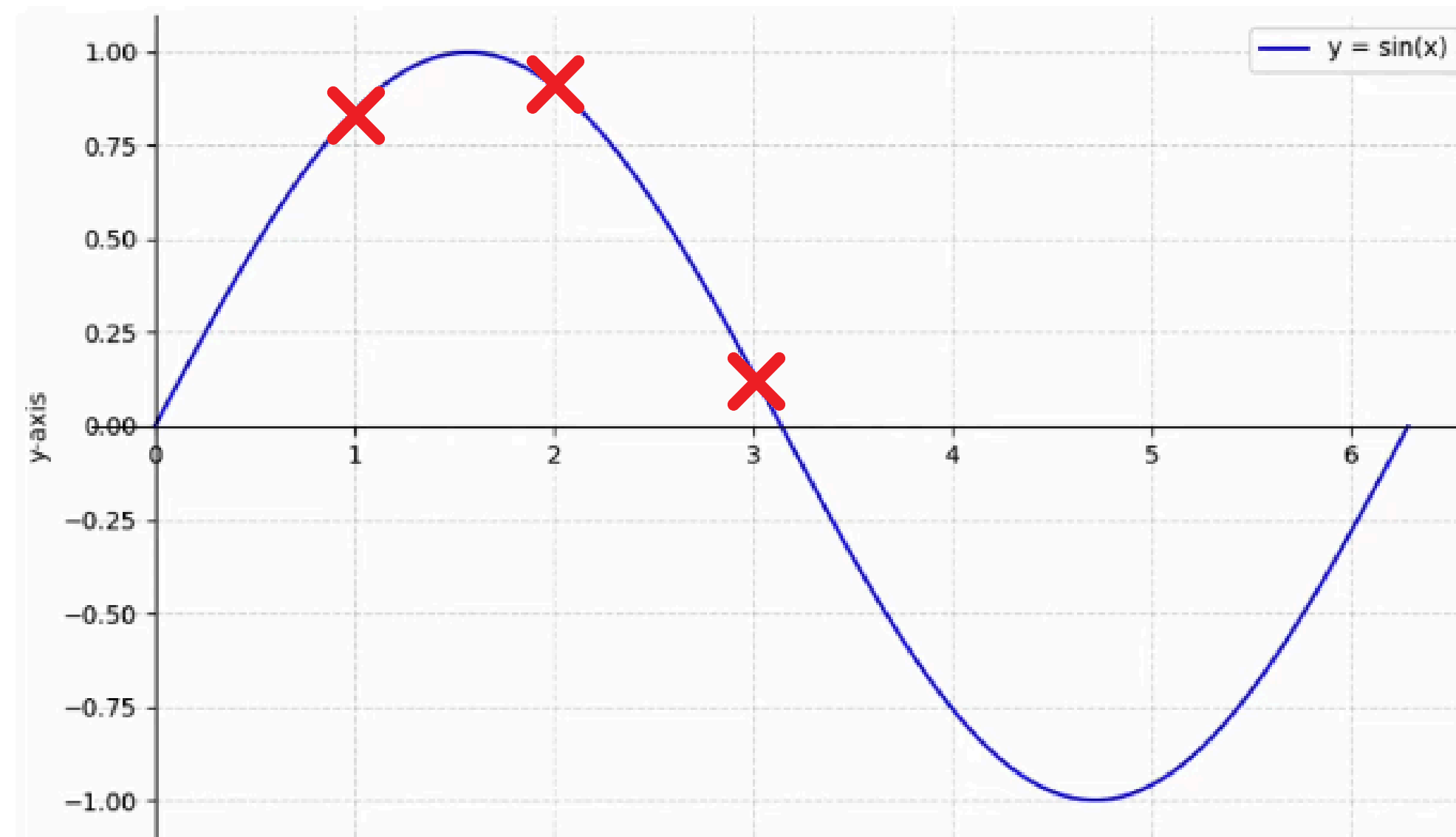
☐ ☐ ☐ ☒



☐ ☐ ☐ ☒ Sin(1)

☐ ☐ ☐ ☒ Sin(2)

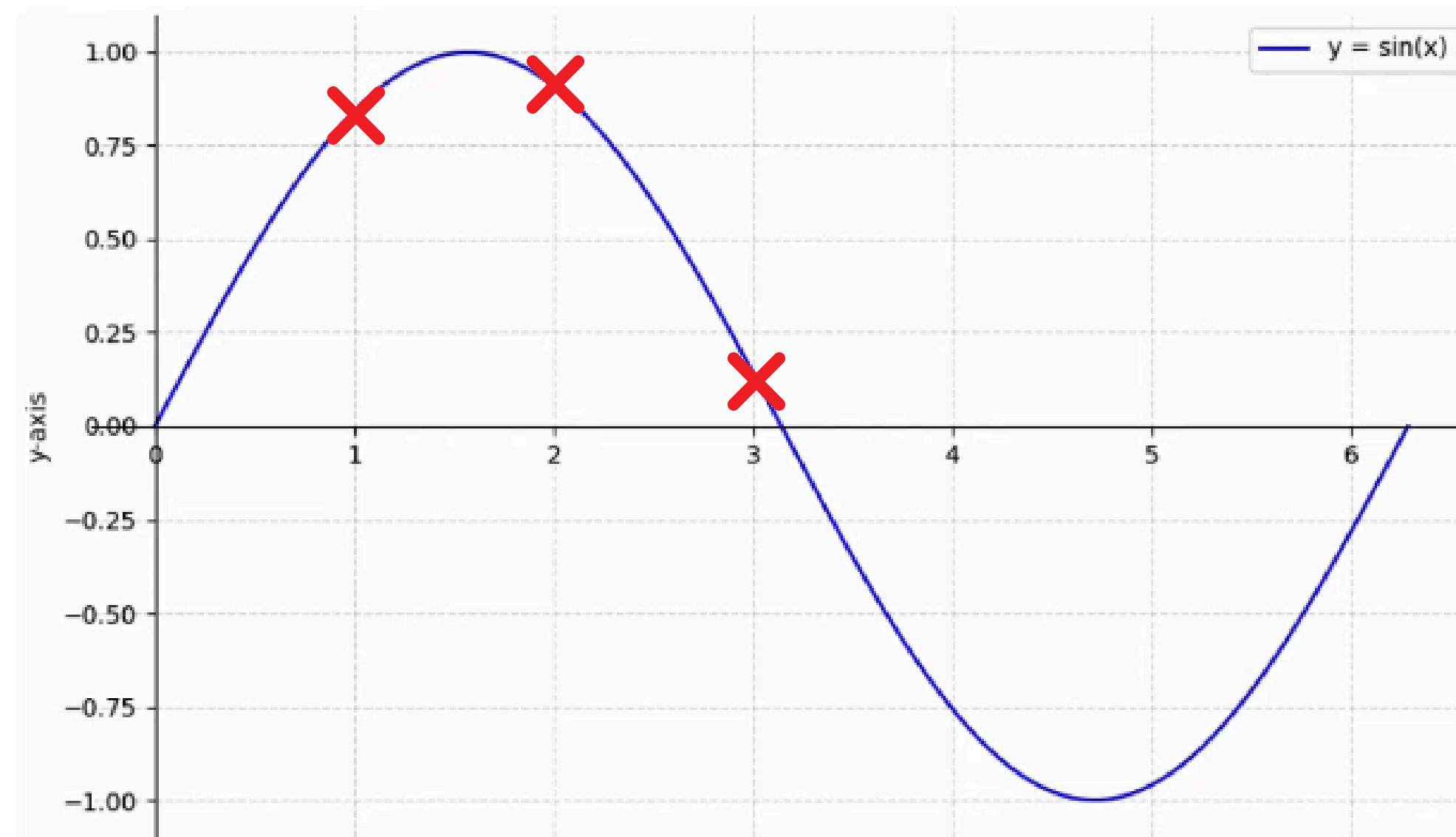
☐ ☐ ☐ ☐



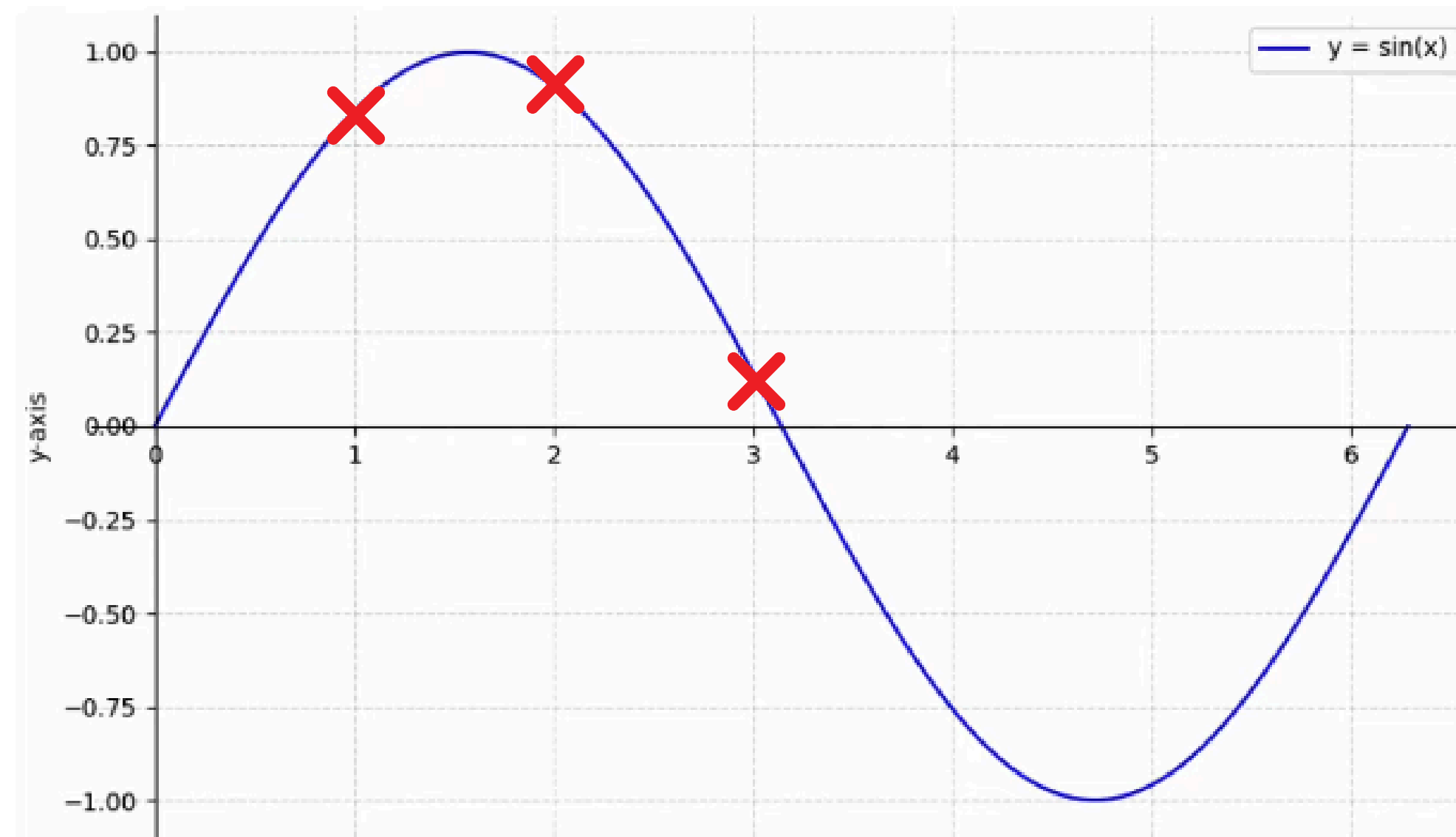
☐ ☐ ☐ ☒ Sin(1)

☐ ☐ ☐ ☒ Sin(2)

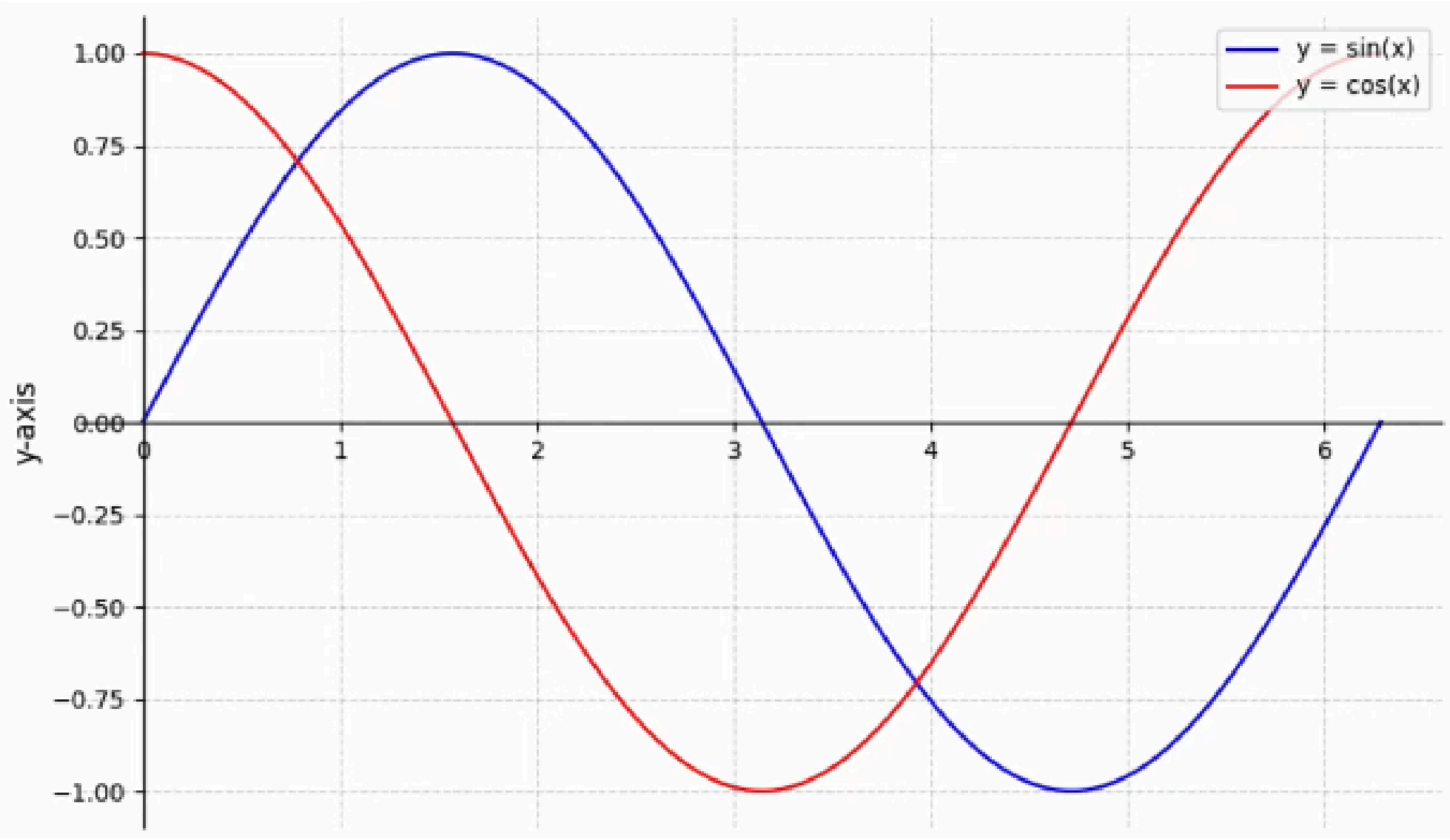
☐ ☐ ☐ ☒ Sin(3)

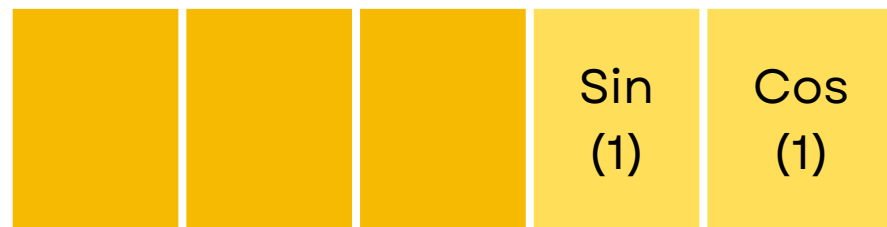
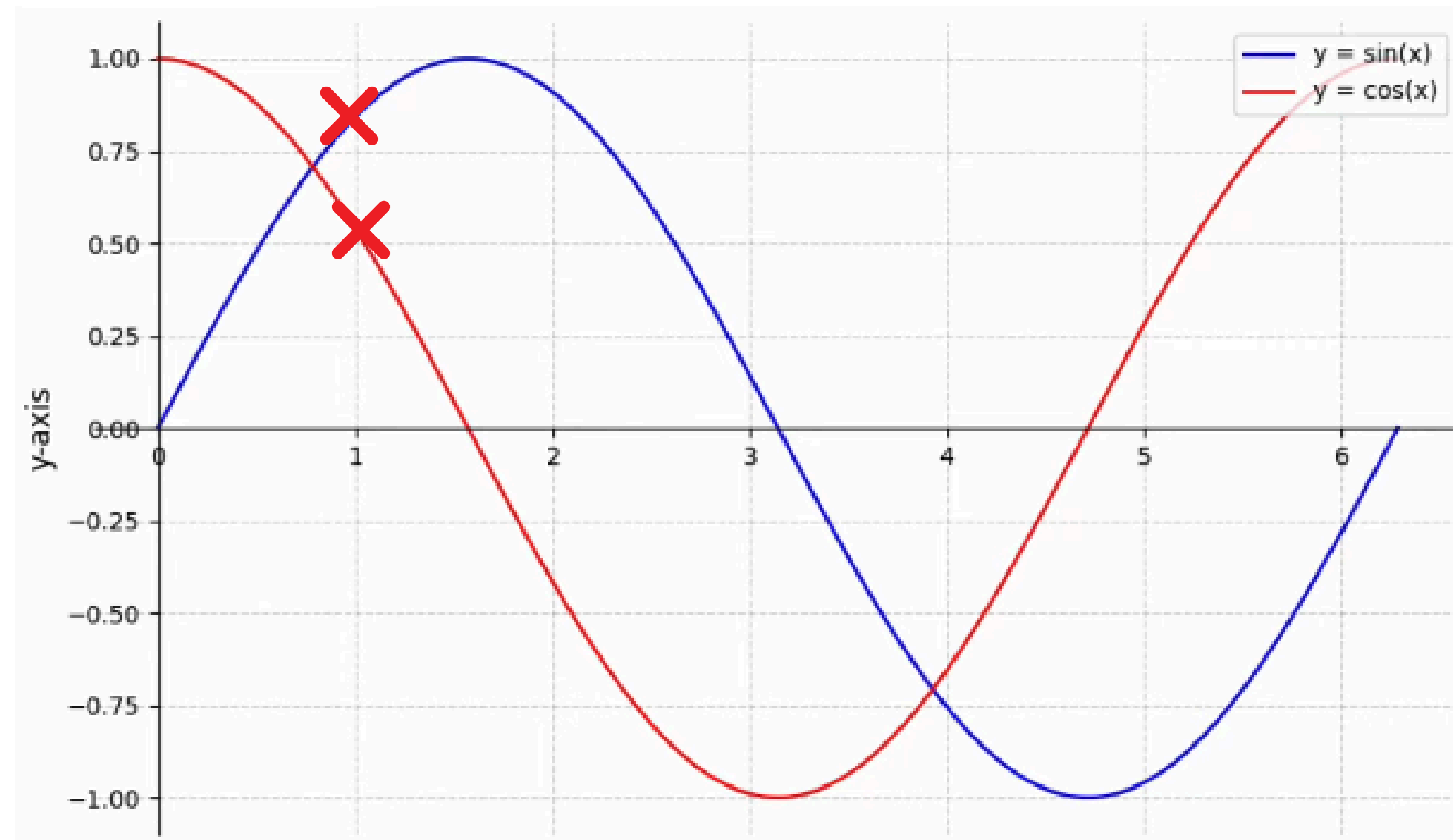


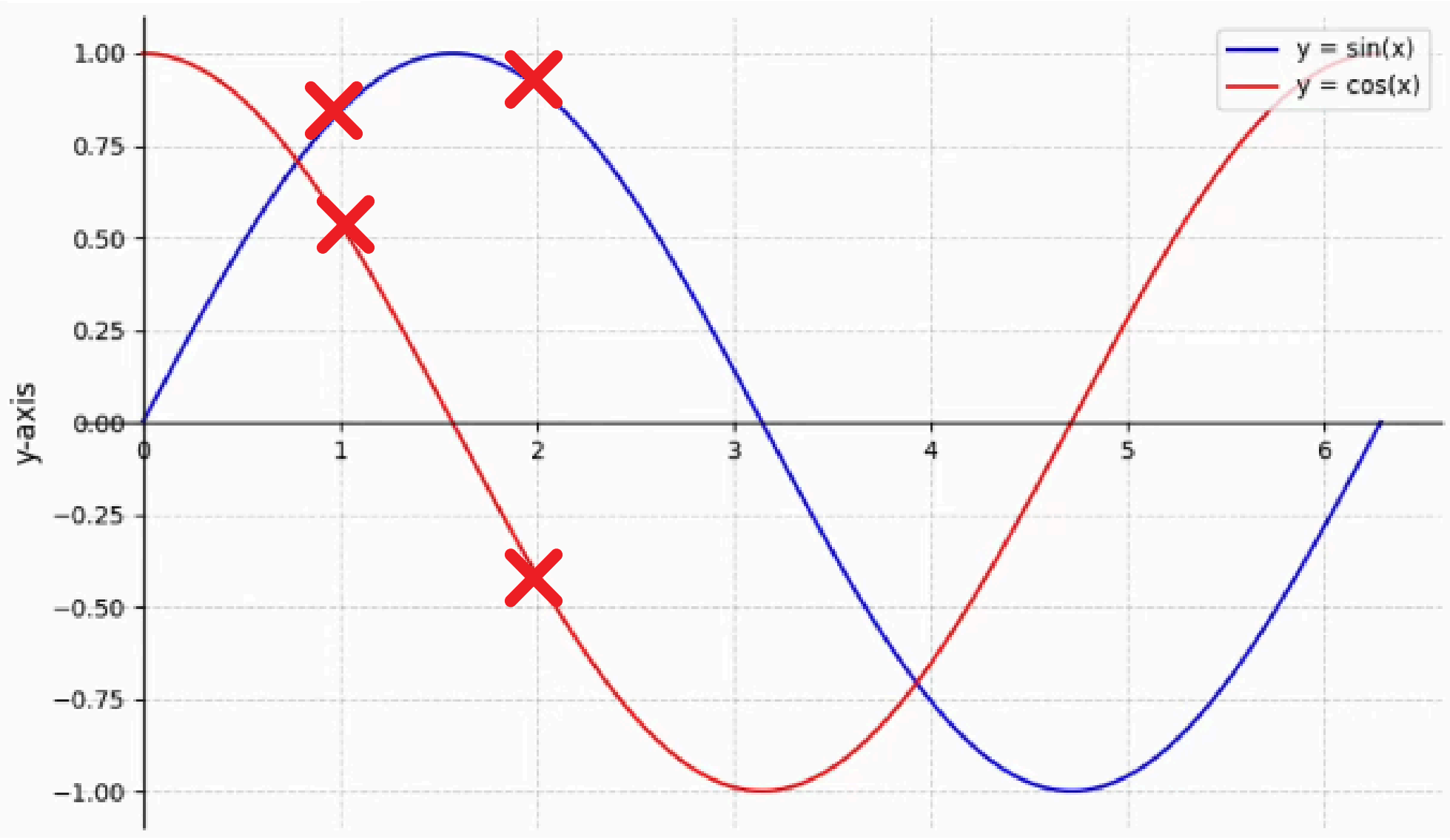
Q: Is it enough? Will we always have unique values always?



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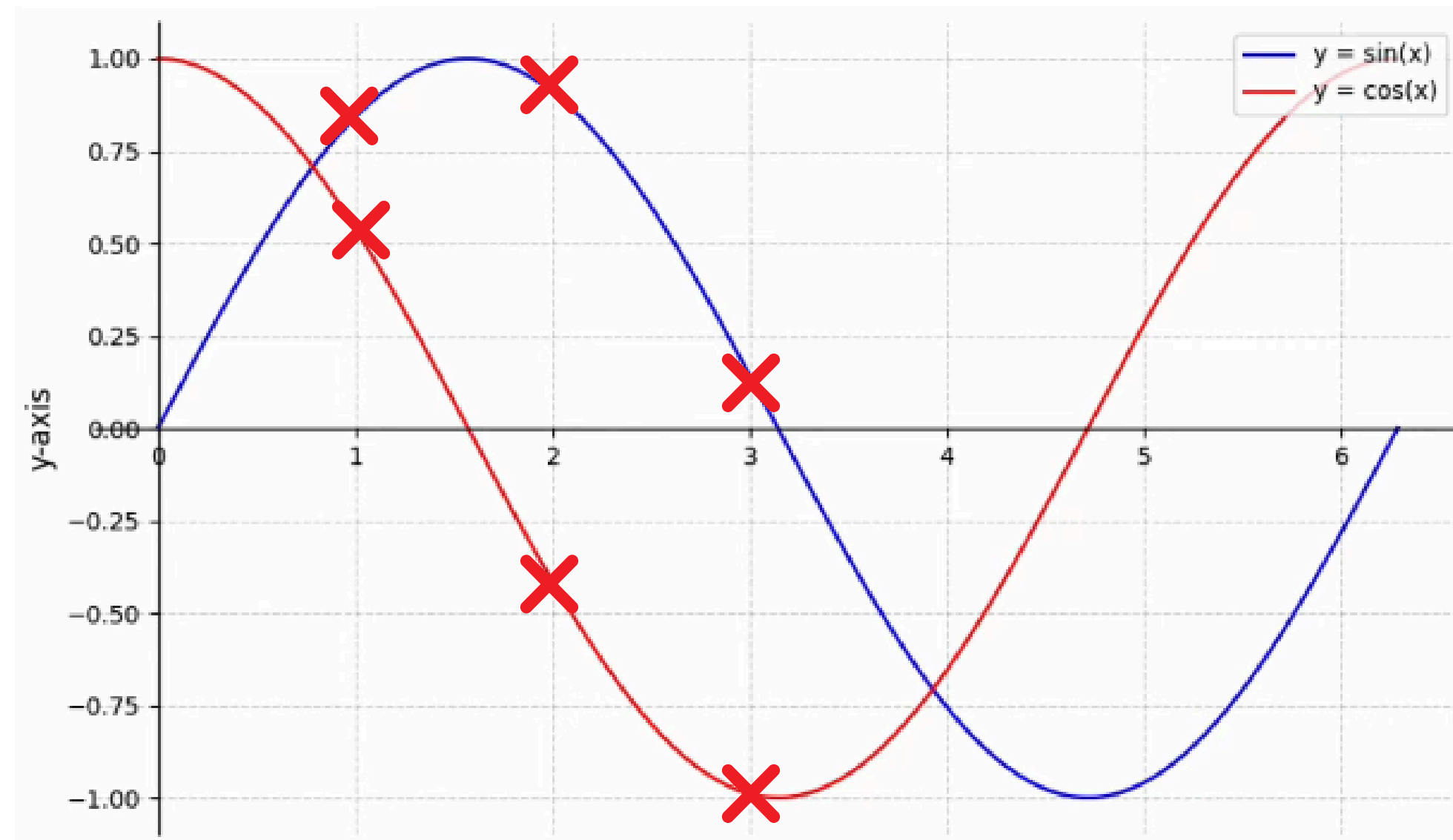


Sin
(1)

Cos
(1)

Sin
(2)

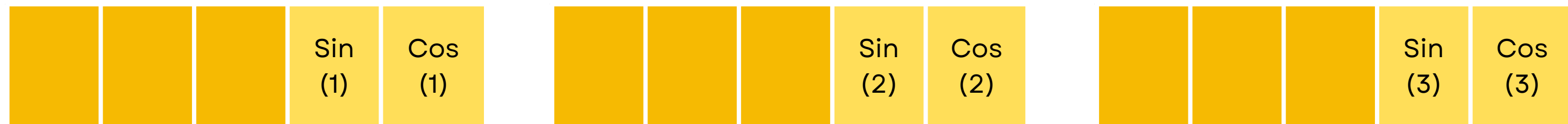
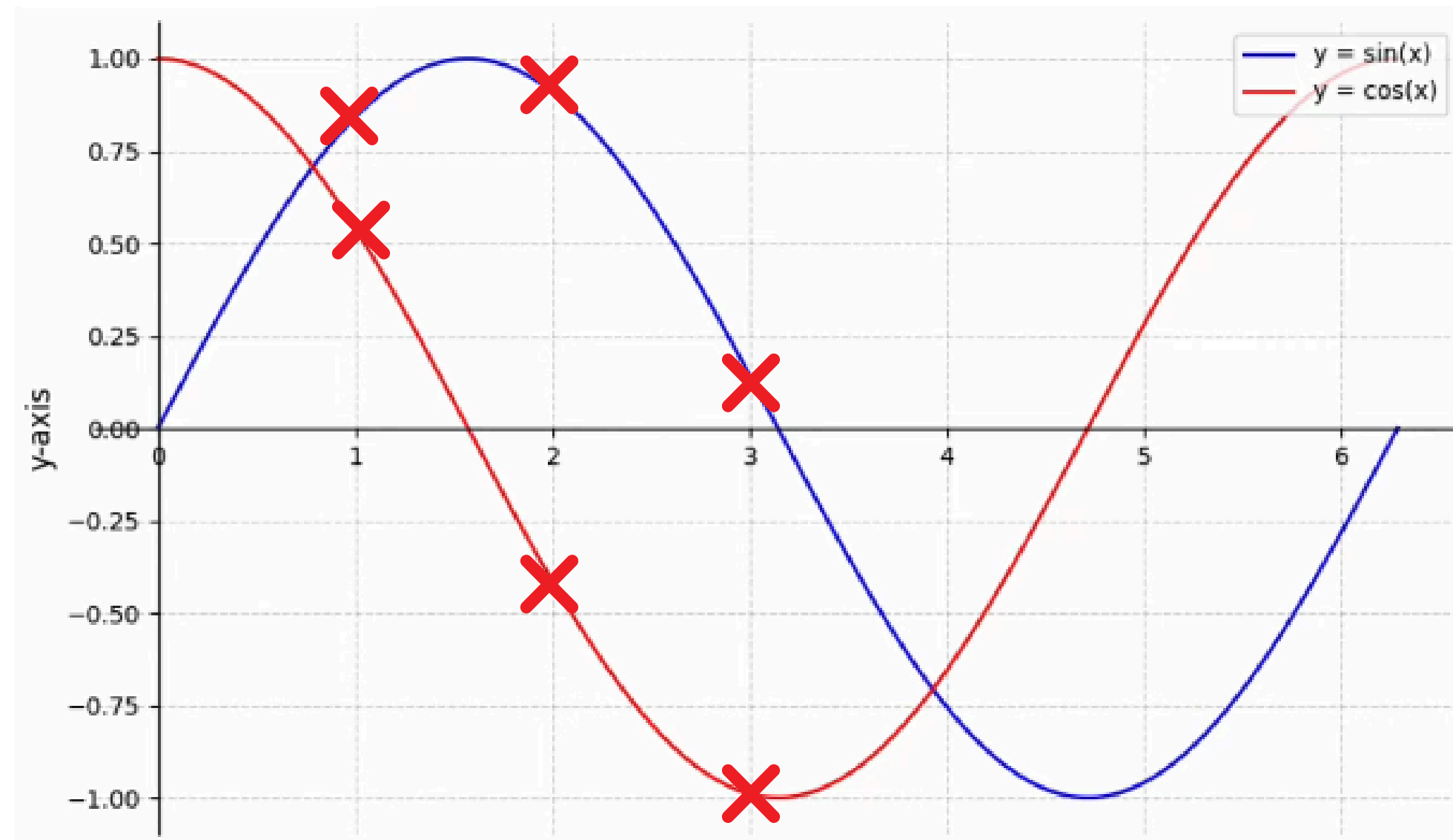
Cos
(2)



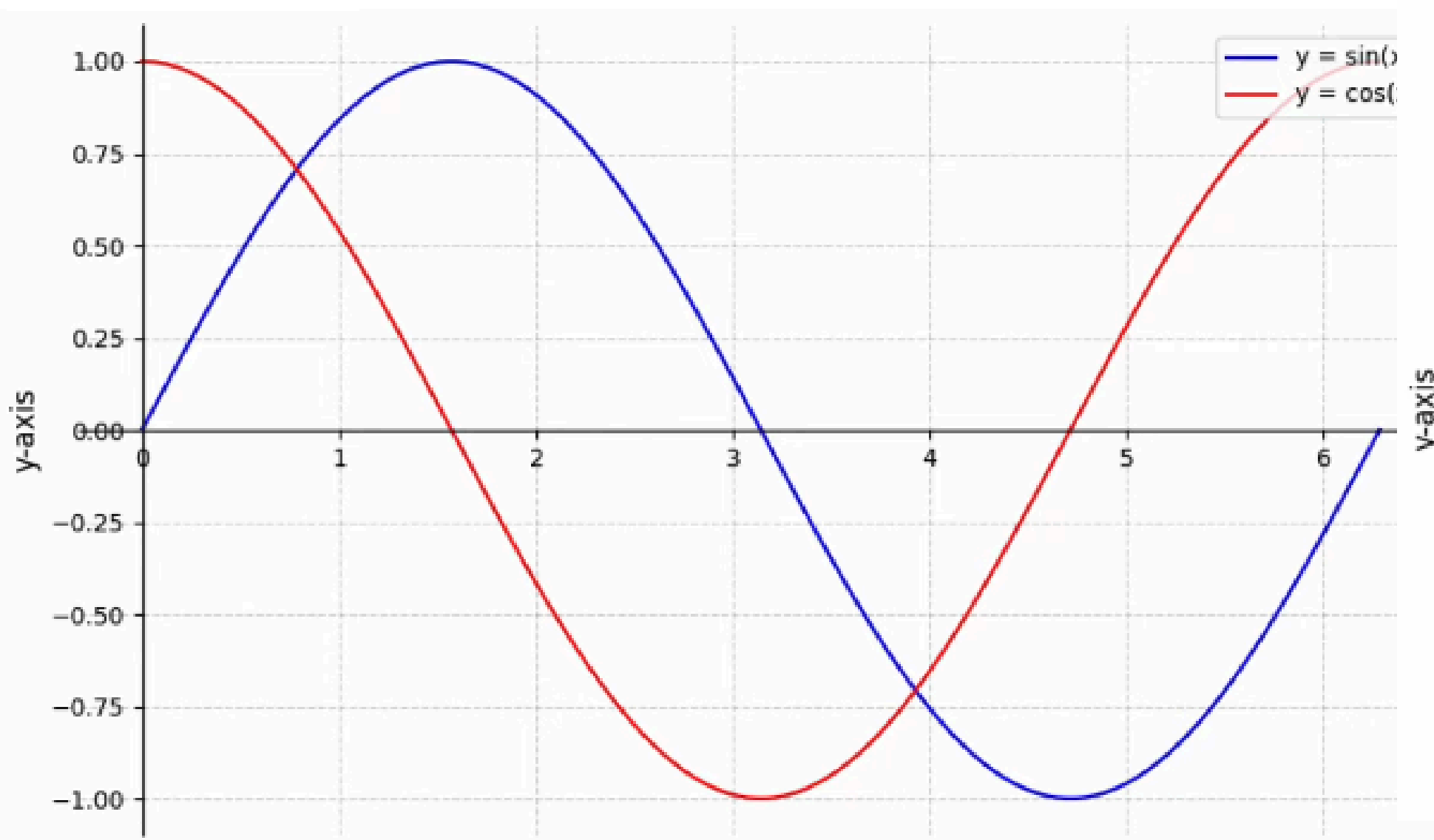
			Sin (1)	Cos (1)
--	--	--	------------	------------

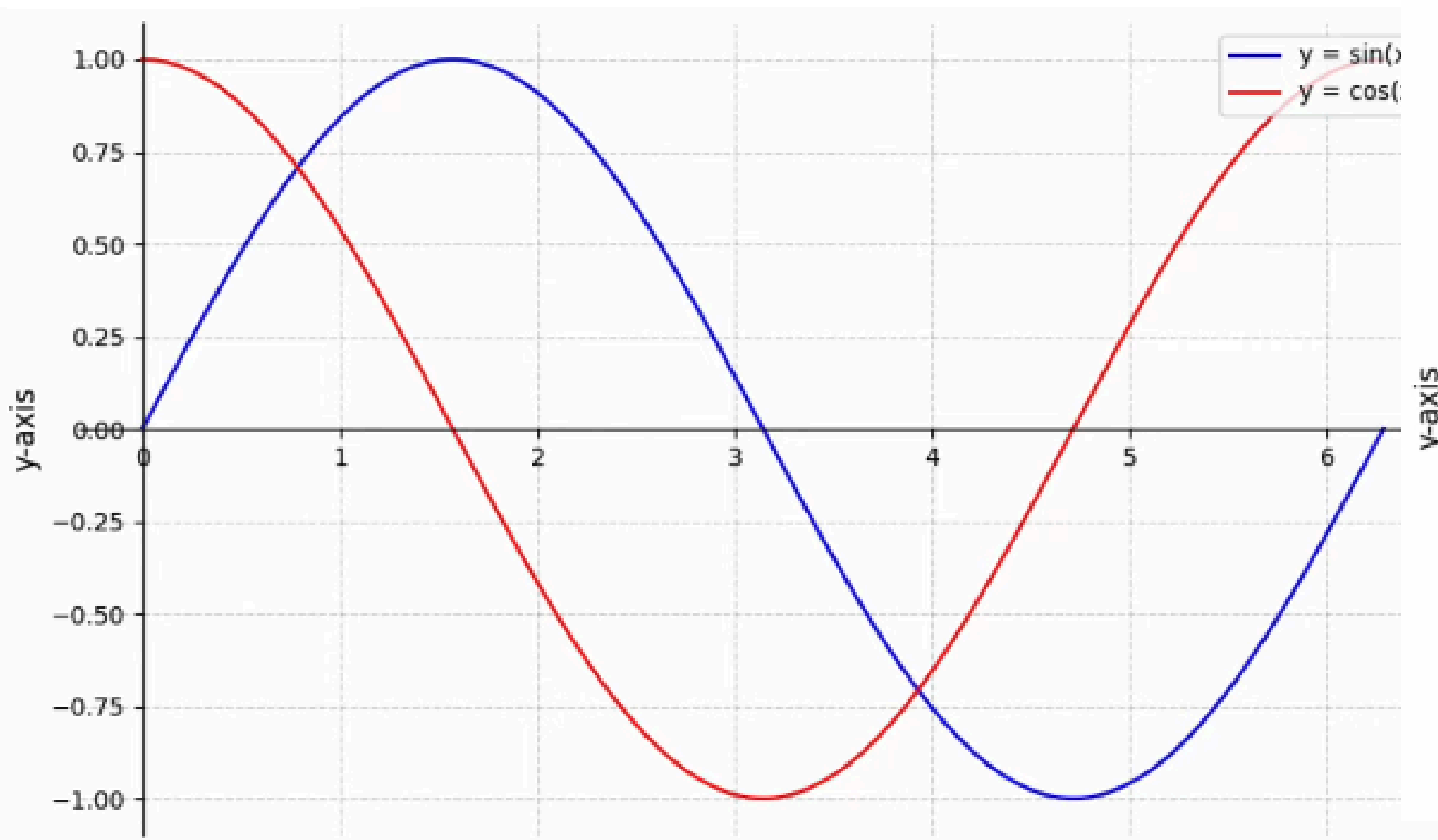
			Sin (2)	Cos (2)
--	--	--	------------	------------

			Sin (3)	Cos (3)
--	--	--	------------	------------

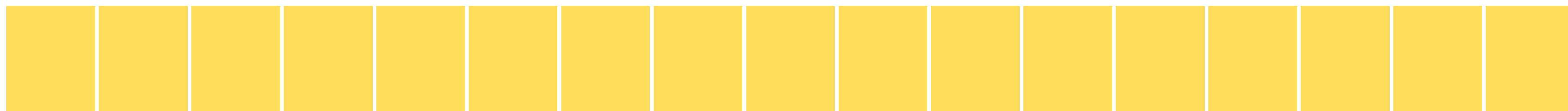


Q: Is it enough? Will we always have unique values always?

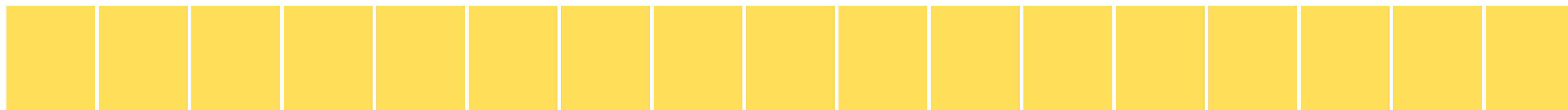




Q: Is it enough? Will we always have unique values always?



512 dims Vector



512 dims Vector

264 pairs of Sine and Cosine



Word Embeddings

Positions



Word Embeddings

Positions

512



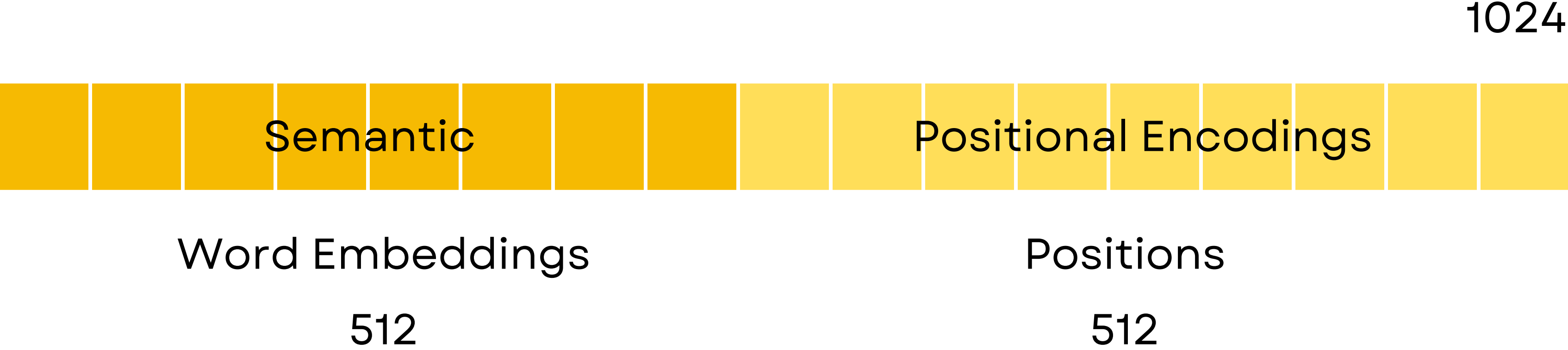
Word Embeddings

512

Positions

512





1024



Word Embeddings

Positions

512

512

$$(1, 512) \xrightarrow{\quad} E \times W_{q,k,v} \xleftarrow{\quad} (512, 64)$$

1024



Word Embeddings

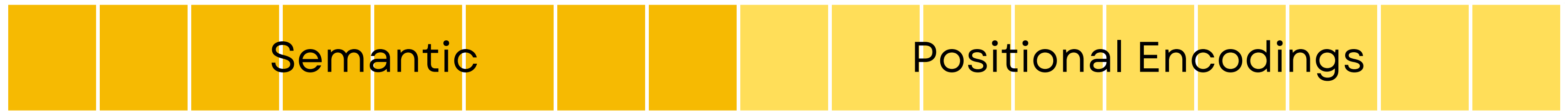
Positions

512

512



1024



Word Embeddings

512

Positions

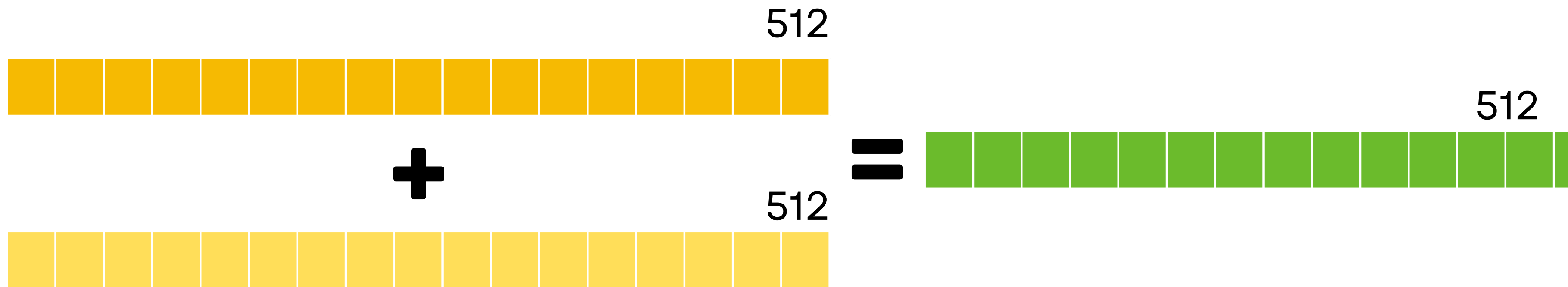
512

$(1, 512) \rightarrow E \times W_{q,k,v} \leftarrow (512, 64)$

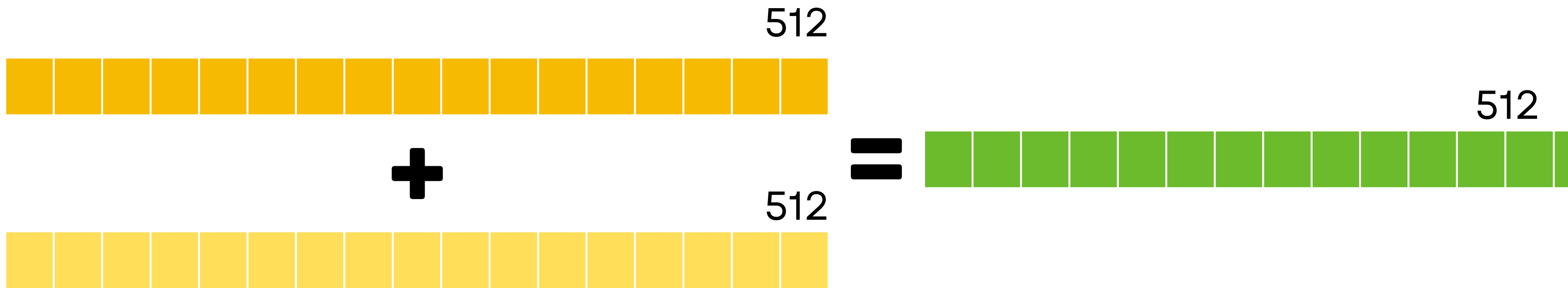
The diagram shows a red 'X' over the equation $E \times W_{q,k,v}$, indicating that this operation is not the correct choice.

$(1, 1024) \rightarrow (E + PE) \times W \leftarrow (1024, 64)$

Q: Is it a feasible choice?

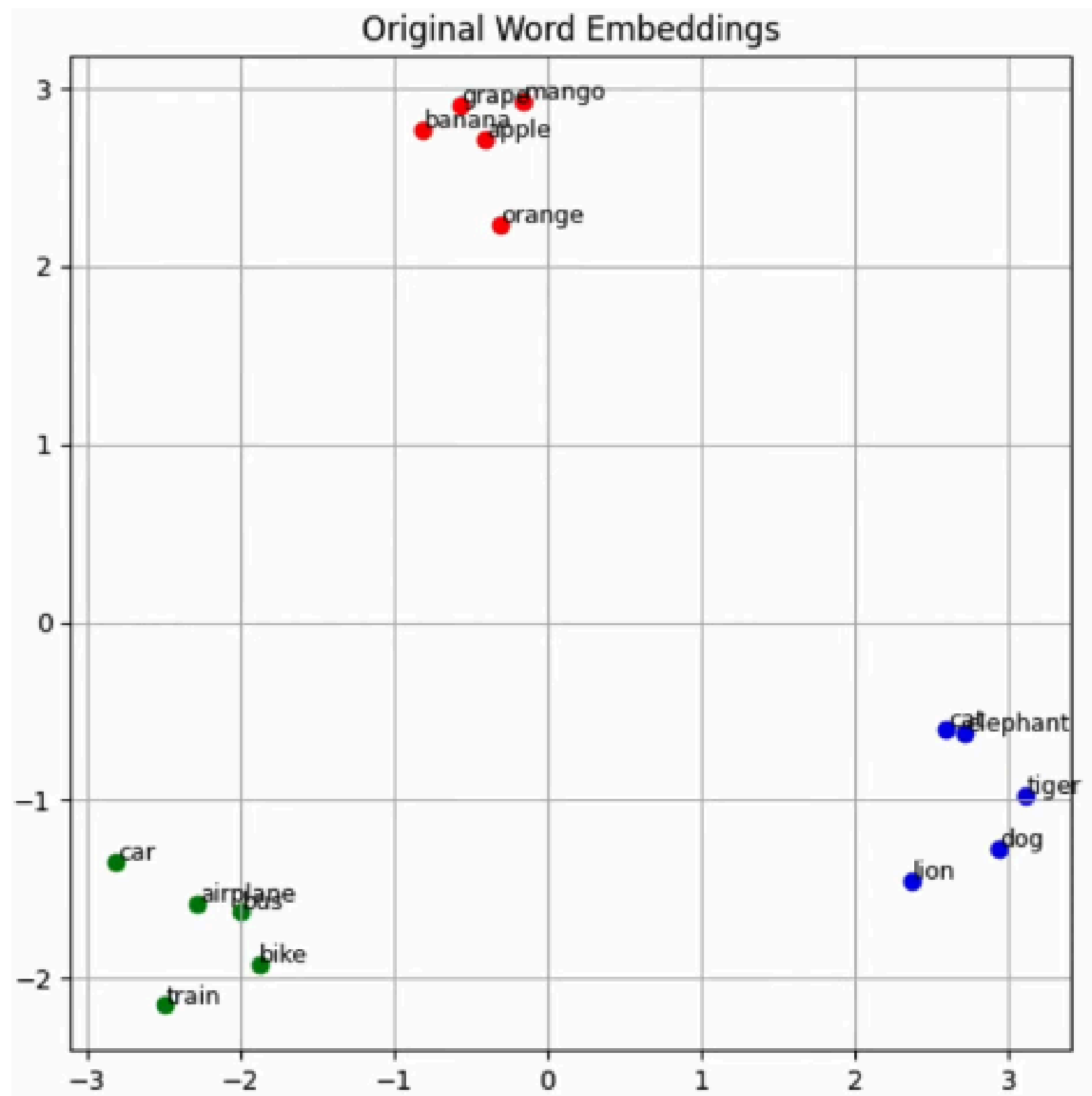


*Element wise addition

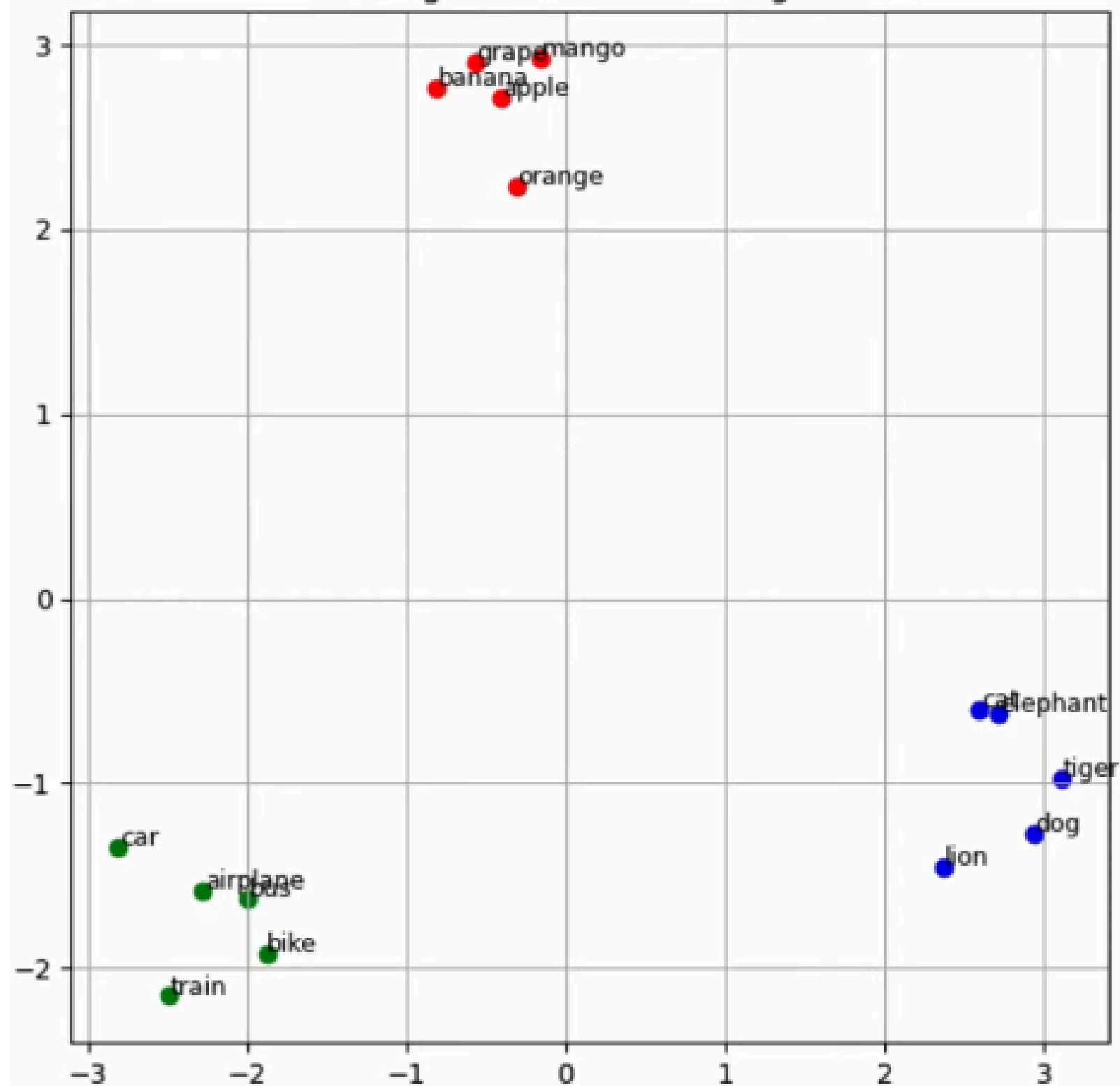


*Element wise addition

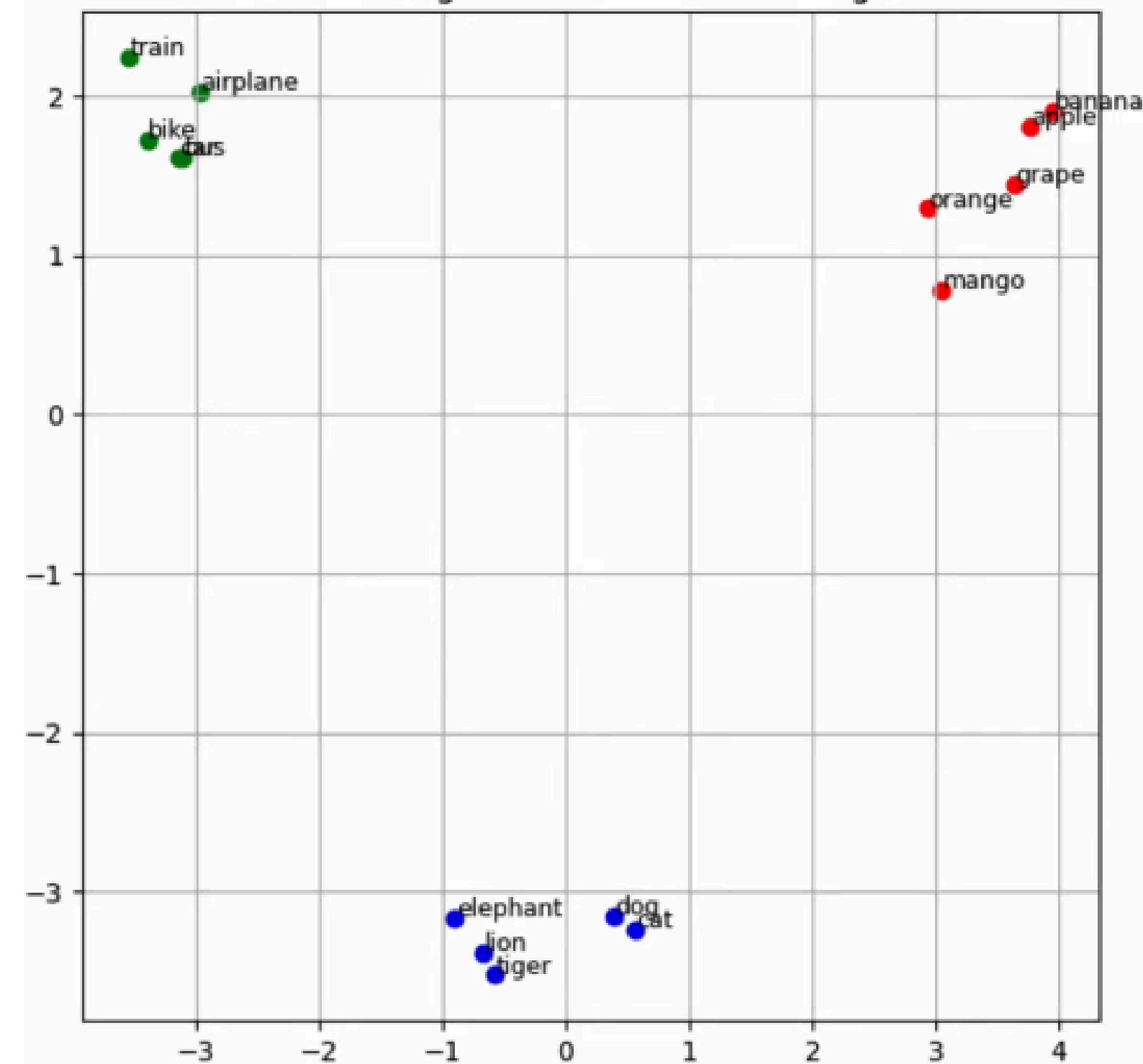
Q: Is it a feasible choice? Wouldn't data be changed now?



Original Word Embeddings



Word Embeddings with Positional Encodings Added





LAYER NORMALIZATION

Why Normalization?

Why Normalization?

Sq. Ft. Area	Number of Rooms	House Price (\$)
1200	3	250,000
1500	4	300,000
1800	4	350,000
1000	2	200,000
2500	5	500,000
800	2	180,000
2200	5	450,000
1400	3	275,000
2000	4	375,000
2700	6	550,000

Why Normalization?

Sq. Ft. Area	Number of Rooms	House Price (\$)
1200	3	250,000
1500	4	300,000
1800	4	350,000
1000	2	200,000
2500	5	500,000
800	2	180,000
2200	5	450,000
1400	3	275,000
2000	4	375,000
2700	6	550,000

$$\frac{X_i - \text{Mean}}{\text{Standard Deviation}}$$

SELF ATTENTION

SELF ATTENTION



[0.1, 0.3, -1.2, , 0.7]

HOW



[0.3, 0.2, -1.4, , 0.2]

ARE



[1.2, -0.3, -0.9, , 0.5]

YOU

HOW'

[0.8, 0.2, 0.2, , 0.7]

ARE'

[0.5, 0.1, 1.2, , 0.5]

YOU'

[1.7, -1.4, 0.8, , 0.1]

SELF ATTENTION

[0.1, 0.3, -1.2, , 0.7]

HOW

[0.3, 0.2, -1.4, , 0.2]

ARE

[1.2, -0.3, -0.9, , 0.5]

YOU

HOW'

[0.8, 0.2, 0.2, , 0.7]

ARE'

[0.5, 0.1, 1.2, , 0.5]

YOU'

[1.7, -1.4, 0.8, , 0.1]

SELF ATTENTION

512 [0.1, 0.3, -1.2, , 0.7]

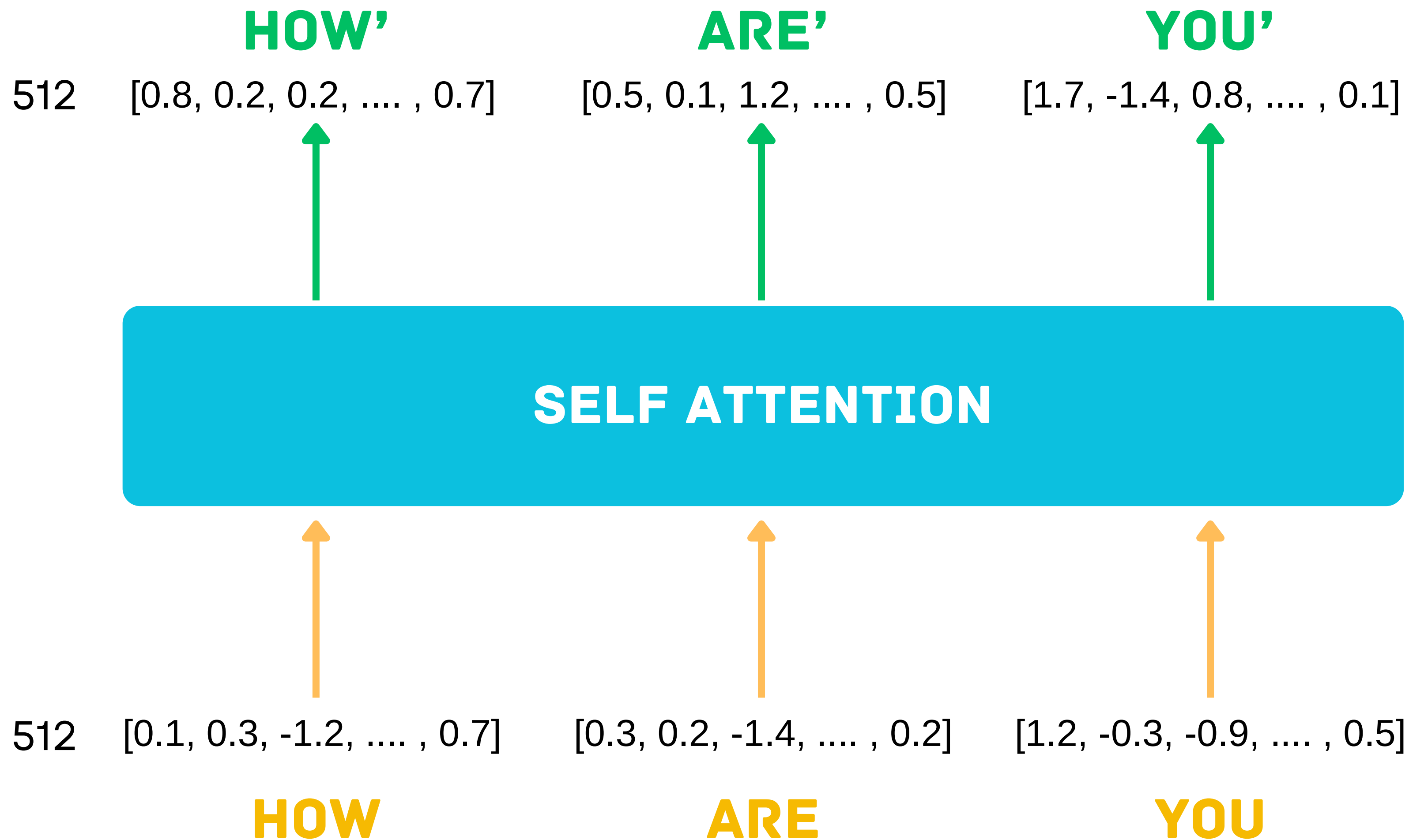
HOW

[0.3, 0.2, -1.4, , 0.2]

ARE

[1.2, -0.3, -0.9, , 0.5]

YOU



SELF ATTENTION

how	[0.1, 0.3, -1.2, , 0.7]	good	[0.3, -1.4, 1.7, , 0.6]
are	[0.3, 0.2, -1.4, , 0.2]	<pad>	[0, 0, 0, , 0]
you	[1.2, -0.3, -0.9, , 0.5]	<pad>	[0, 0, 0, , 0]

SELF ATTENTION

{	how	[0.1, 0.3, -1.2, , 0.7]	good	[0.3, -1.4, 1.7, , 0.6]	}	2
	are	[0.3, 0.2, -1.4, , 0.2]	<pad>	[0, 0, 0, , 0]		
	you	[1.2, -0.3, -0.9, , 0.5]	<pad>	[0, 0, 0, , 0]		

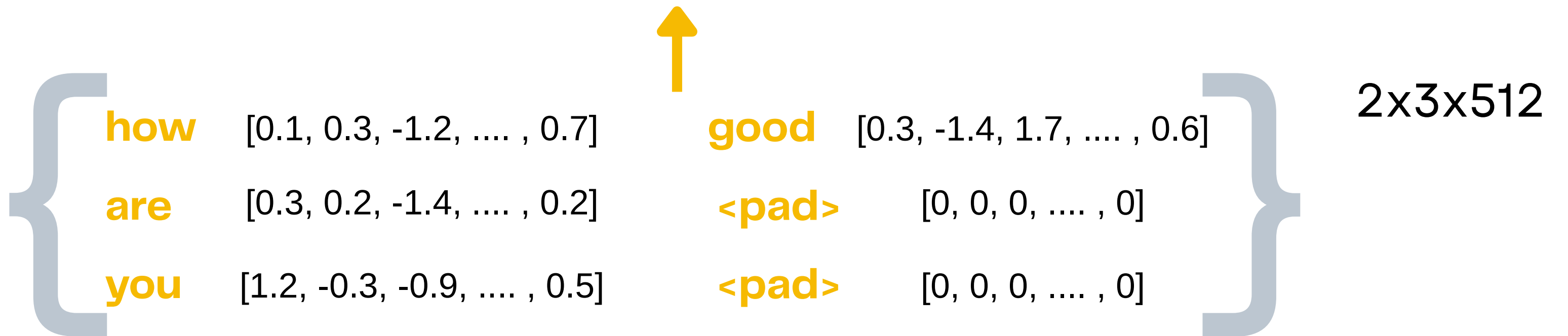
SELF ATTENTION

{	how	[0.1, 0.3, -1.2, , 0.7]	good	[0.3, -1.4, 1.7, , 0.6]	}	2x3
	are	[0.3, 0.2, -1.4, , 0.2]	<pad>	[0, 0, 0, , 0]		
	you	[1.2, -0.3, -0.9, , 0.5]	<pad>	[0, 0, 0, , 0]		

SELF ATTENTION

{	how	[0.1, 0.3, -1.2, , 0.7]	good	[0.3, -1.4, 1.7, , 0.6]	}	2x3x512
	are	[0.3, 0.2, -1.4, , 0.2]	<pad>	[0, 0, 0, , 0]		
	you	[1.2, -0.3, -0.9, , 0.5]	<pad>	[0, 0, 0, , 0]		

SELF ATTENTION



how'	[0.8, 0.2, 0.2, , 0.7]	good'	[0.1, 1.1, -0.3, , -0.2]
are'	[0.5, 0.1, 1.2, , 0.5]	<pad>	[0, 0, 0, , 0]
you'	[1.7, -1.4, 0.8, , 0.1]	<pad>	[0, 0, 0, , 0]

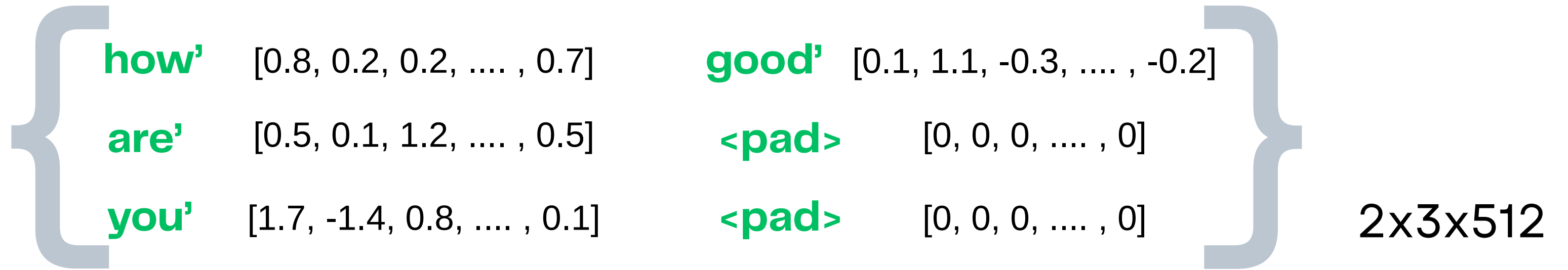


SELF ATTENTION

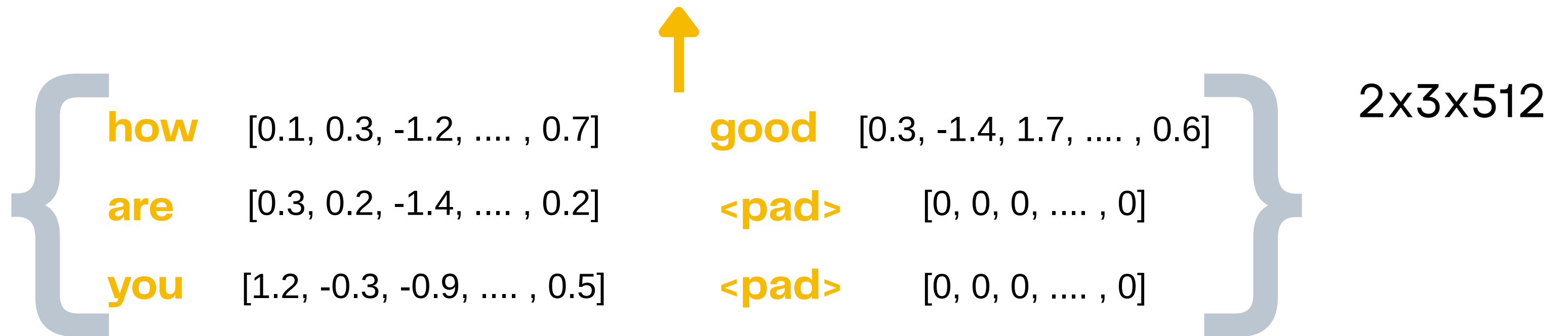


how	[0.1, 0.3, -1.2, , 0.7]	good	[0.3, -1.4, 1.7, , 0.6]
are	[0.3, 0.2, -1.4, , 0.2]	<pad>	[0, 0, 0, , 0]
you	[1.2, -0.3, -0.9, , 0.5]	<pad>	[0, 0, 0, , 0]

2x3x512



SELF ATTENTION



{	how'	[0.8, 0.2, 0.2, , 0.7]	good'	[0.1, 1.1, -0.3, , -0.2]	}
	are'	[0.5, 0.1, 1.2, , 0.5]	<pad>	[0, 0, 0, , 0]	
	you'	[1.7, -1.4, 0.8, , 0.1]	<pad>	[0, 0, 0, , 0]	

how' [0.8, 0.2, 0.2, , 0.7]

are' [0.5, 0.1, 1.2, , 0.5]

you' [1.7, -1.4, 0.8, , 0.1]

good' [0.1, 1.1, -0.3, , -0.2]

<pad> [0, 0, 0, , 0]

<pad> [0, 0, 0, , 0]

how' [0.8, 0.2, 0.2, , 0.7]

are' [0.5, 0.1, 1.2, , 0.5]

you' [1.7, -1.4, 0.8, , 0.1]

good' [0.1, 1.1, -0.3, , -0.2]

<pad> [0, 0, 0, , 0]

<pad> [0, 0, 0, , 0]

6x512

how'	[0.8, 0.2, 0.2, , 0.7]
are'	[0.5, 0.1, 1.2, , 0.5]
you'	[1.7, -1.4, 0.8, , 0.1]
good'	[0.1, 1.1, -0.3, , -0.2]
<pad>	[0 , 0 , 0 , , 0]
<pad>	[0 , 0 , 0 , , 0]

6x512

how'	[0.8, 0.2, 0.2, , 0.7]	6x512
are'	[0.5, 0.1, 1.2, , 0.5]	
you'	[1.7, -1.4, 0.8, , 0.1]	
good'	[0.1, 1.1, -0.3, , -0.2]	
<pad>	[0 , 0 , 0 , , 0]	
<pad>	[0 , 0 , 0 , , 0]	

Mean₁, Sigma₁

how'	[0.8, 0.2, 0.2,, 0.7]
are'	[0.5, 0.1, 1.2,, 0.5]
you'	[1.7, -1.4, 0.8,, 0.1]
good'	[0.1, 1.1, -0.3,, -0.2]
<pad>	[0, 0, 0,, 0]
<pad>	[0, 0, 0,, 0]

6x512

Mean₁, Sigma₁ Mean₅₁₂, Sigma₅₁₂





Batch Normalization

how'	[0.8, 0.2, 0.2, ..., 0.7]
are'	[0.5, 0.1, 1.2, ..., 0.5]
you'	[1.7, -1.4, 0.8, ..., 0.1]
good'	[0.1, 1.1, -0.3, ..., -0.2]
<pad>	[0, 0, 0, ..., 0]
<pad>	[0, 0, 0, ..., 0]

6x512

Mean₁, Sigma₁ Mean₅₁₂, Sigma₅₁₂

~~Batch
Normalization~~

how'	[0.8, 0.2, 0.2, , 0.7]	← Mean ₁ , Sigma ₁
are'	[0.5, 0.1, 1.2, , 0.5]	
you'	[1.7, -1.4, 0.8, , 0.1]	
good'	[0.1, 1.1, -0.3, , -0.2]	
<pad>	[0 , 0 , 0 , , 0]	
<pad>	[0 , 0 , 0 , , 0]	6x512

how' [0.8, 0.2, 0.2, , 0.7] ← Mean₁, Sigma₁
are' [0.5, 0.1, 1.2, , 0.5]
you' [1.7, -1.4, 0.8, , 0.1] ← Mean₃, Sigma₃
good' [0.1, 1.1, -0.3, , -0.2]
<pad> [0 , 0 , 0 , , 0]
<pad> [0 , 0 , 0 , , 0] 6x512

how' [0.8, 0.2, 0.2, , 0.7] ← Mean₁, Sigma₁
are' [0.5, 0.1, 1.2, , 0.5]
you' [1.7, -1.4, 0.8, , 0.1] ← Mean₃, Sigma₃
good' [0.1, 1.1, -0.3, , -0.2]
<pad> [0, 0, 0, , 0]
<pad> [0, 0, 0, , 0] 6x512

Layer
Normalization

how'	[0.8, 0.2, 0.2, , 0.7]	← Mean ₁ , Sigma ₁
are'	[0.5, 0.1, 1.2, , 0.5]	
you'	[1.7, -1.4, 0.8, , 0.1]	← Mean ₃ , Sigma ₃
good'	[0.1, 1.1, -0.3, , -0.2]	
<pad>	[0, 0, 0, , 0]	
<pad>	[0, 0, 0, , 0]	

6x512

Layer
Normalization





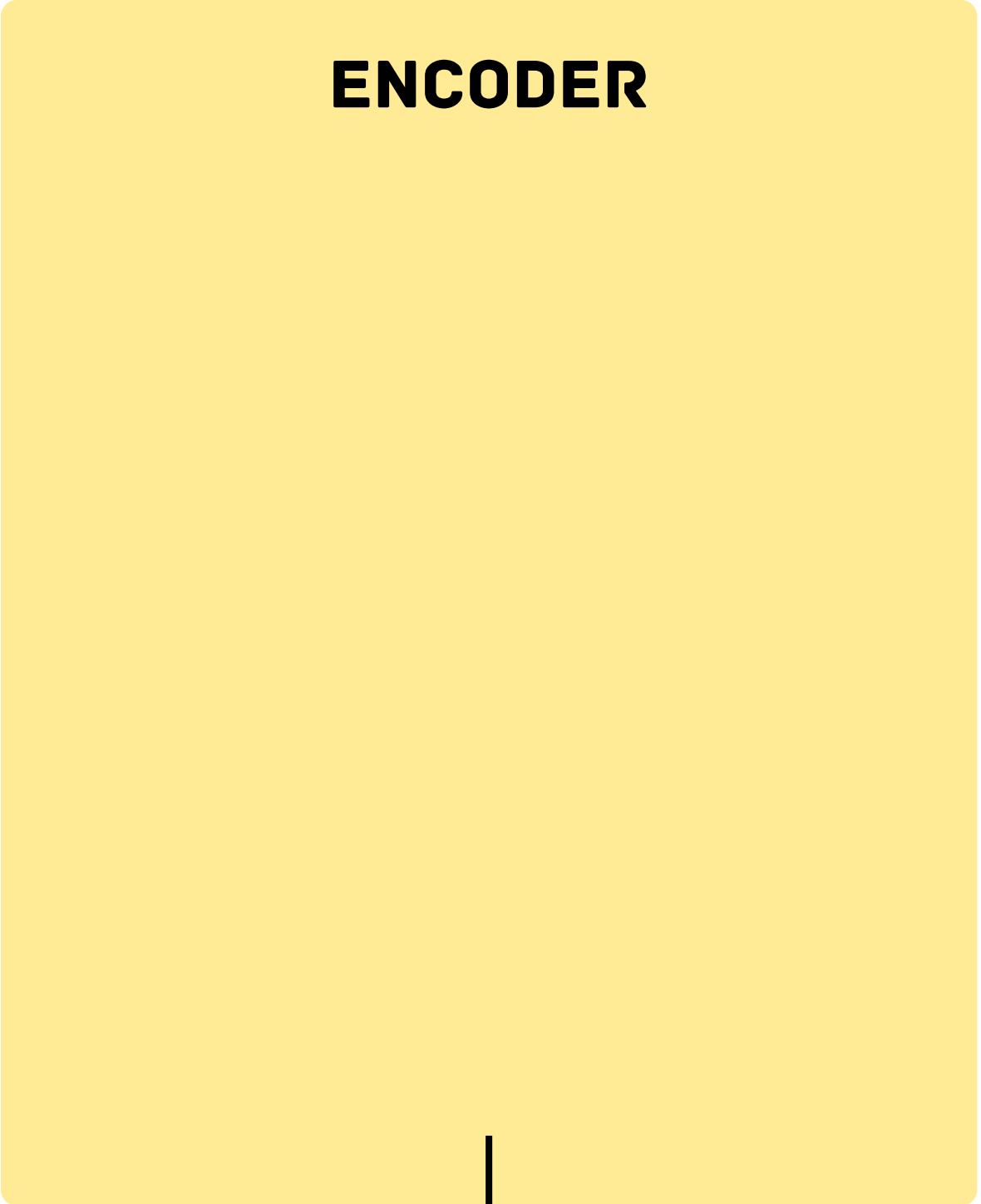
ENCODER ARCHITECTURE

Putting it all Together

ENCODER

ENCODER

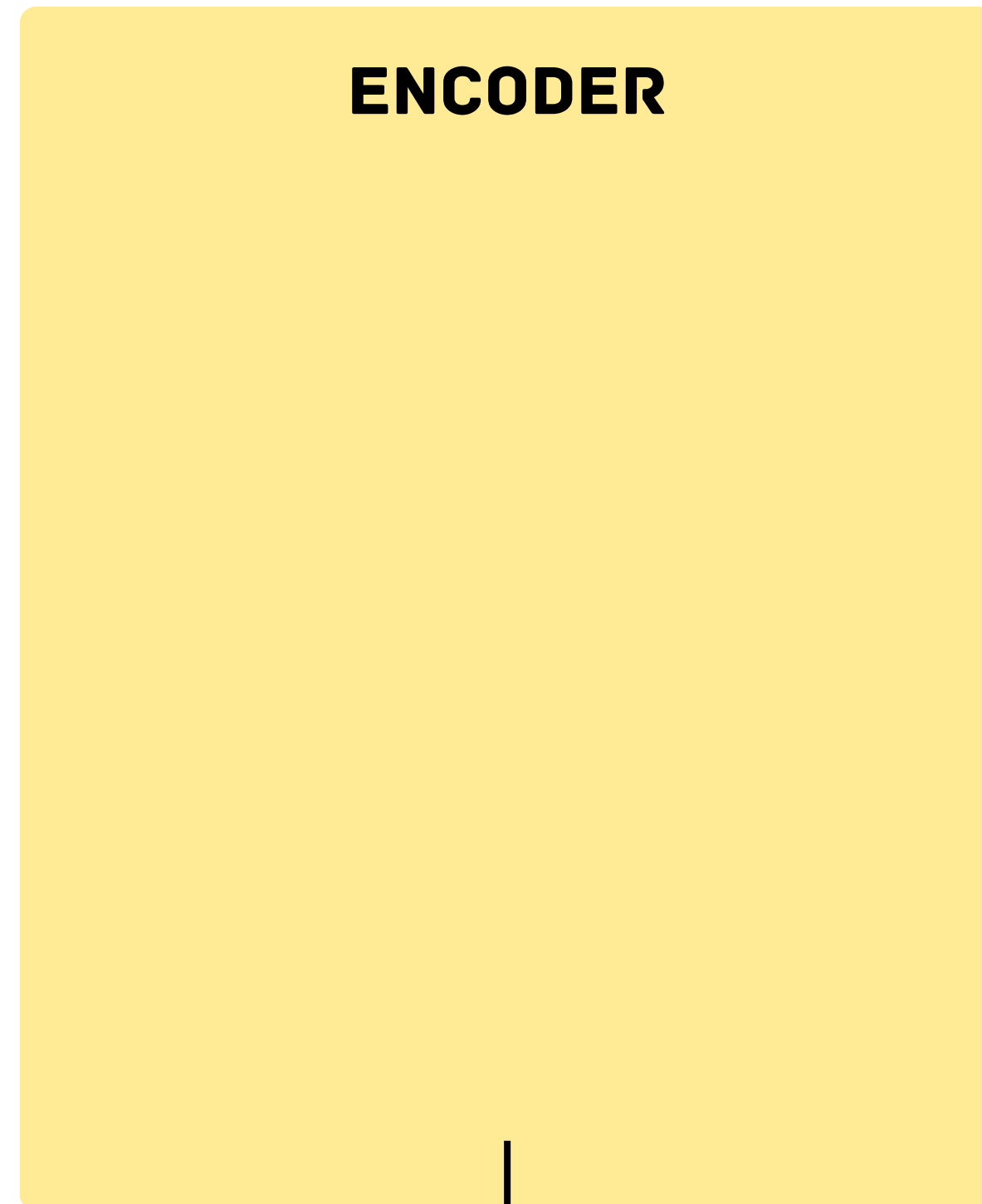
Input



Input

Word Embeddings

King	0.12	0.43	-1.1	0.3	0.4	0.2
Queen	0.22	0.36	-0.9	0.29	0.42	-1.3
Man	-1.5	-0.8	1.0	0.8	0.62	0.23
Woman	-1.3	-0.65	0.98	0.76	0.60	-1.2

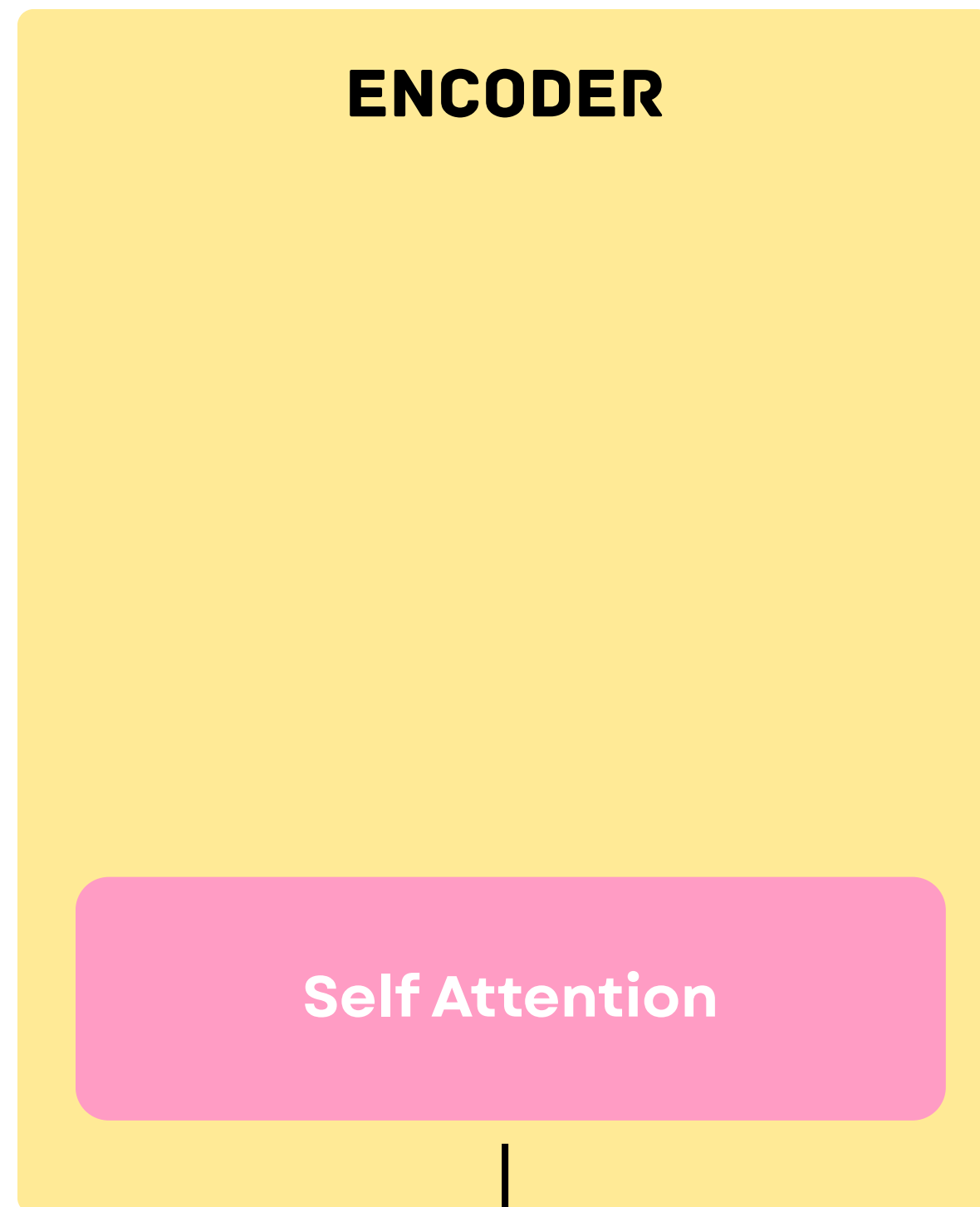


Input

Word Embeddings

King	0.12	0.43	-1.1	0.3	0.4	0.2
Queen	0.22	0.36	-0.9	0.29	0.42	-1.3
Man	-1.5	-0.8	1.0	0.8	0.62	0.23
Woman	-1.3	-0.65	0.98	0.76	0.60	-1.2

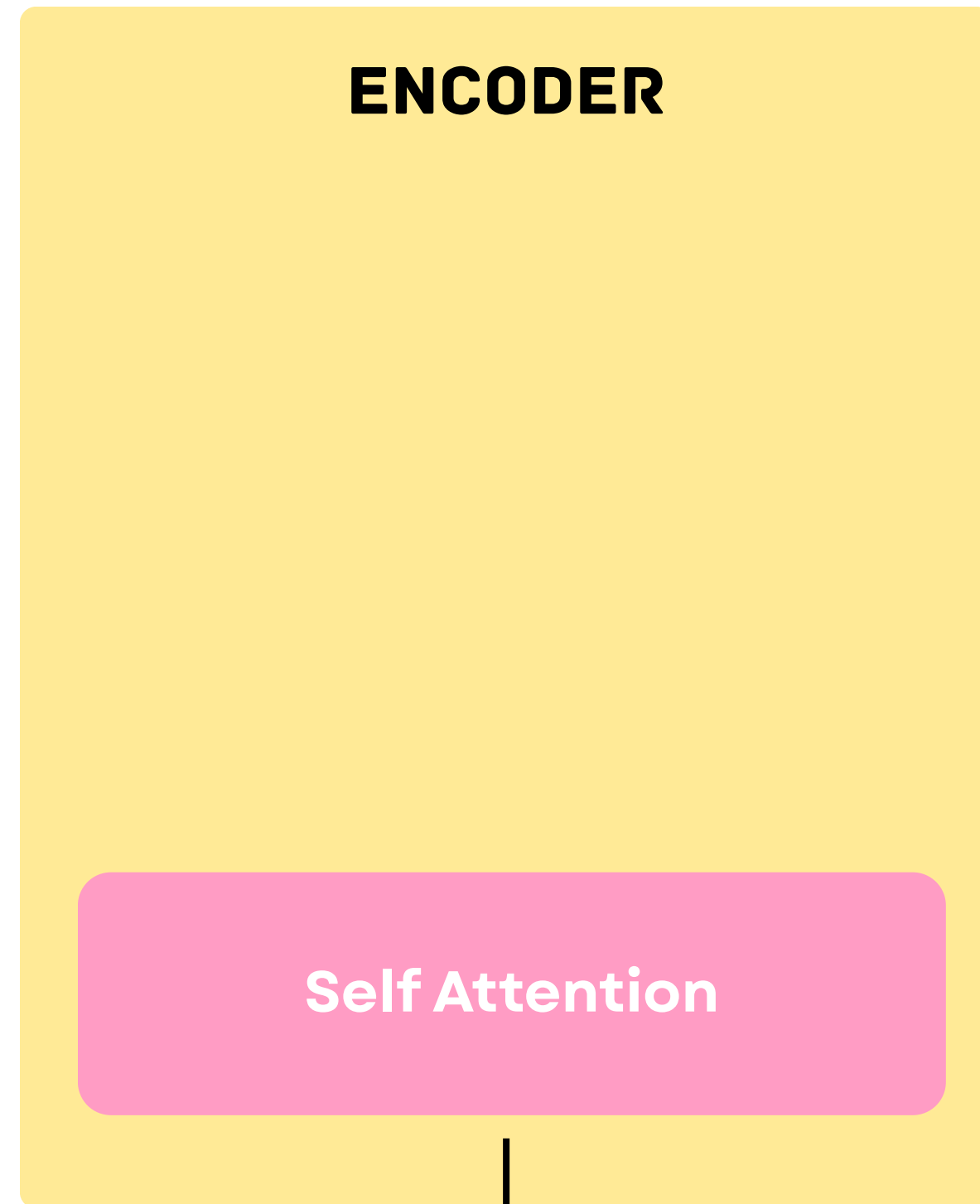
Q: Is there a problem here?



ENCODER

Self Attention

Input



Input

Q: Is there a problem here?

ENCODER

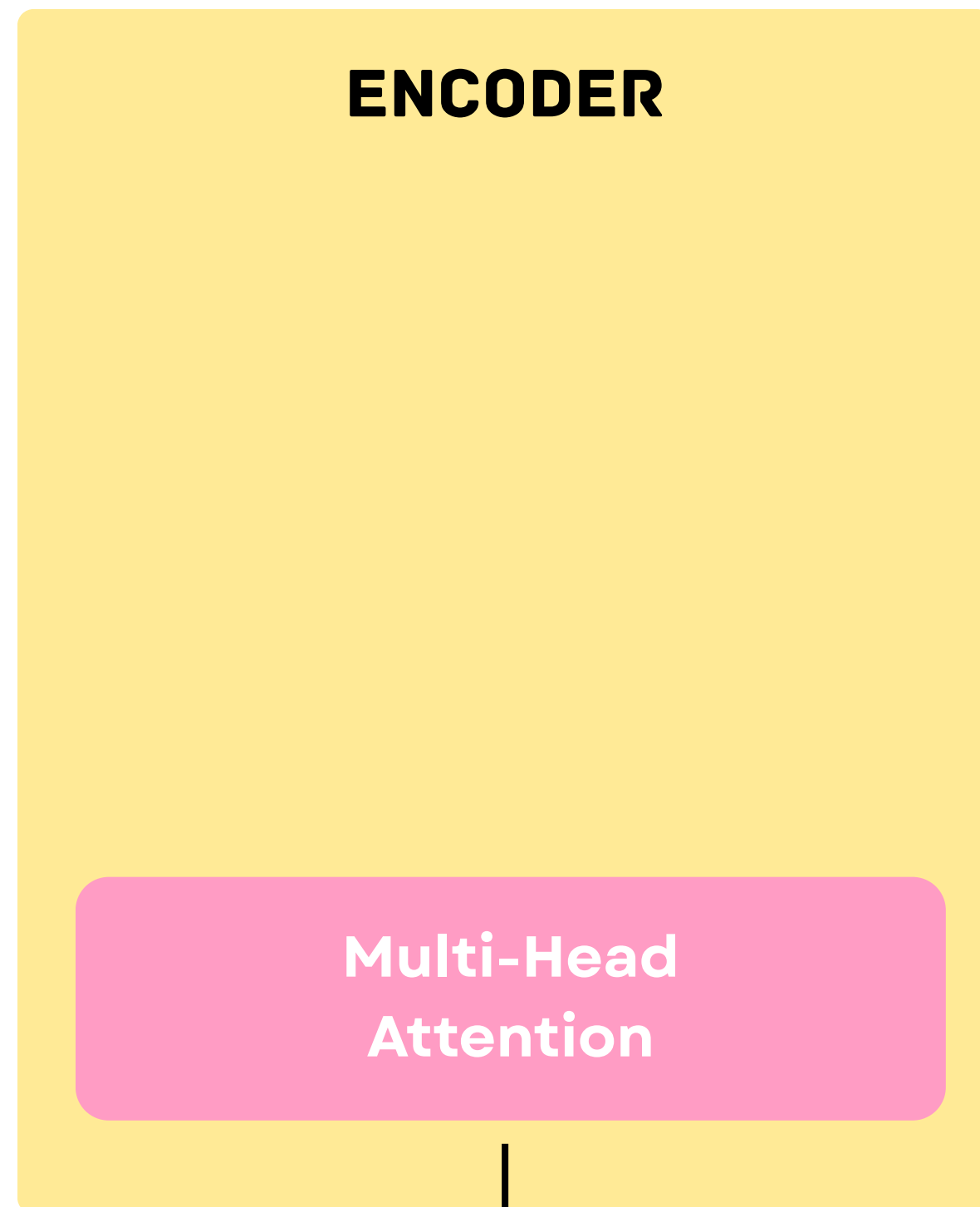
**Multi-Head
Attention**

Input



```
graph TD; Input --> MHA[Multi-Head Attention]; MHA --> Encoder[ENCODER];
```

The diagram illustrates a component of a neural network architecture. A large yellow rounded rectangle is labeled 'ENCODER' at the top. Inside the bottom of this rectangle is a smaller pink rounded rectangle labeled 'Multi-Head Attention'. A black arrow points upwards from the word 'Input' to the bottom of the pink rectangle.



ENCODER

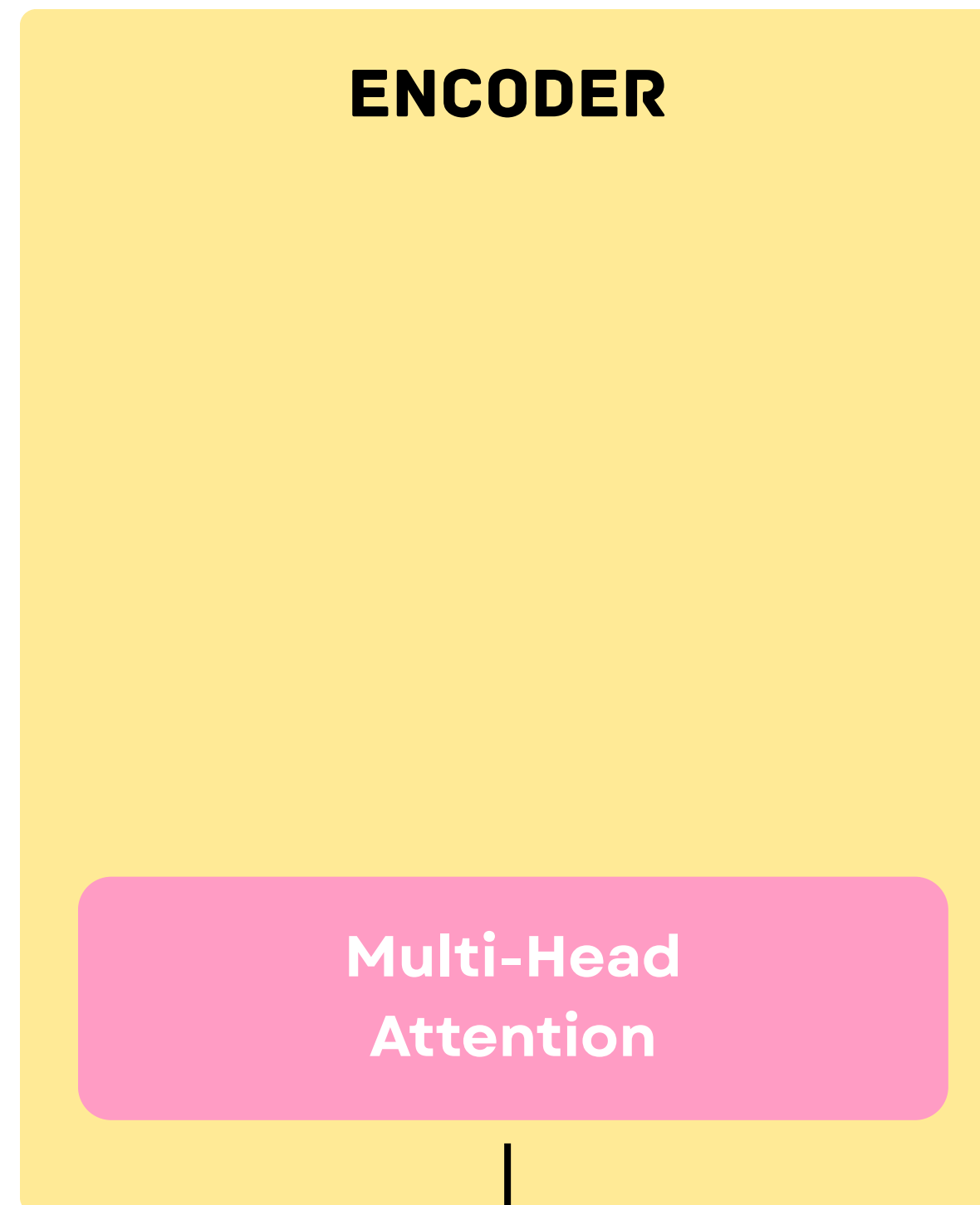
**Multi-Head
Attention**

Input

**Self
Attention**

**Self
Attention**

**Self
Attention**



ENCODER

**Multi-Head
Attention**

Input

**Self
Attention**

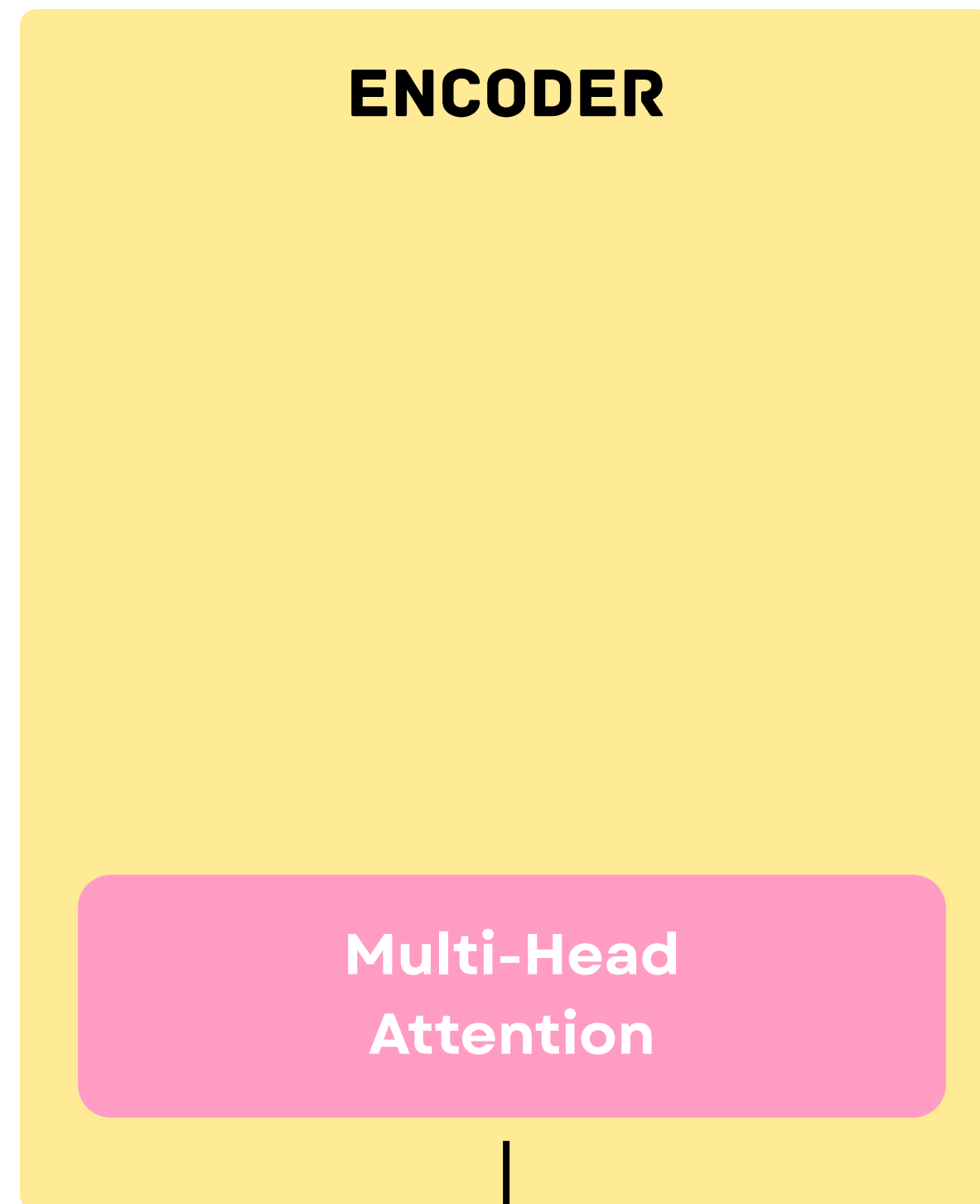
**Self
Attention**

**Self
Attention**

RAM

RAM

RAM



ENCODER

**Multi-Head
Attention**

Input

**Self
Attention**

**Self
Attention**

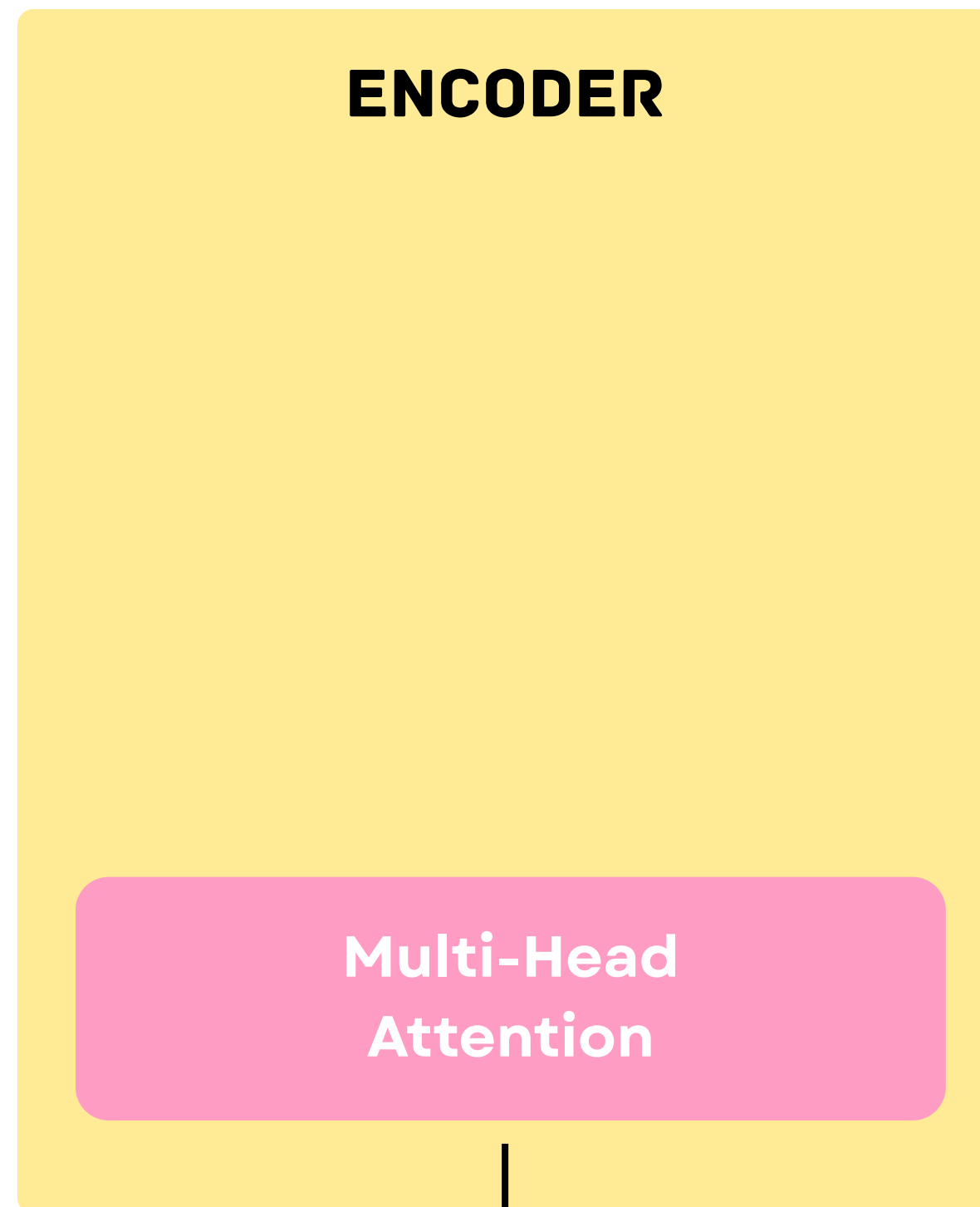
**Self
Attention**

Spatial Relationship

Subject verb object

Timing

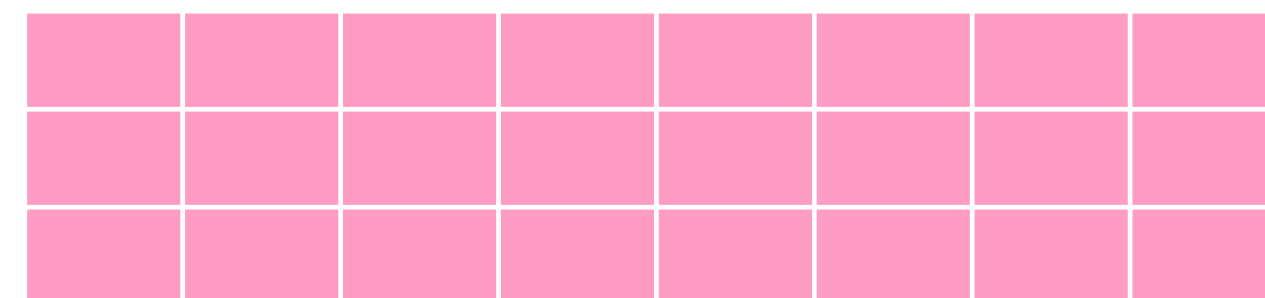
Possession



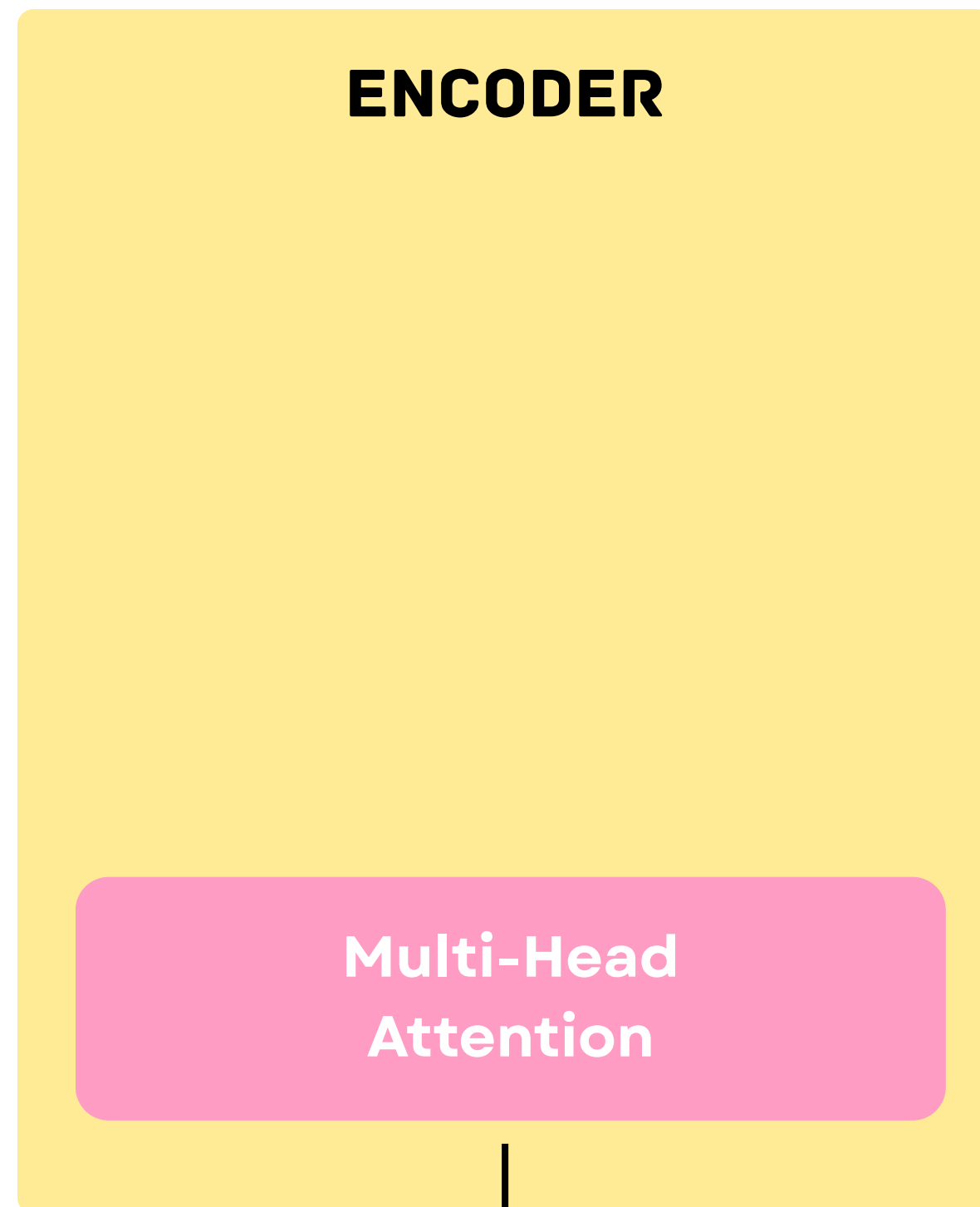
ENCODER

**Multi-Head
Attention**

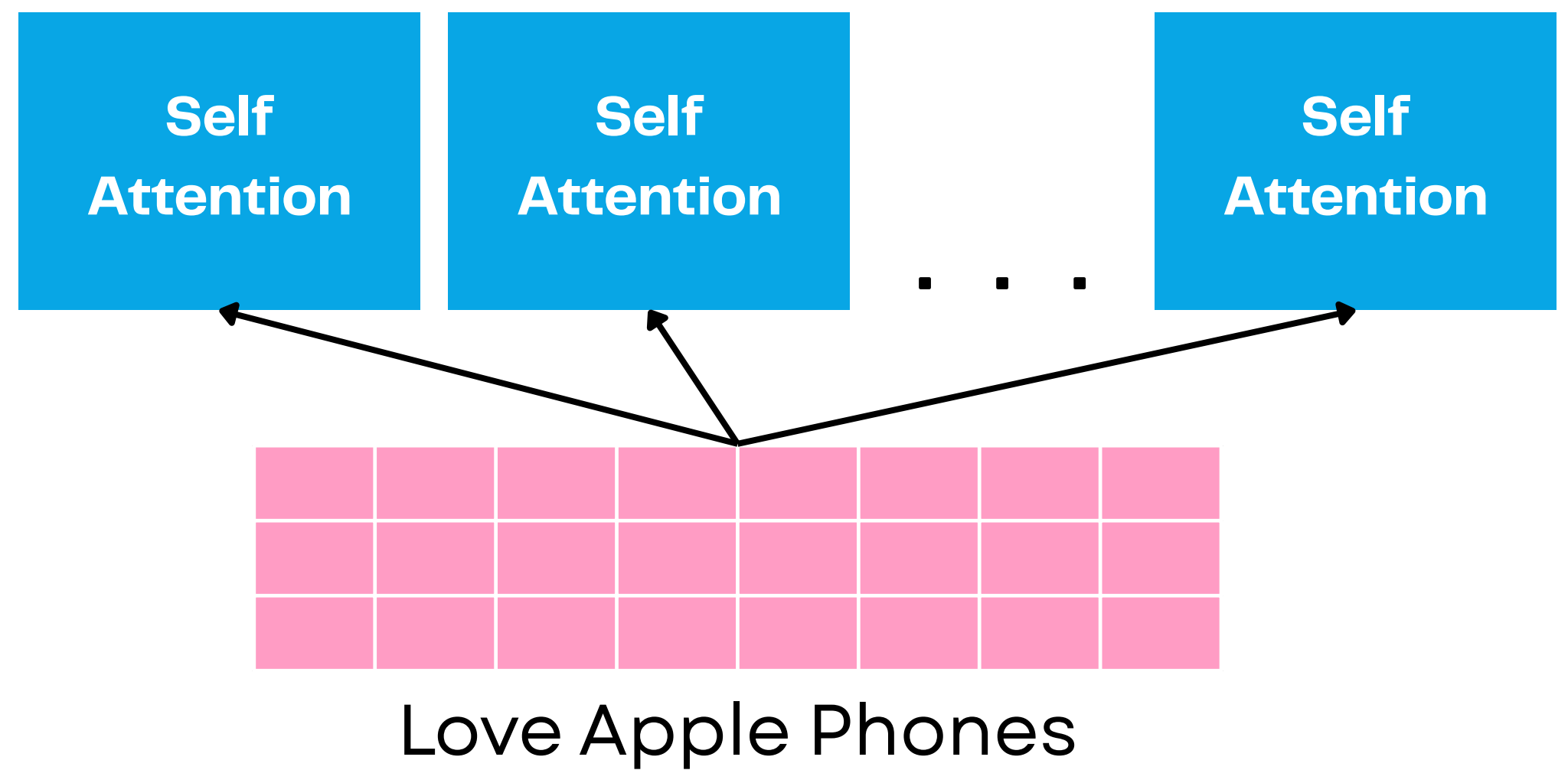
Input

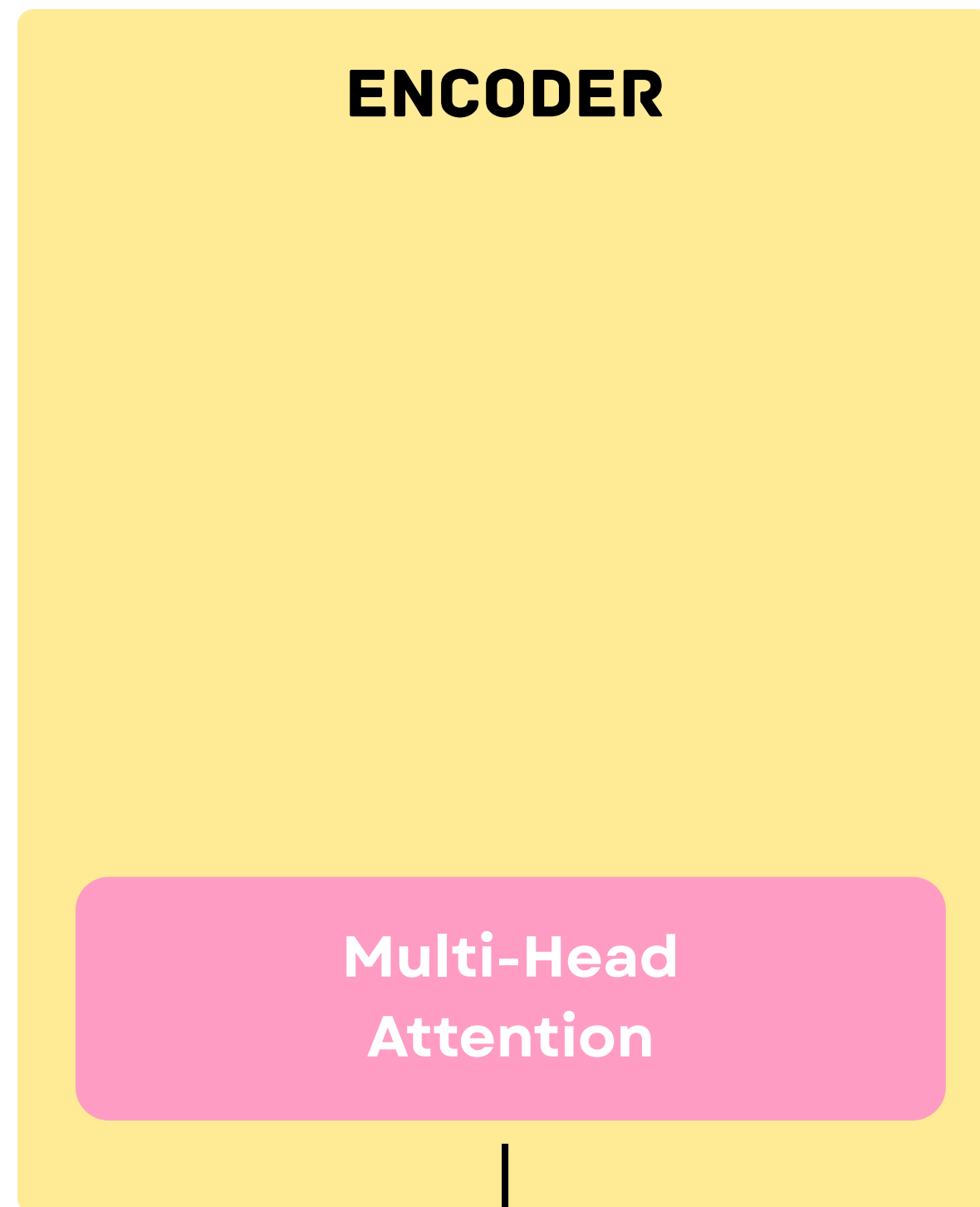


Love Apple Phones

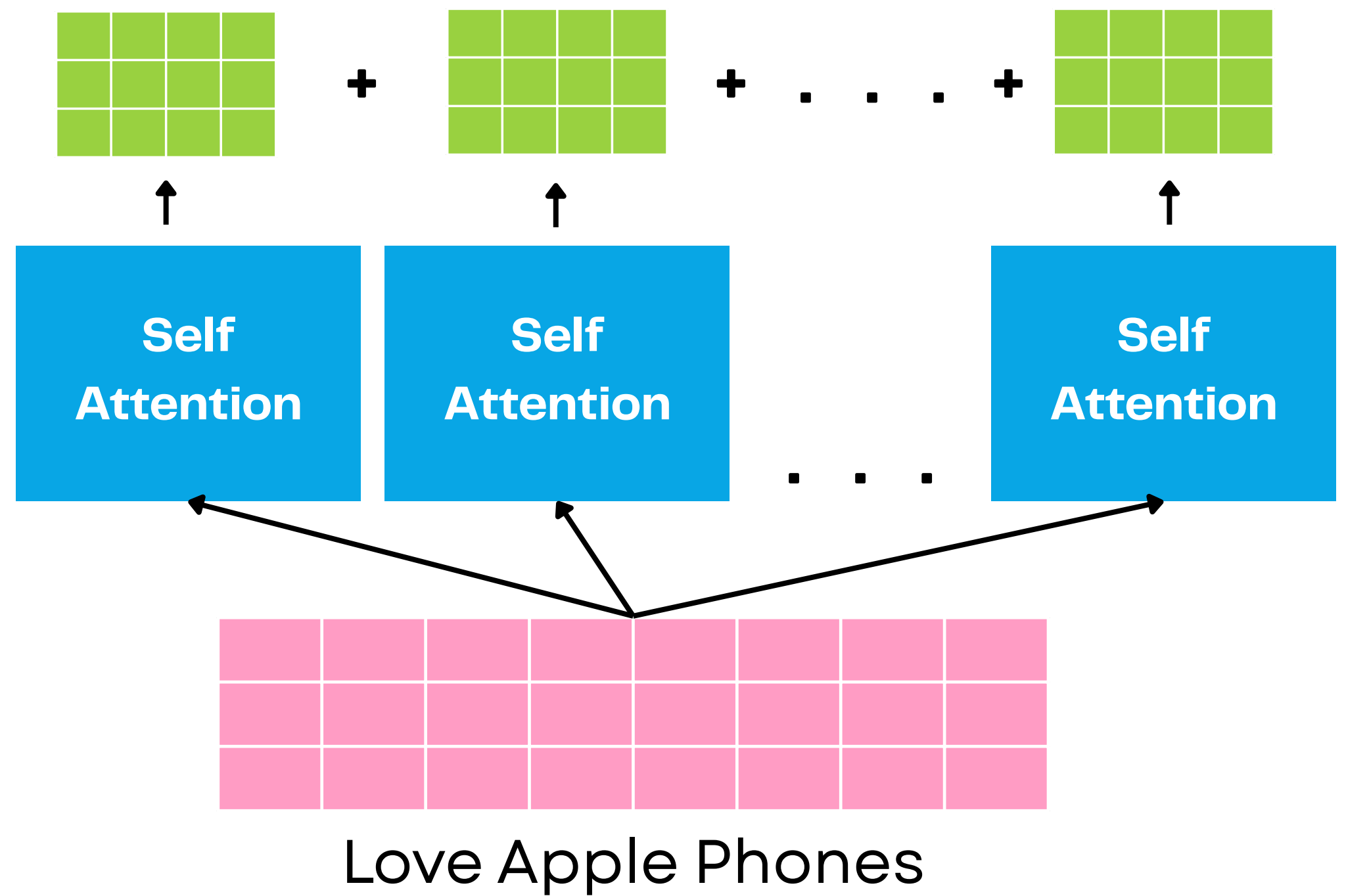


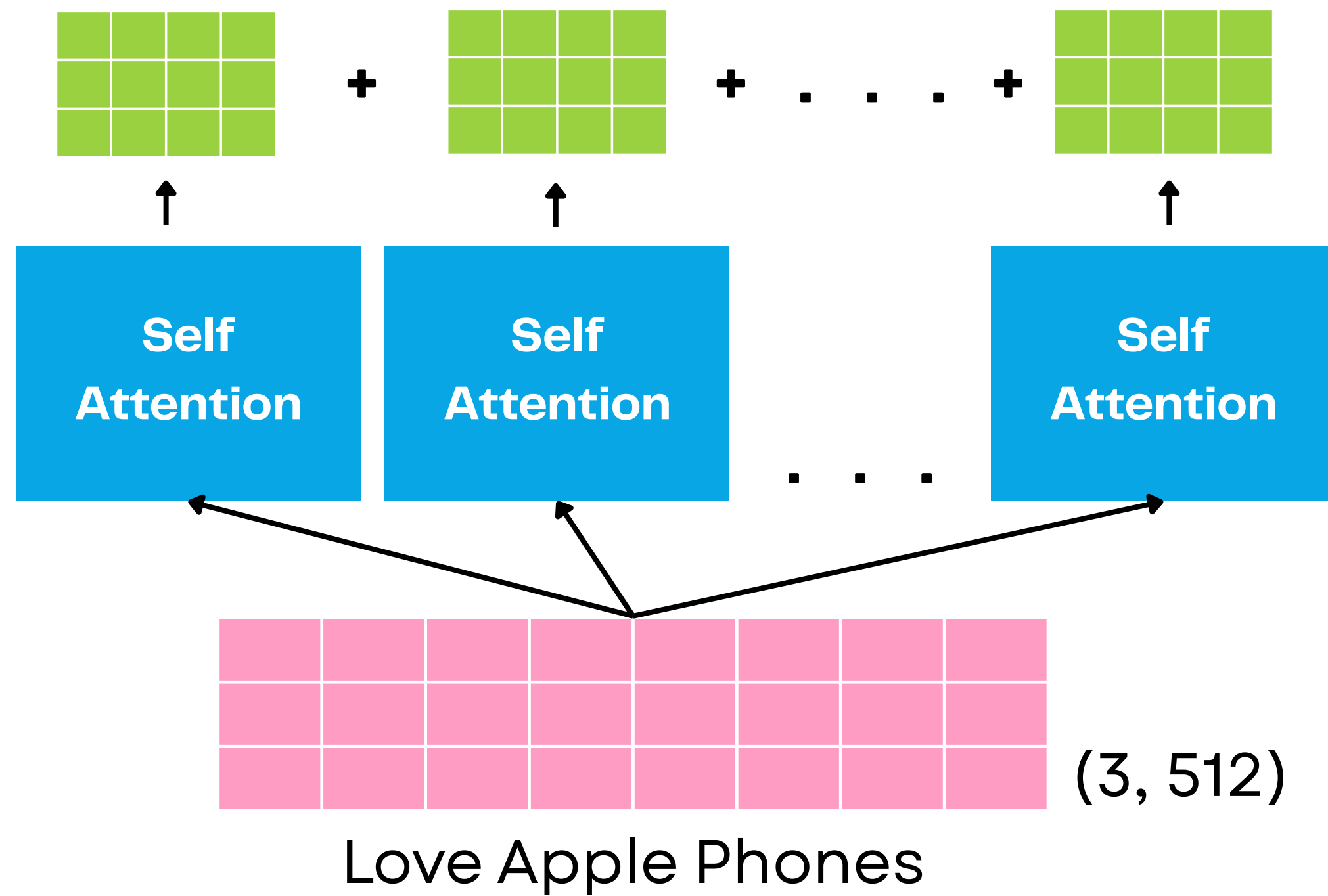
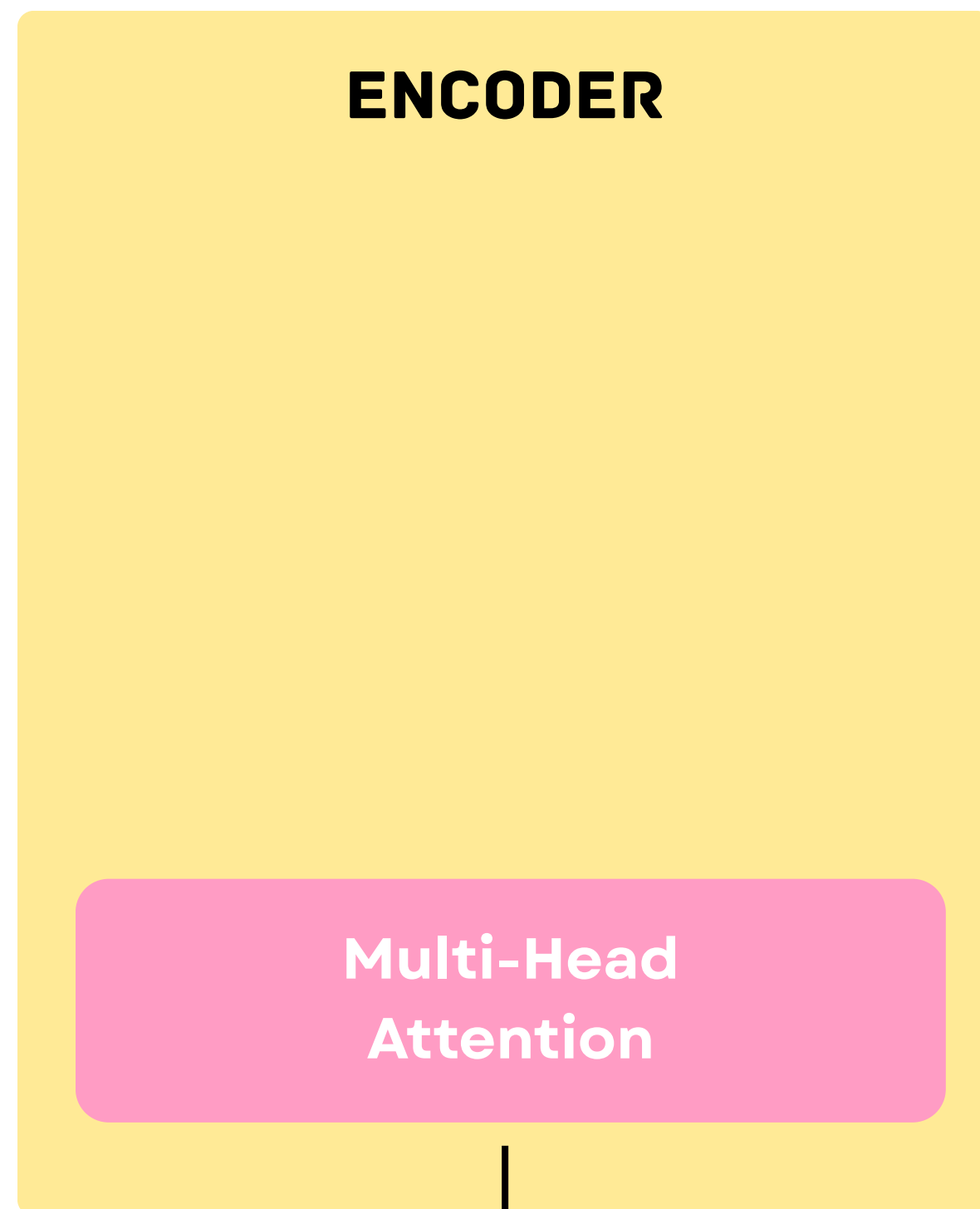
Input

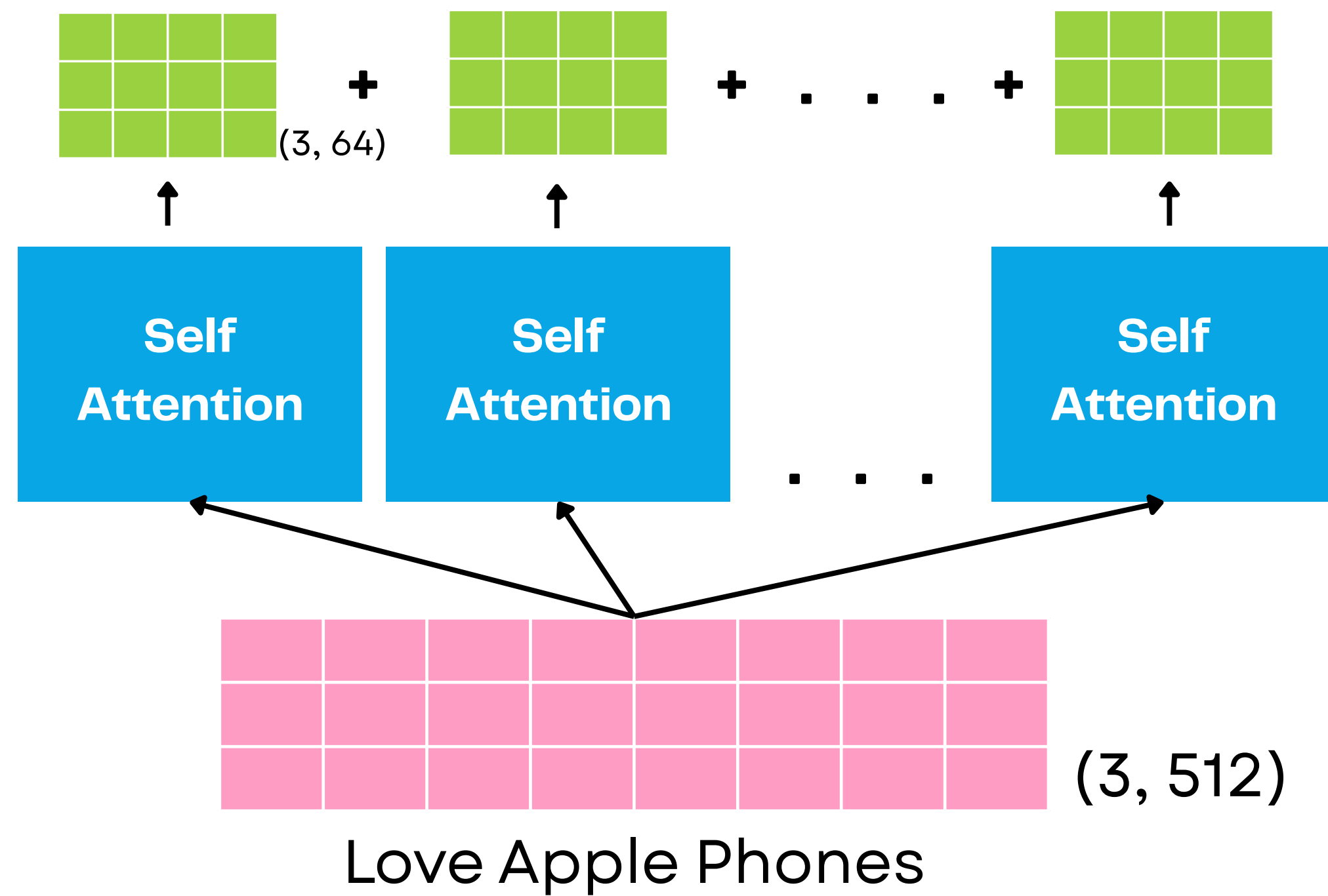
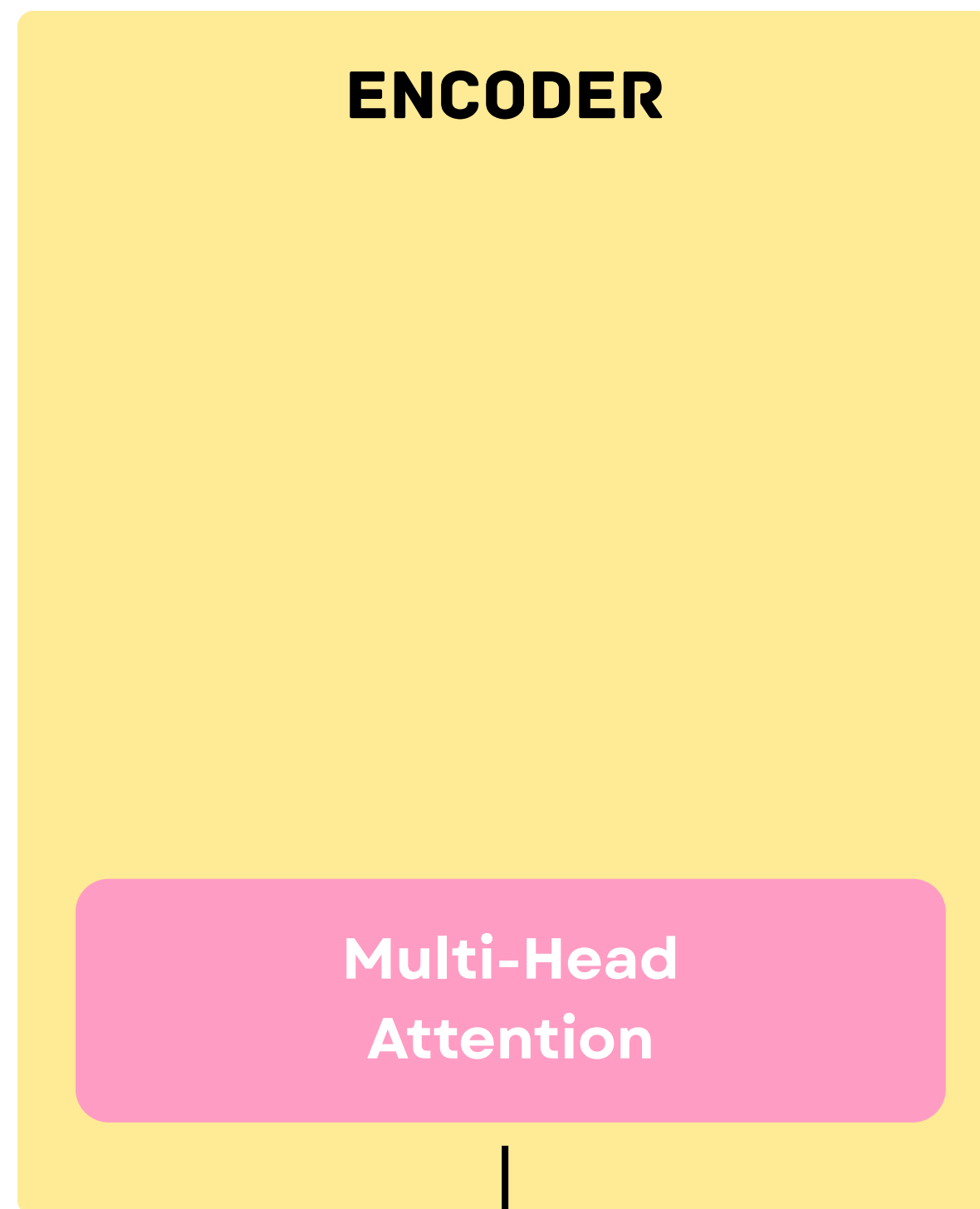


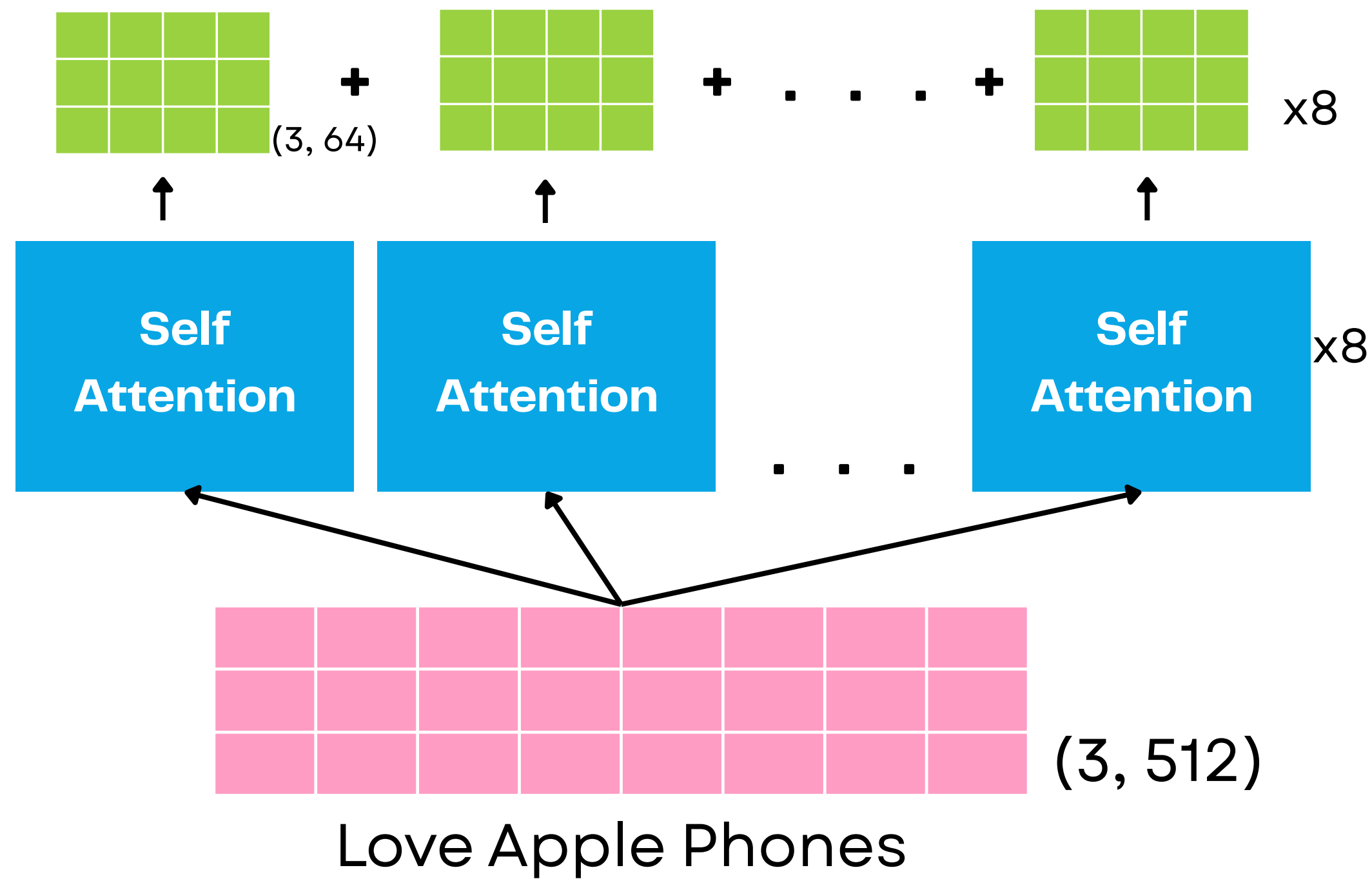
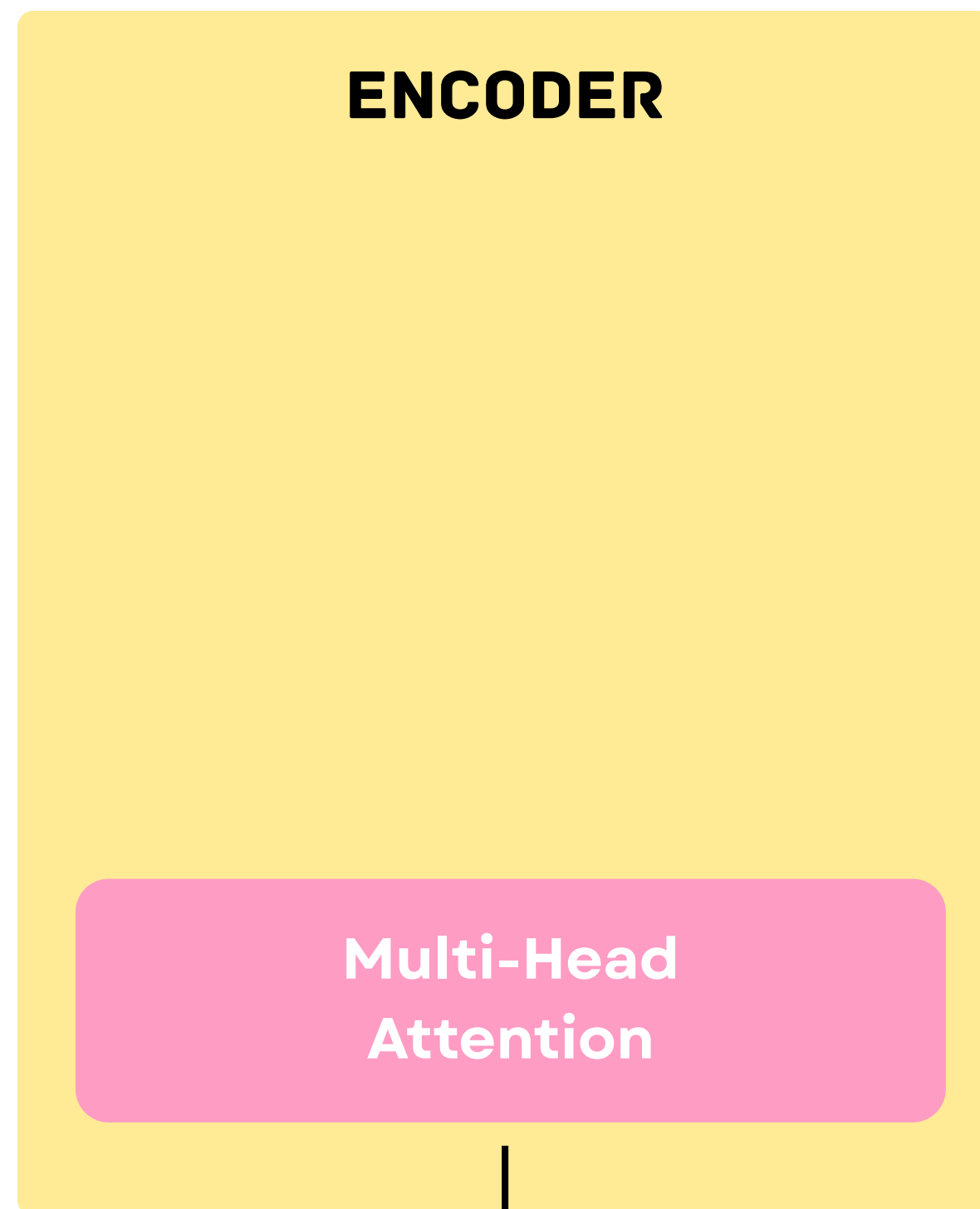


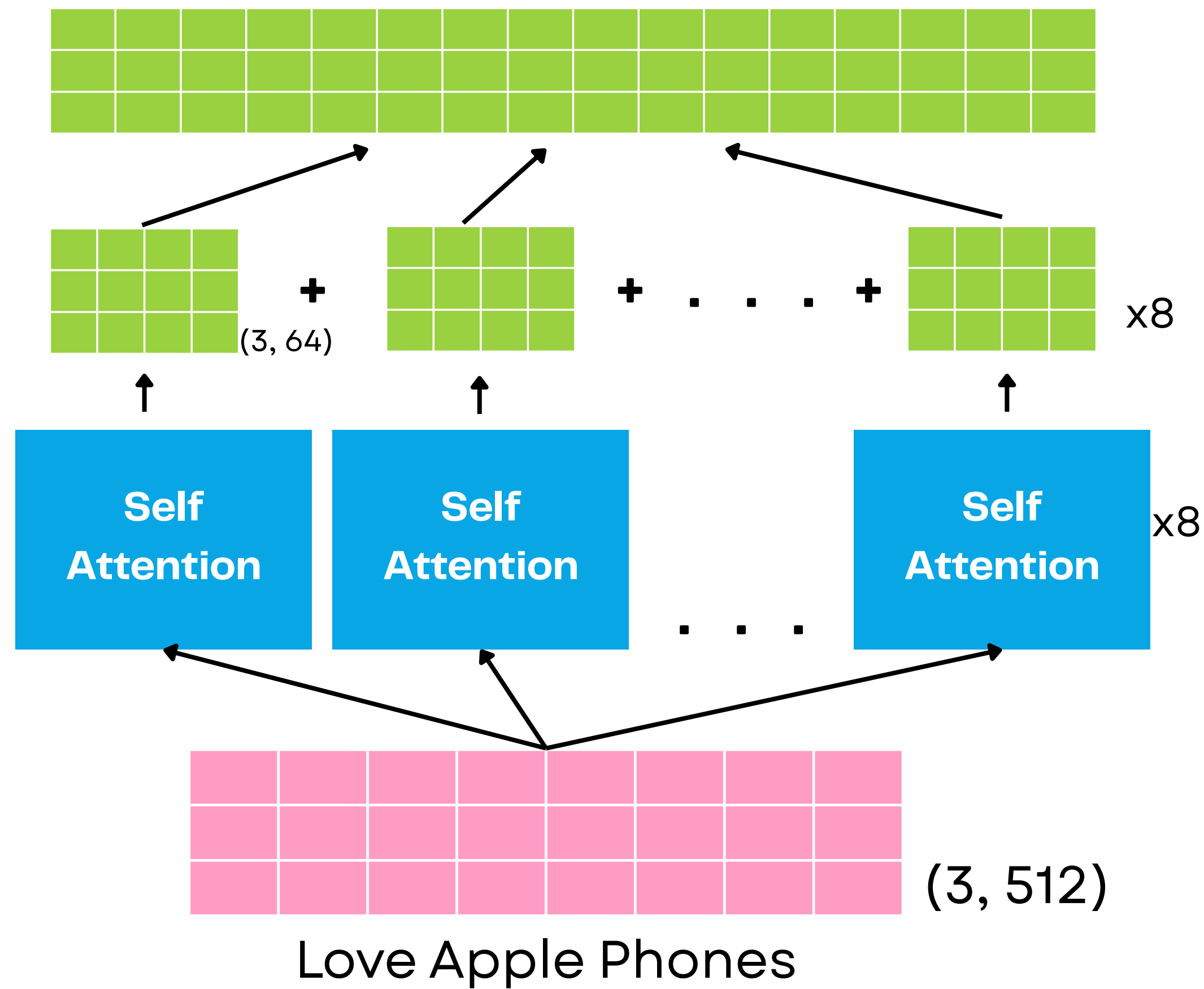
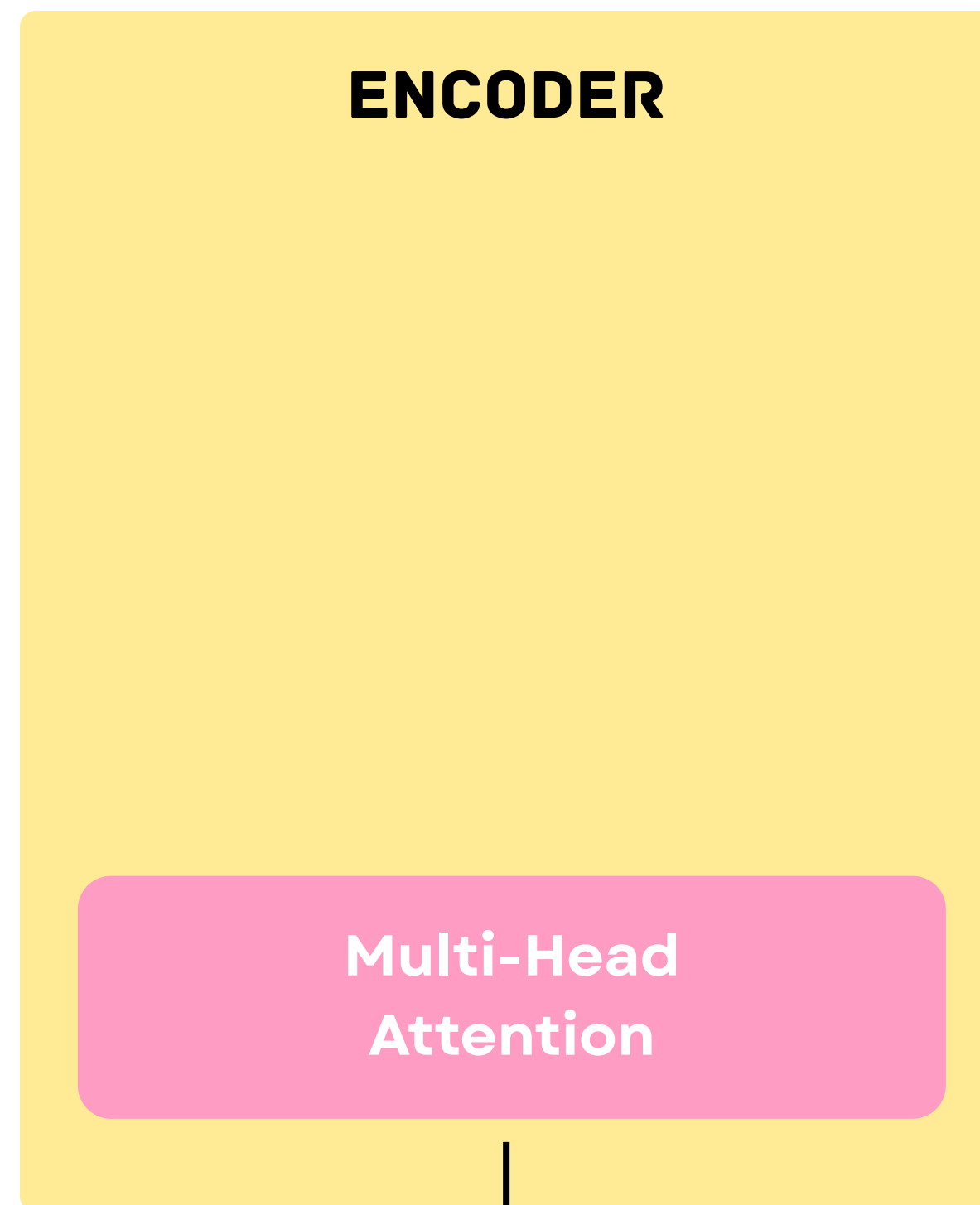
Input

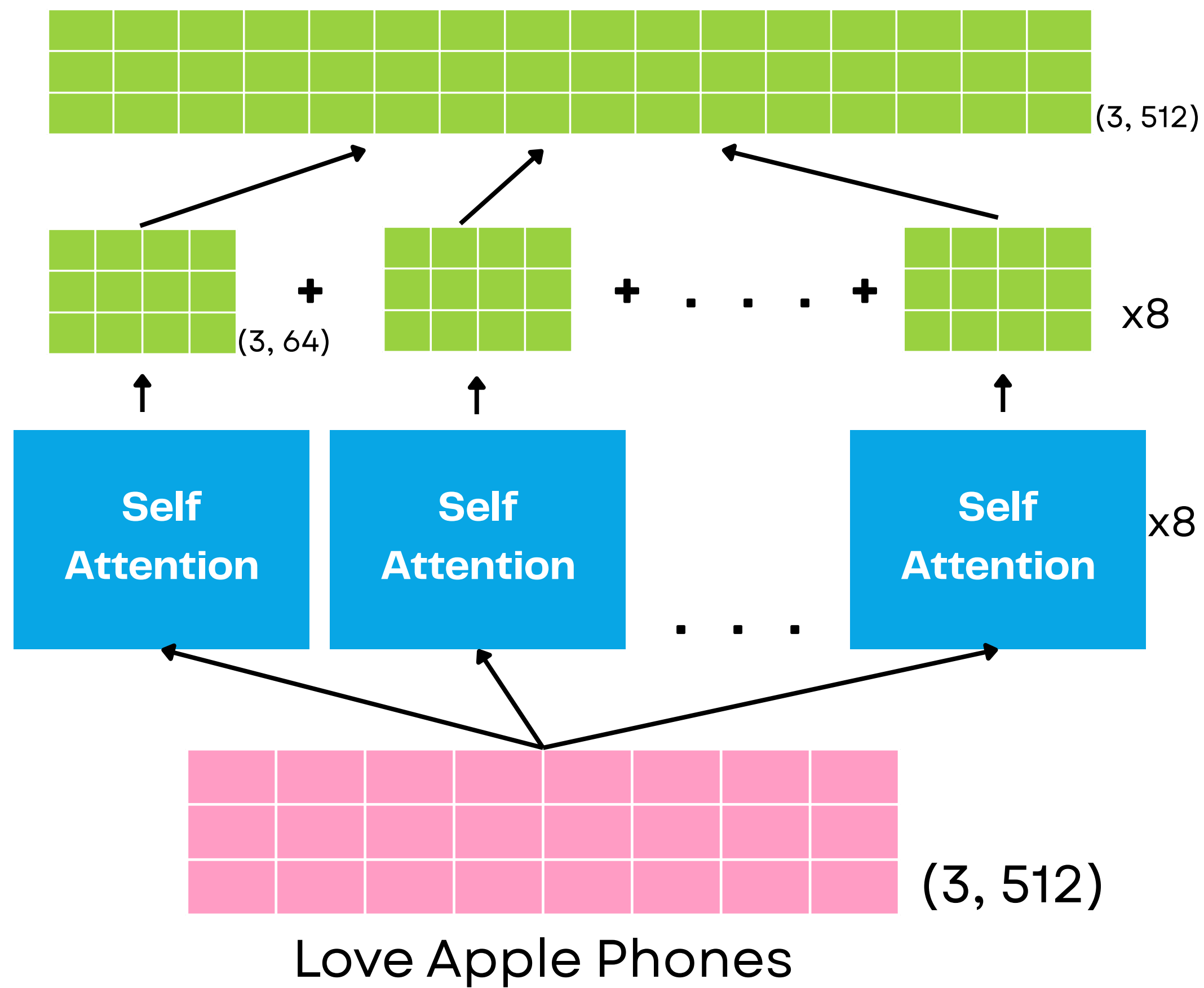
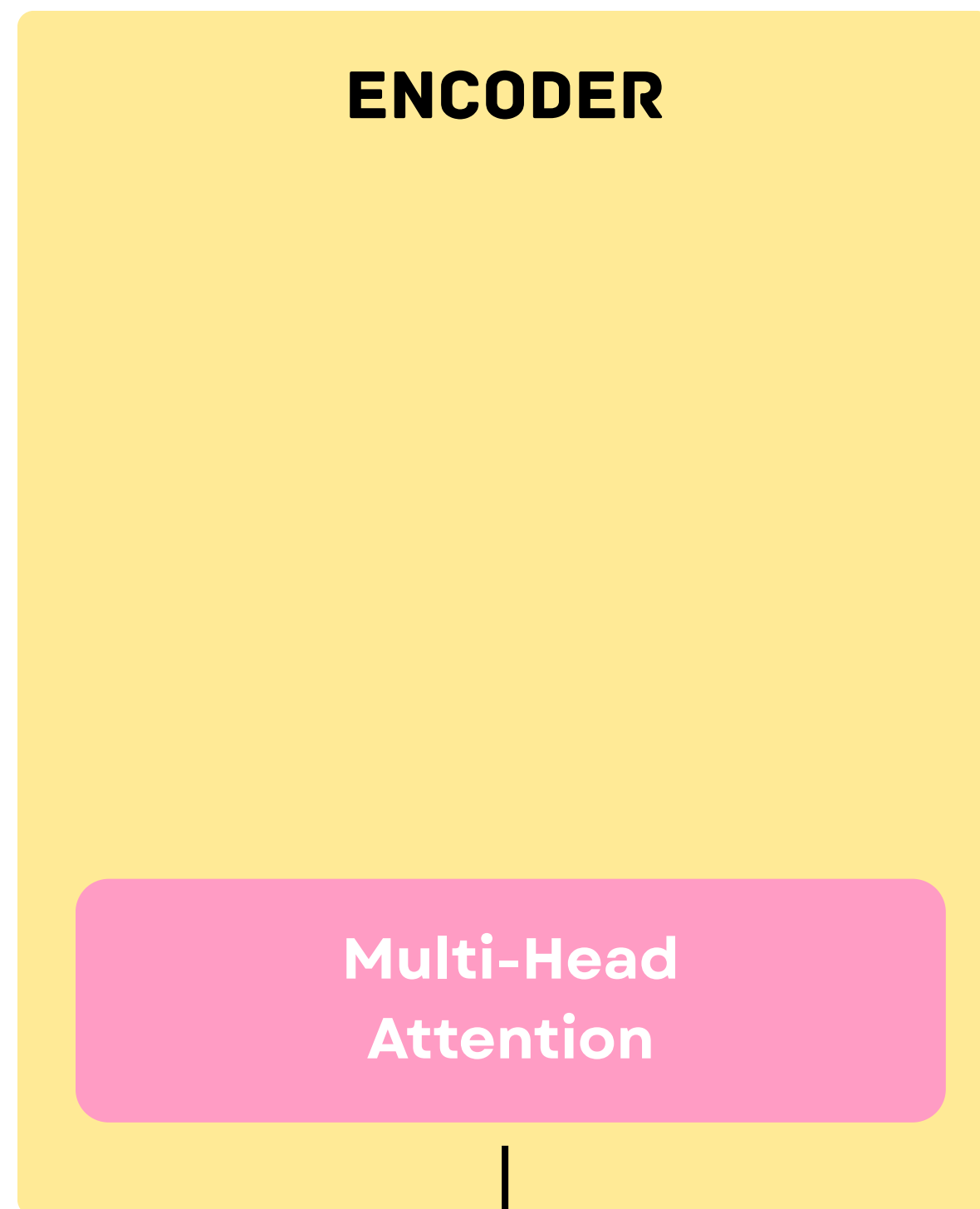


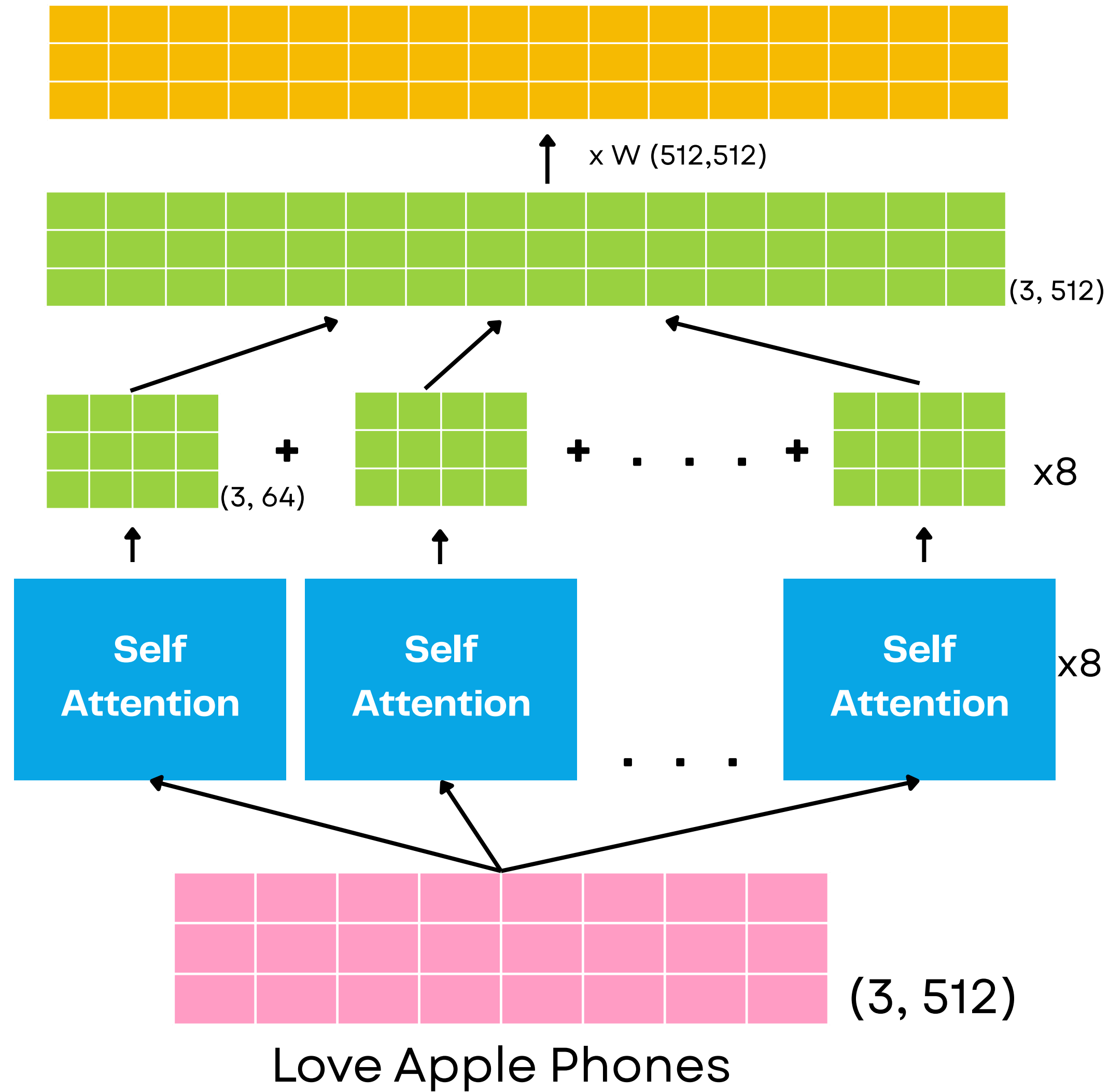
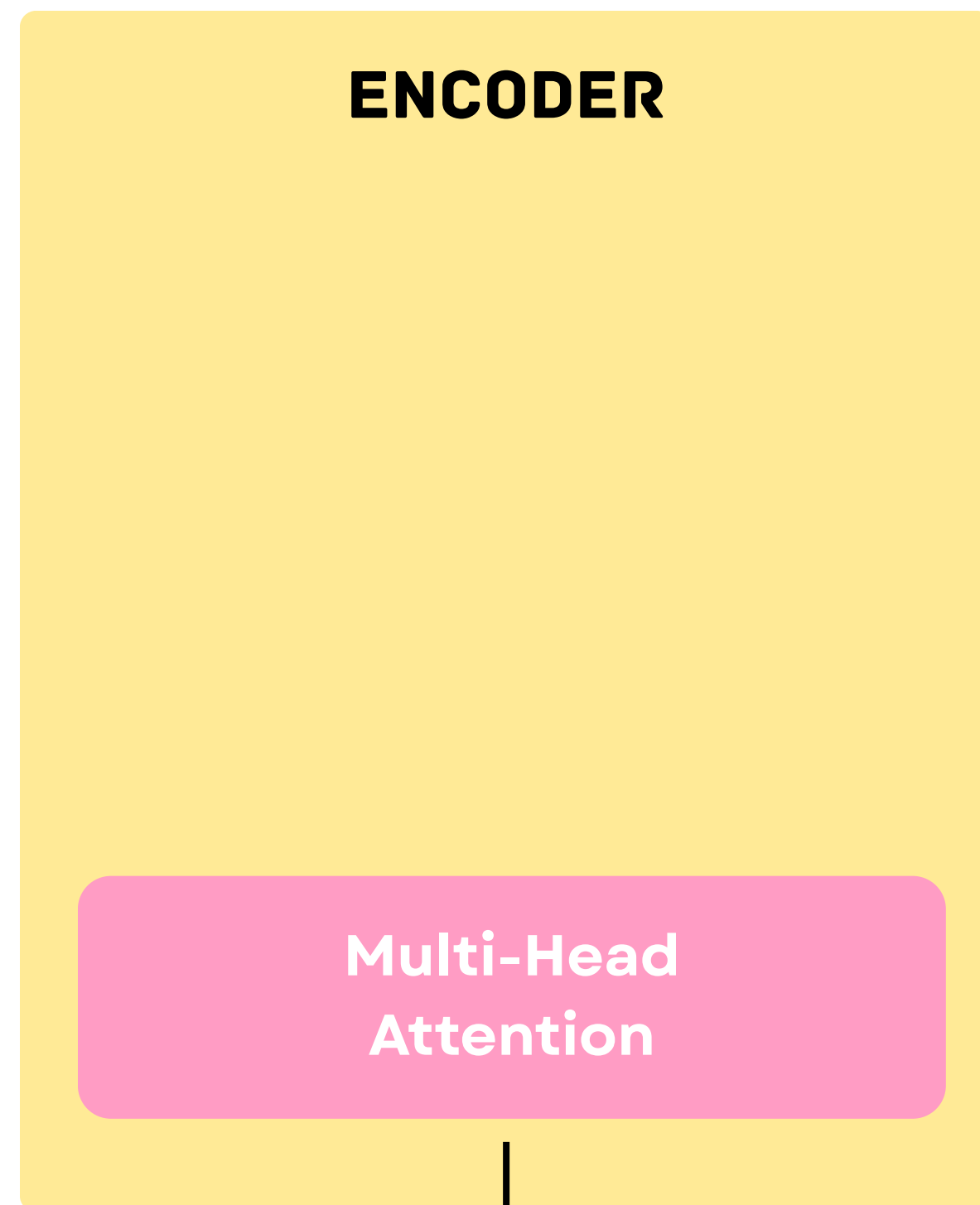


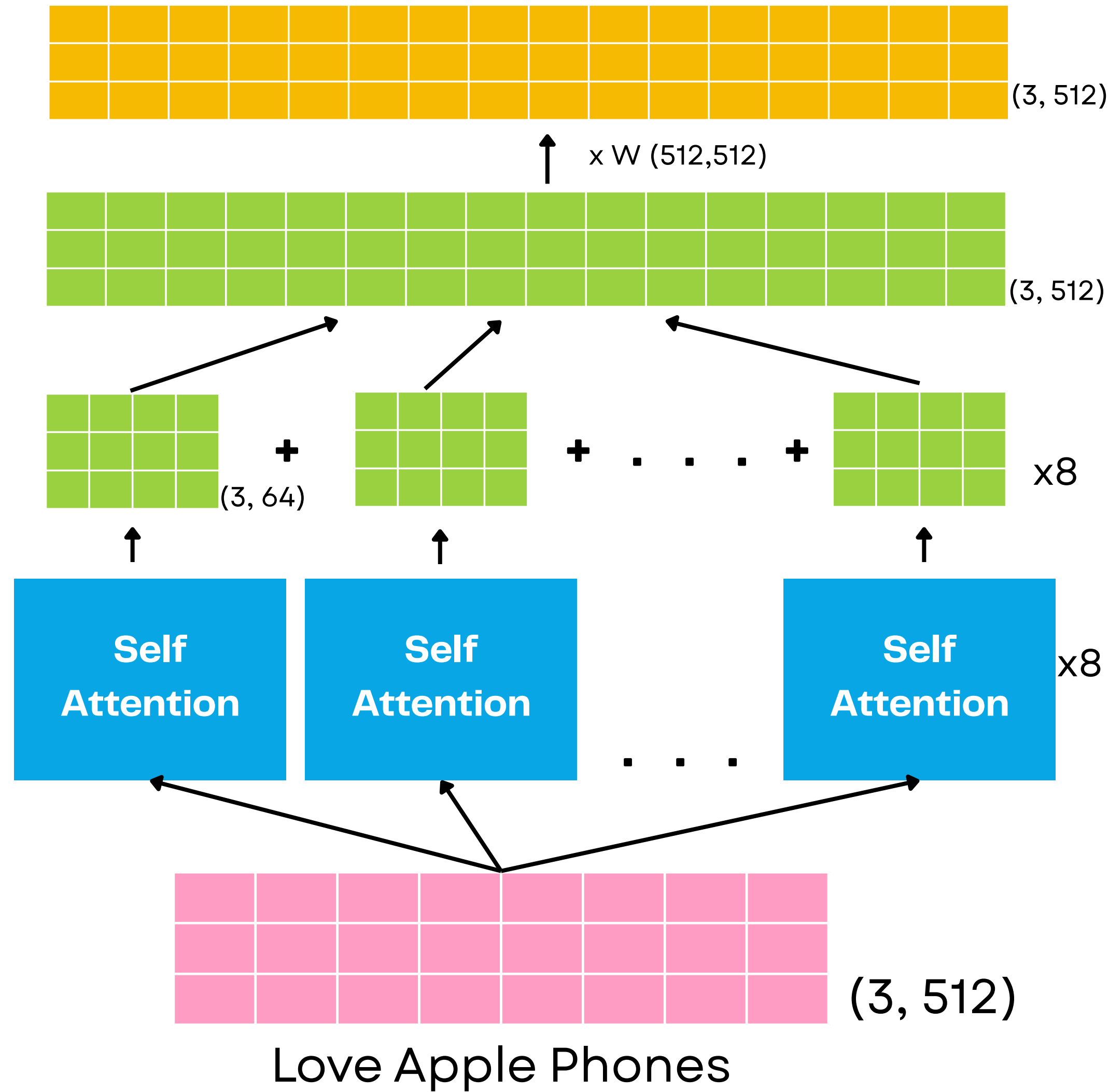
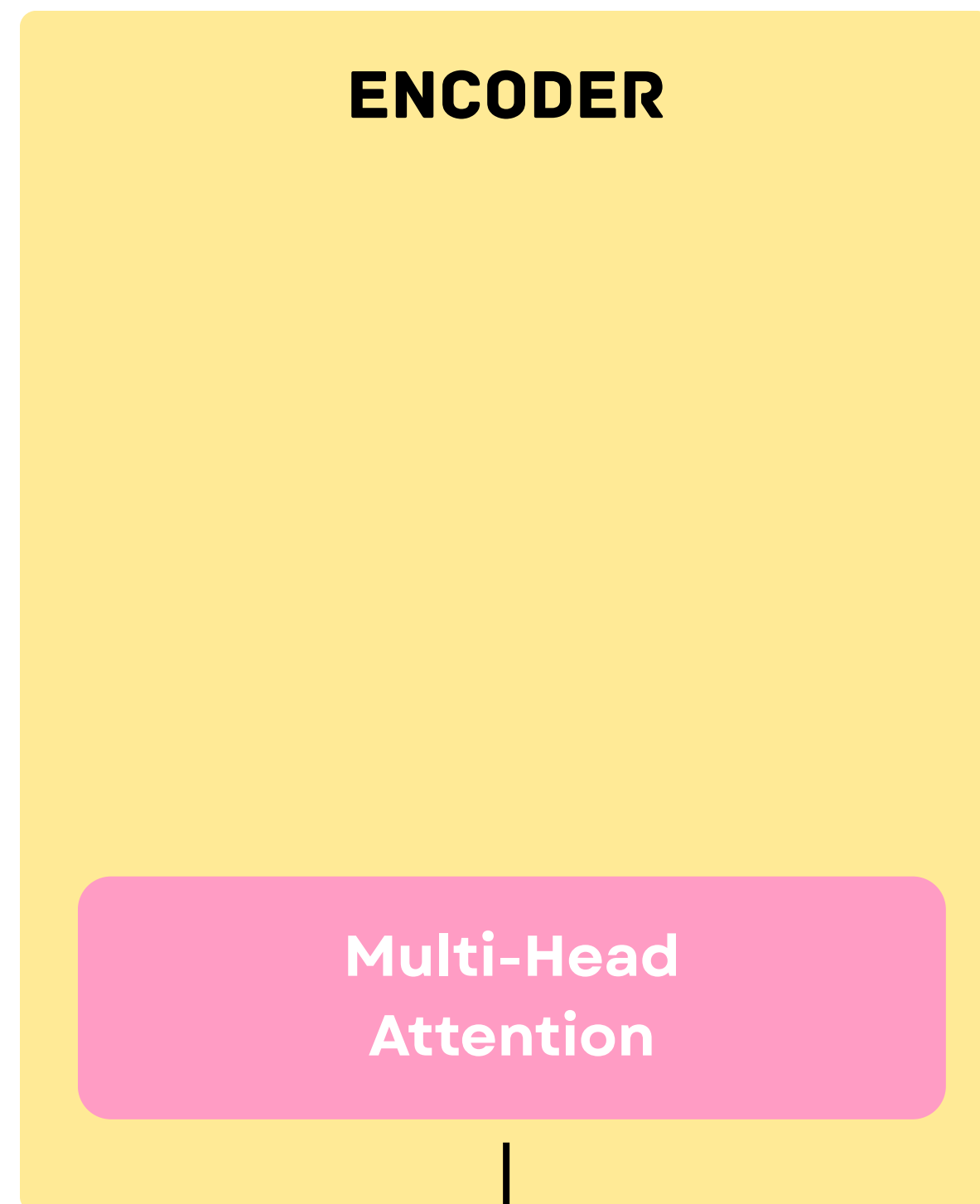


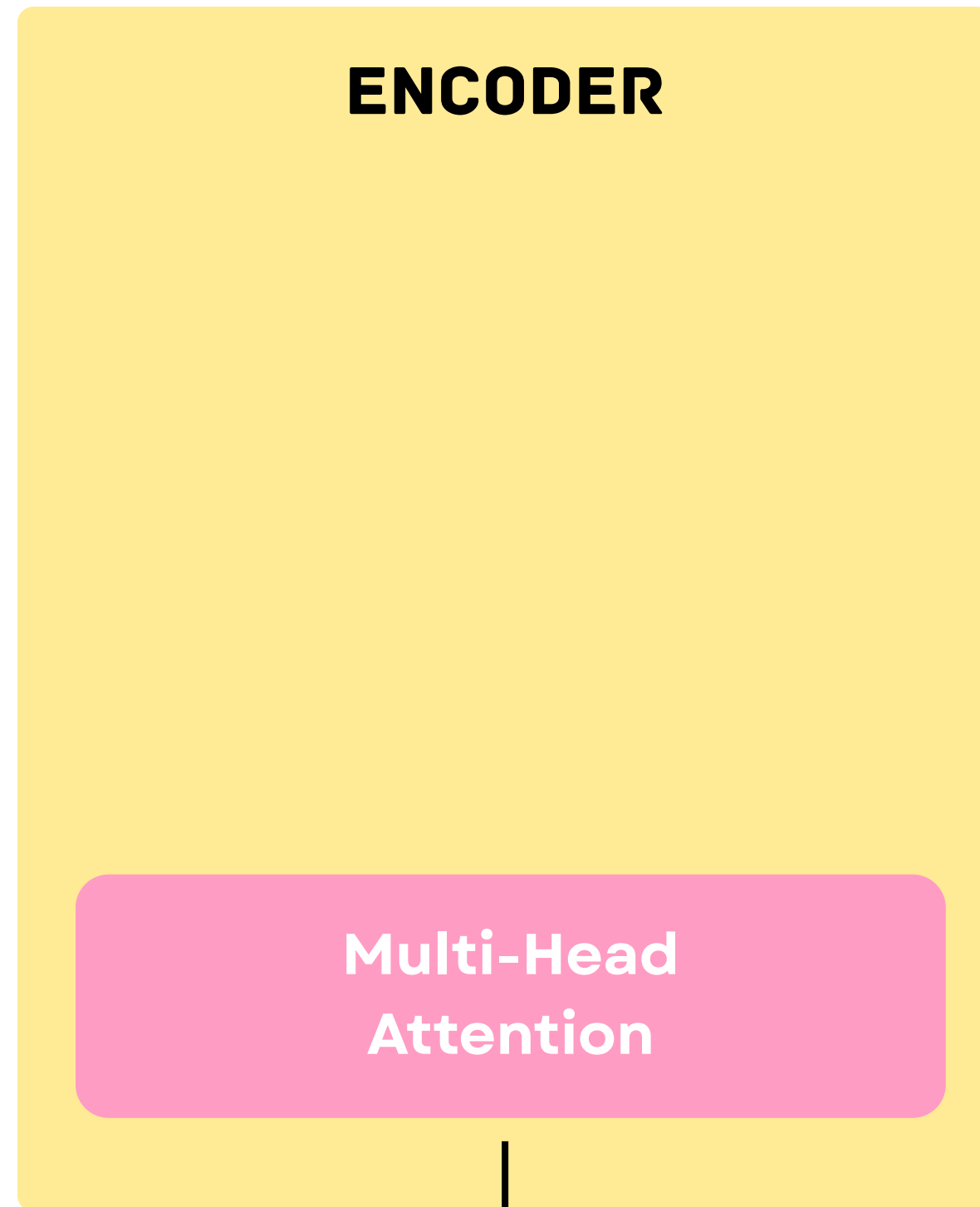










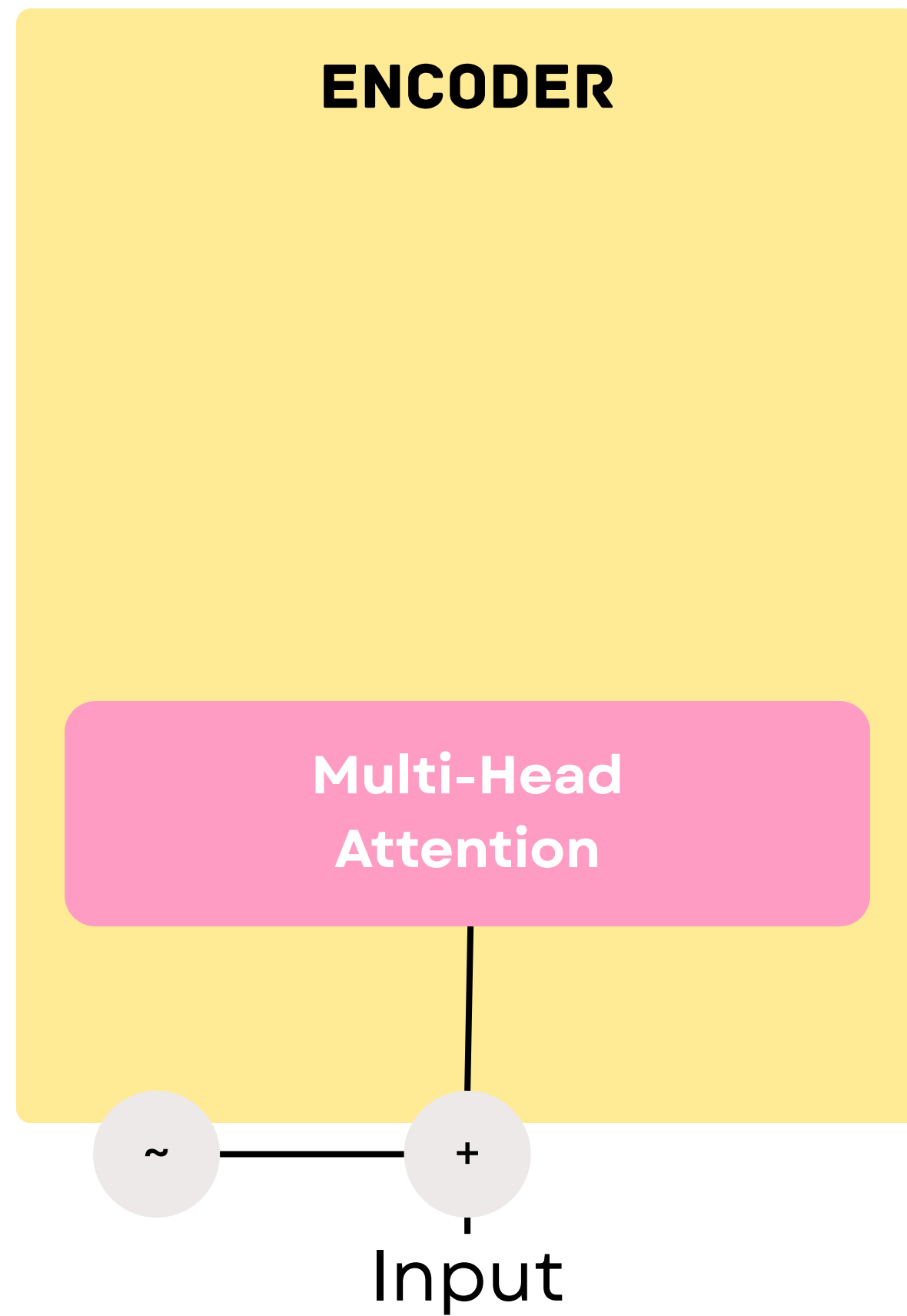


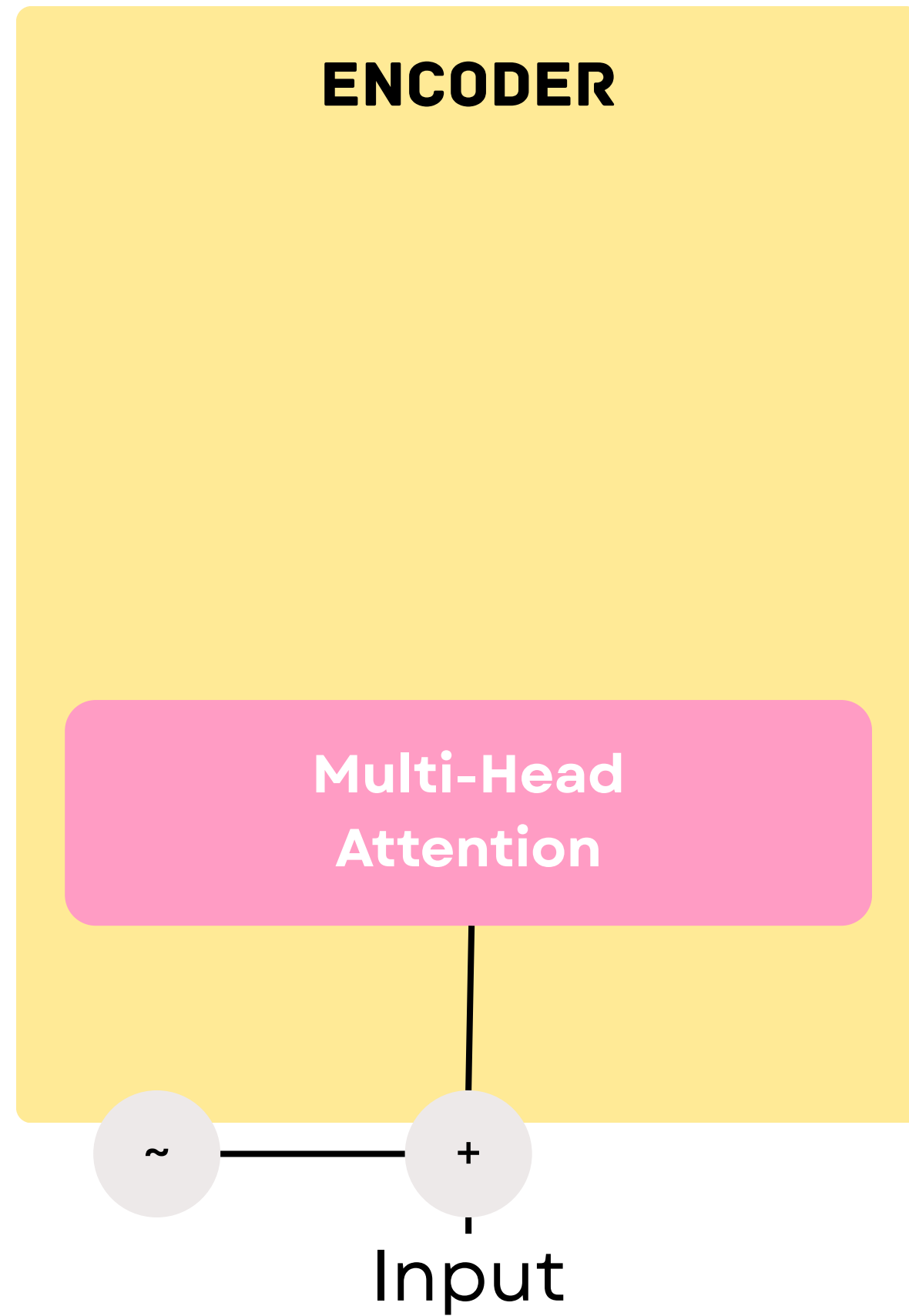
ENCODER

**Multi-Head
Attention**

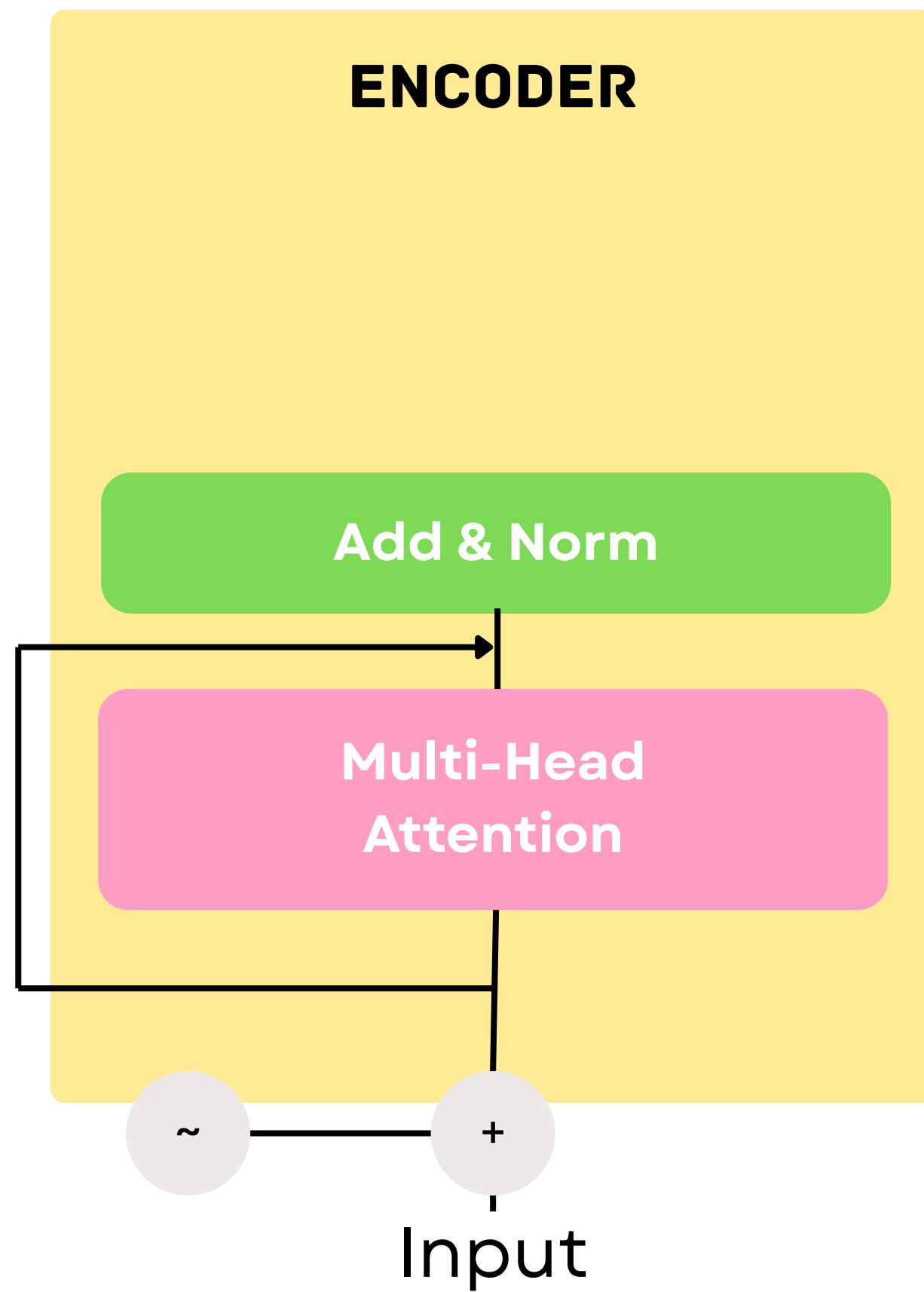
Input

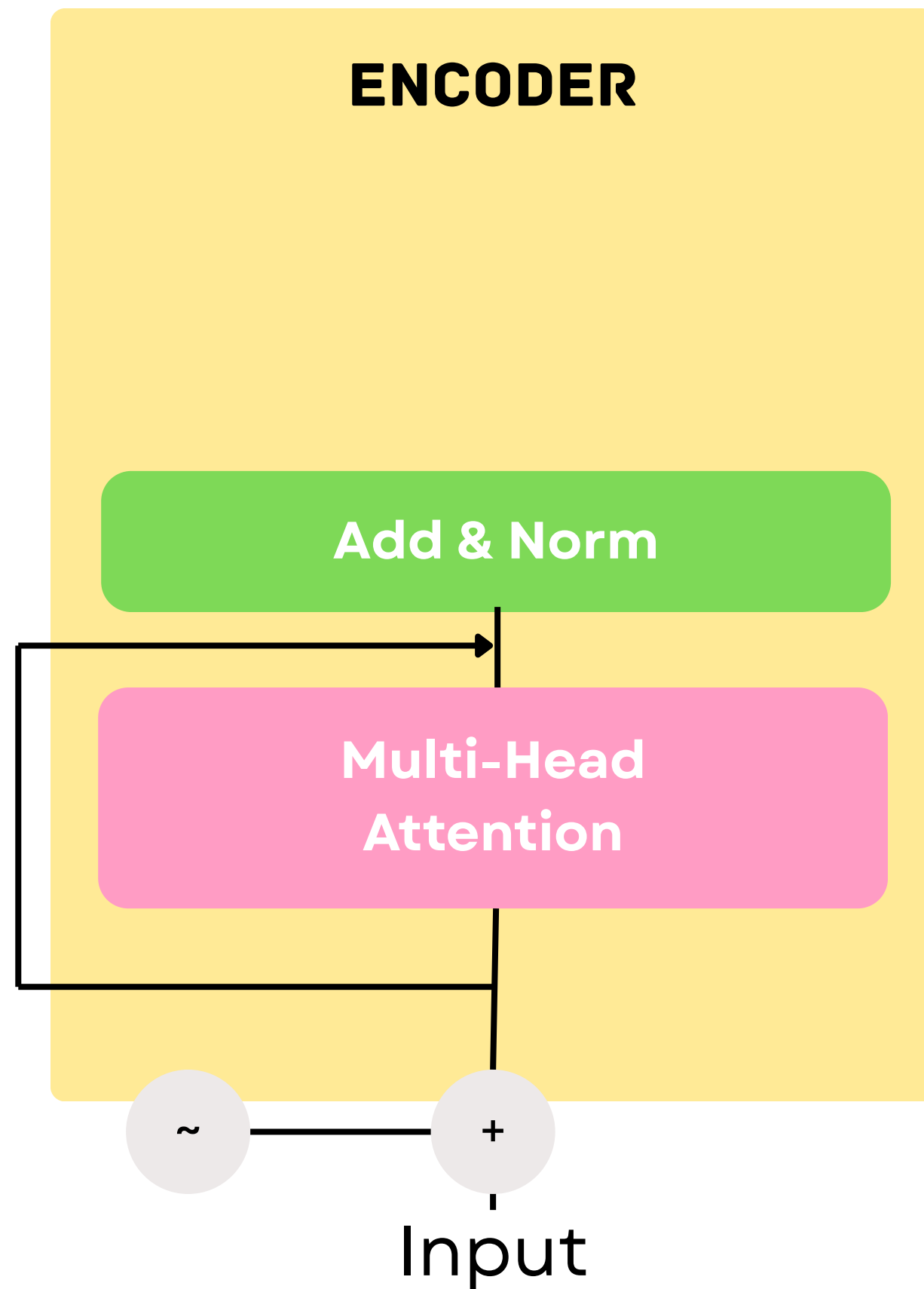
Q: Is there a problem here?





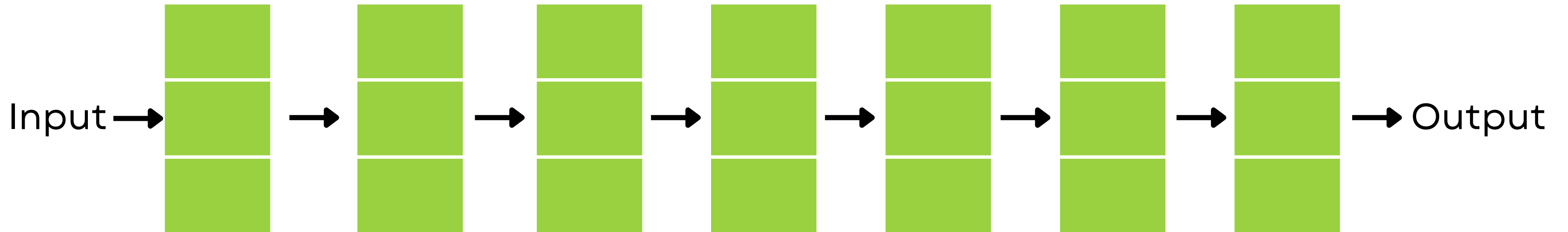
Q: Is there a problem here?



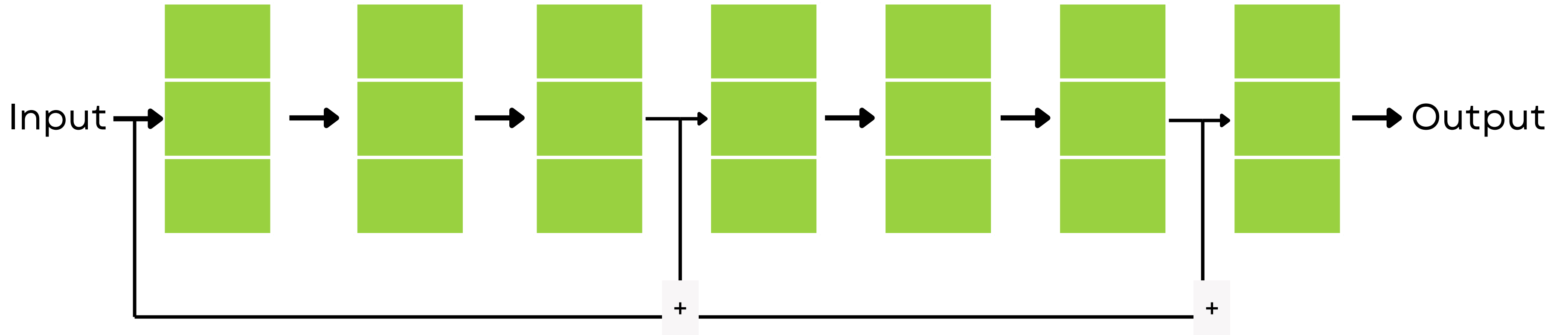


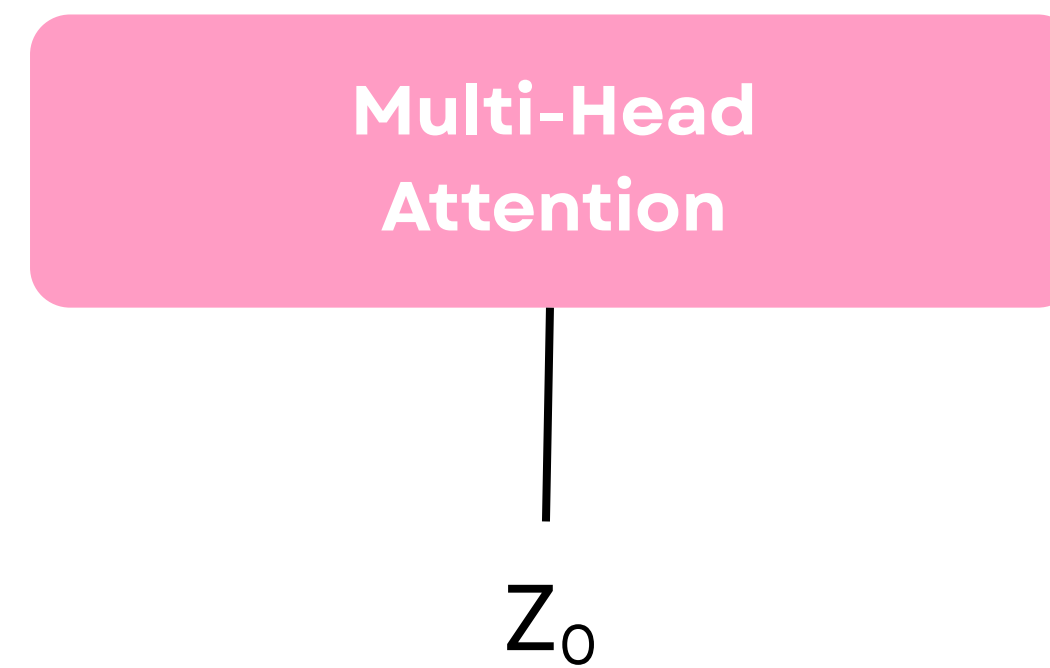
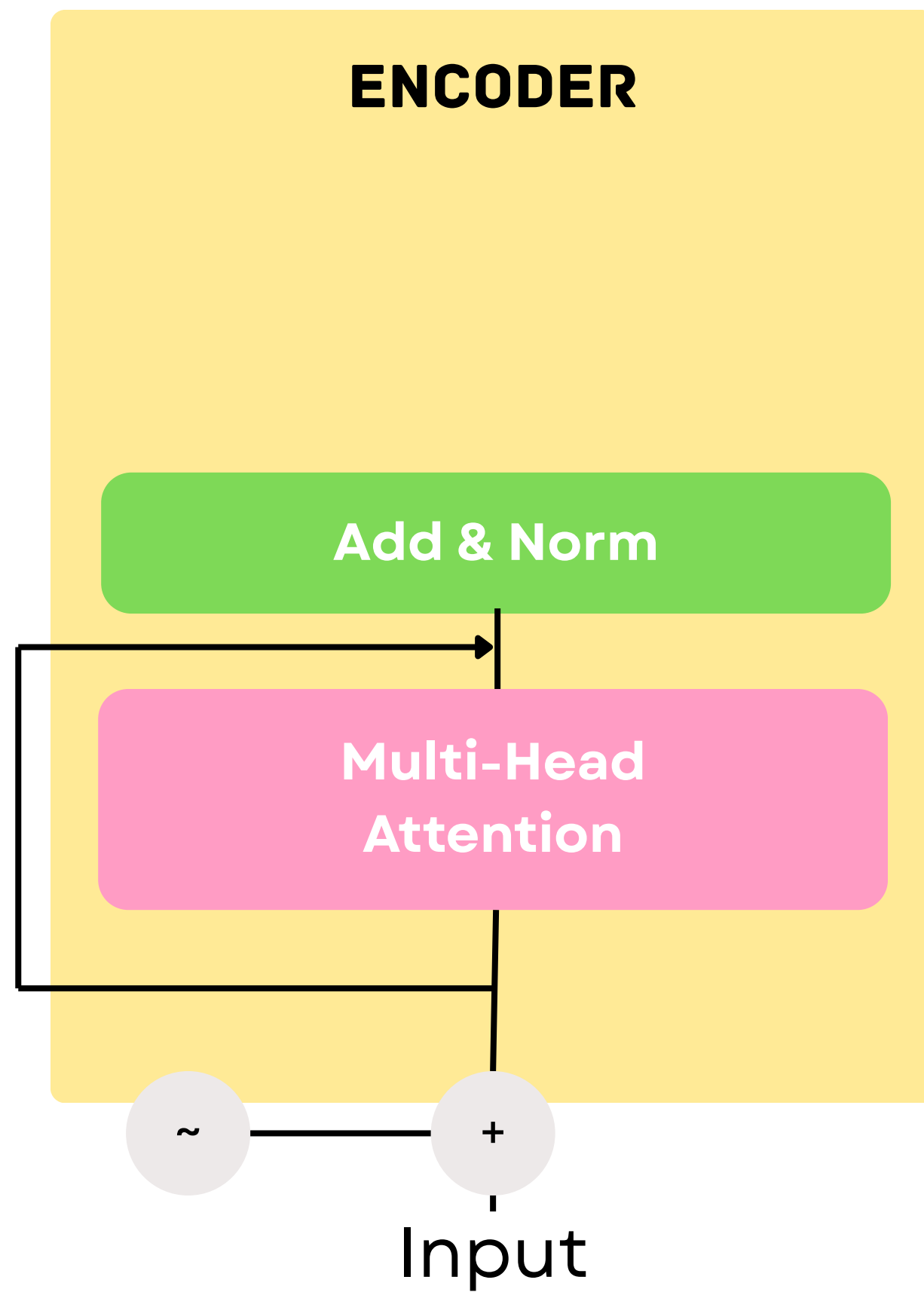
- Residual Connection
- Layer Normalization

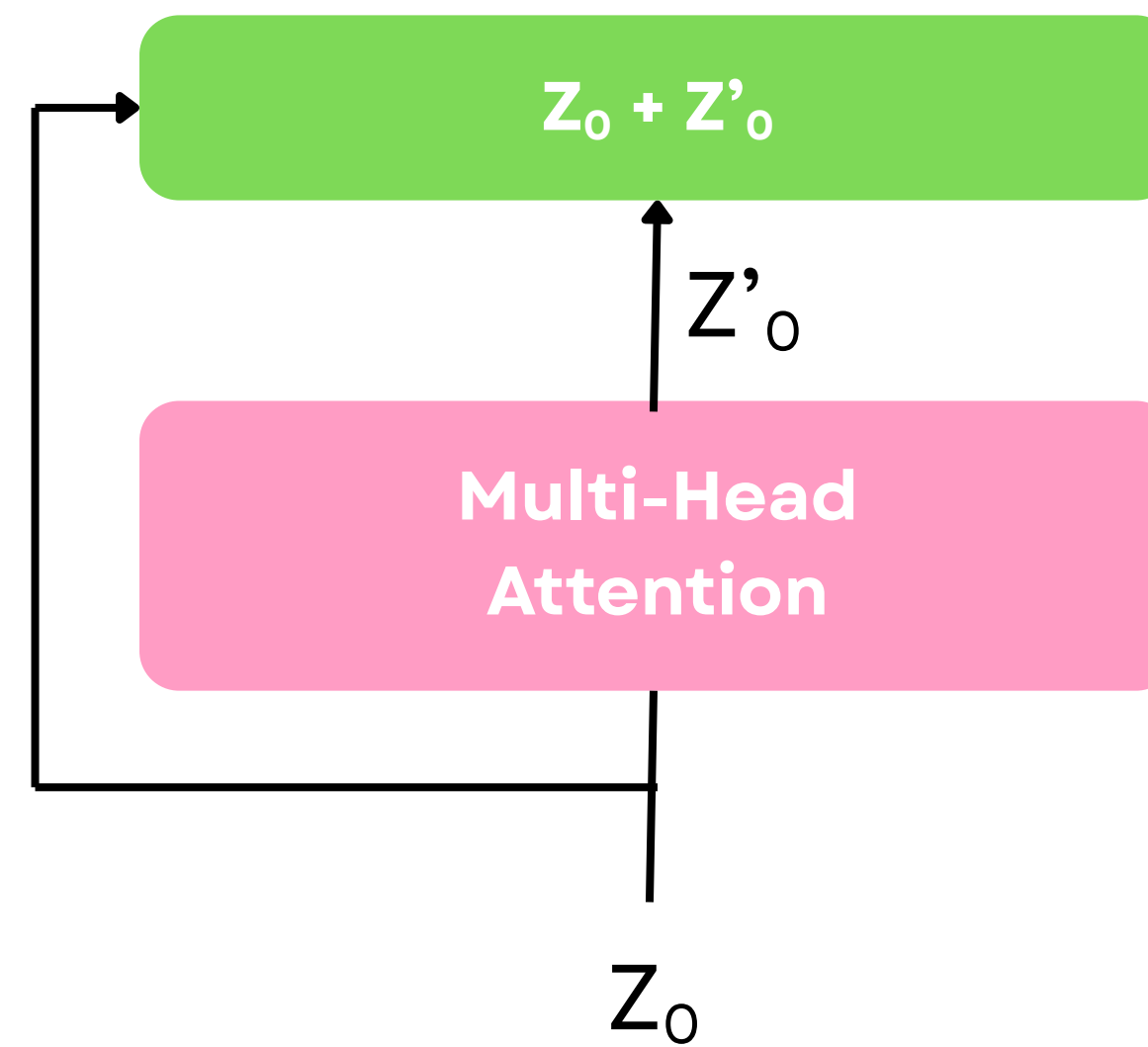
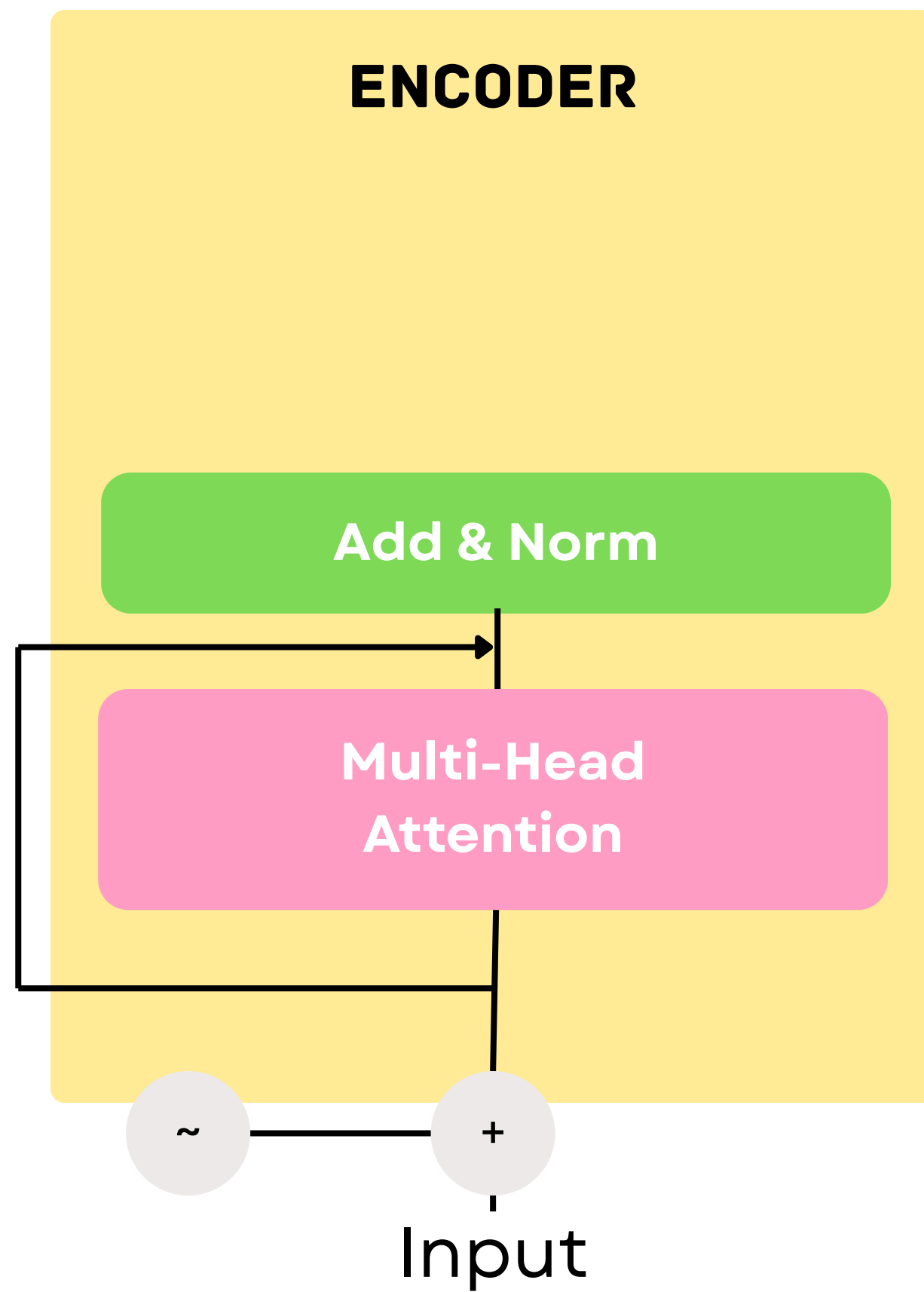
Residual Connection

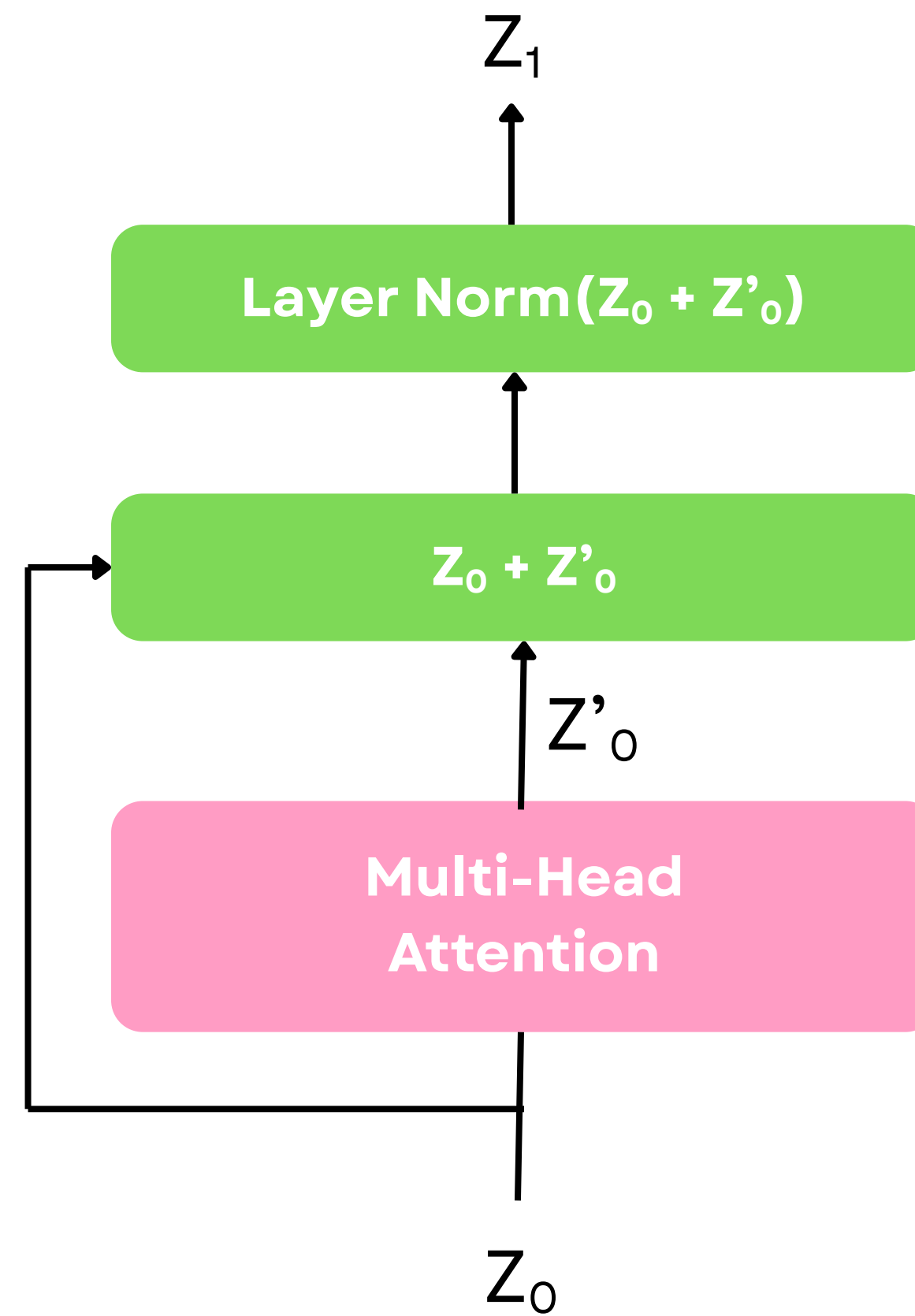
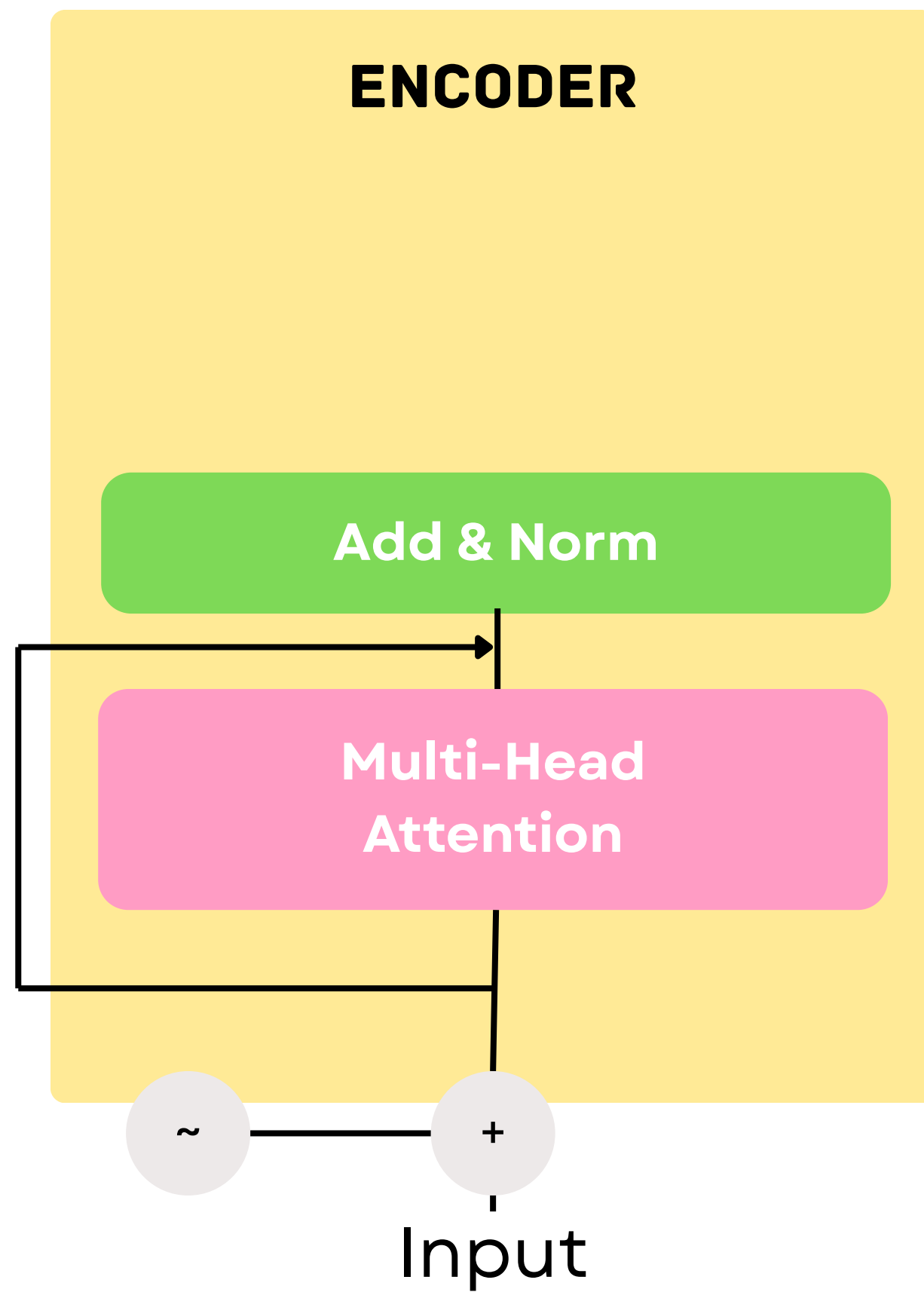


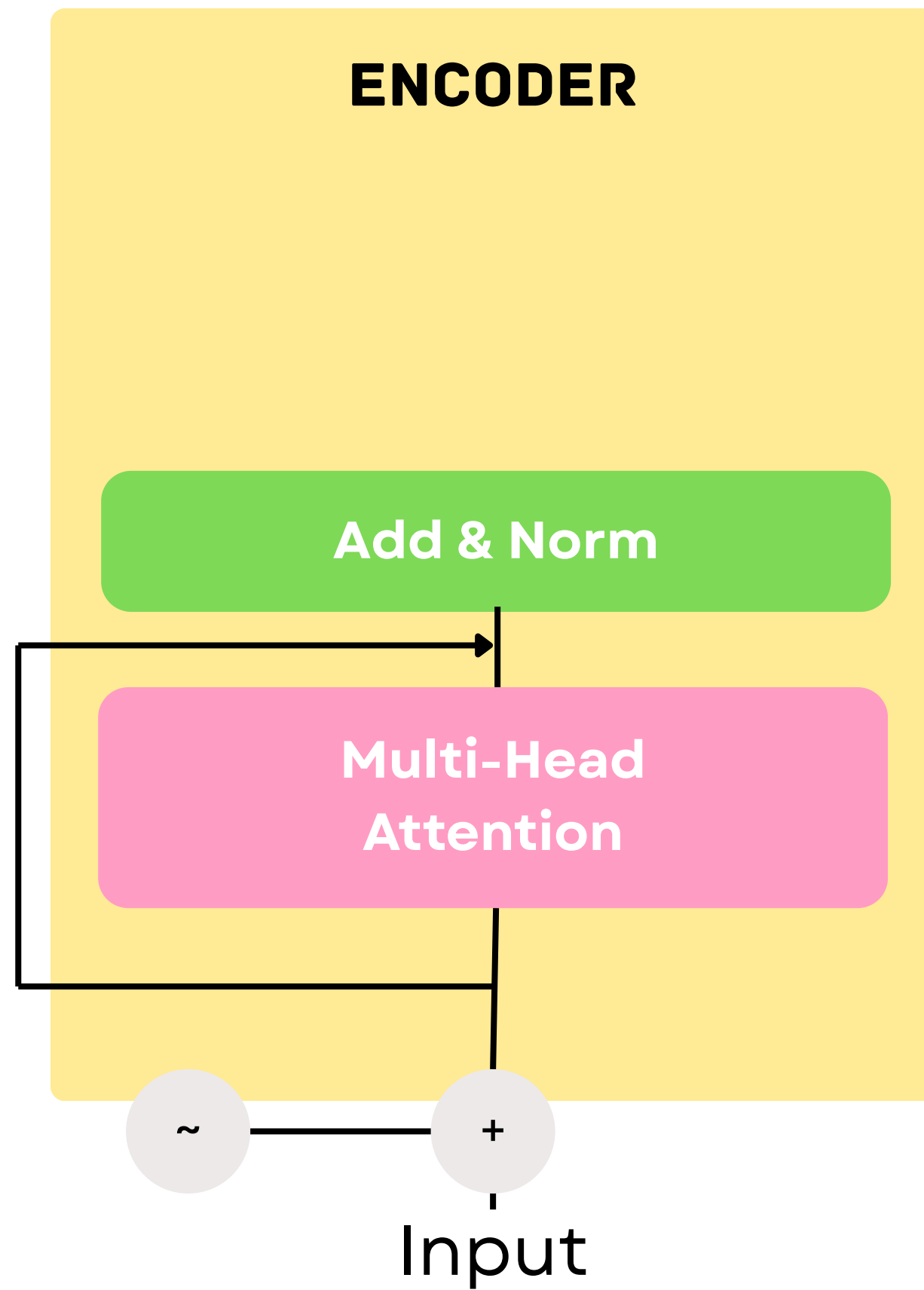
Residual Connection



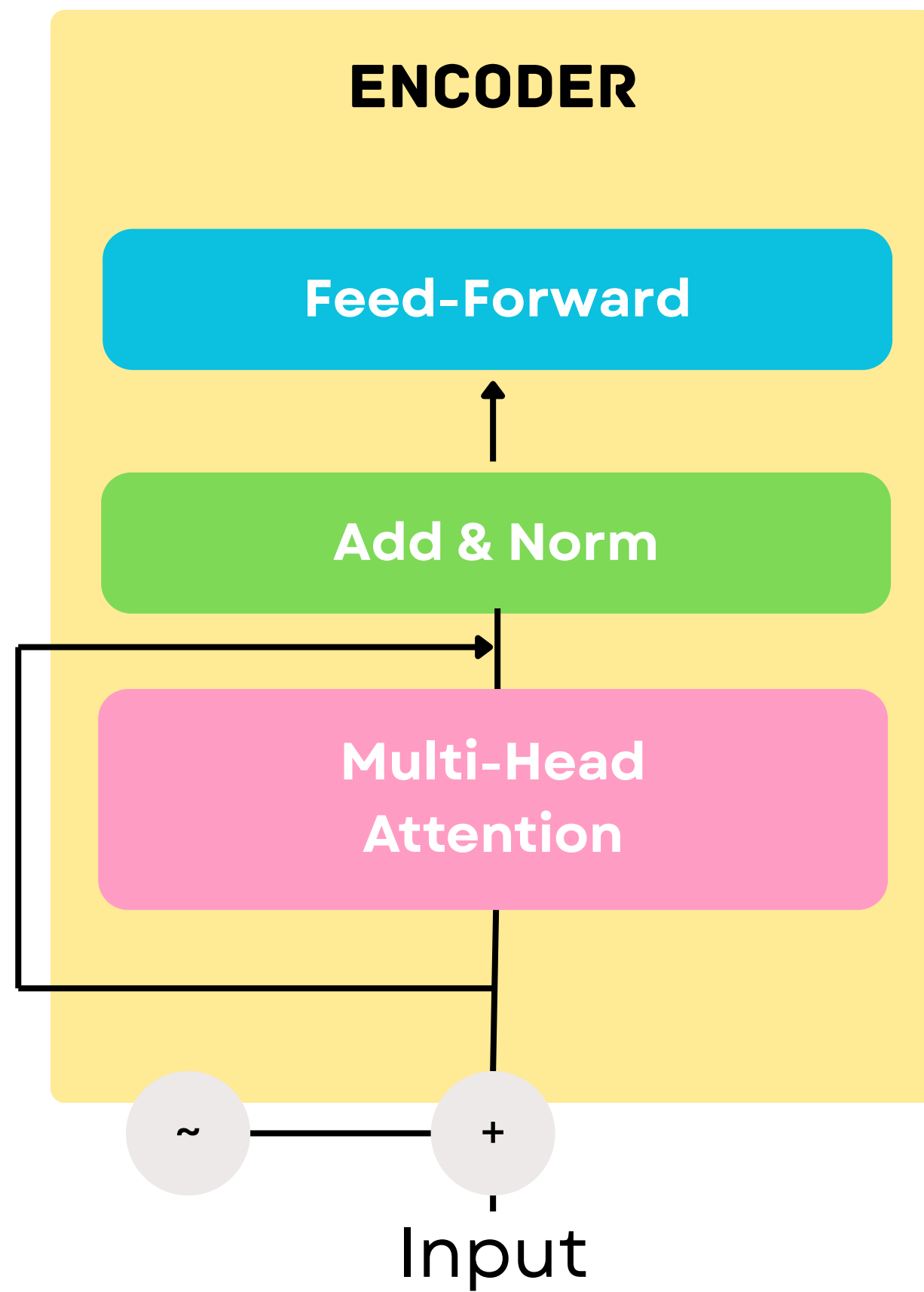


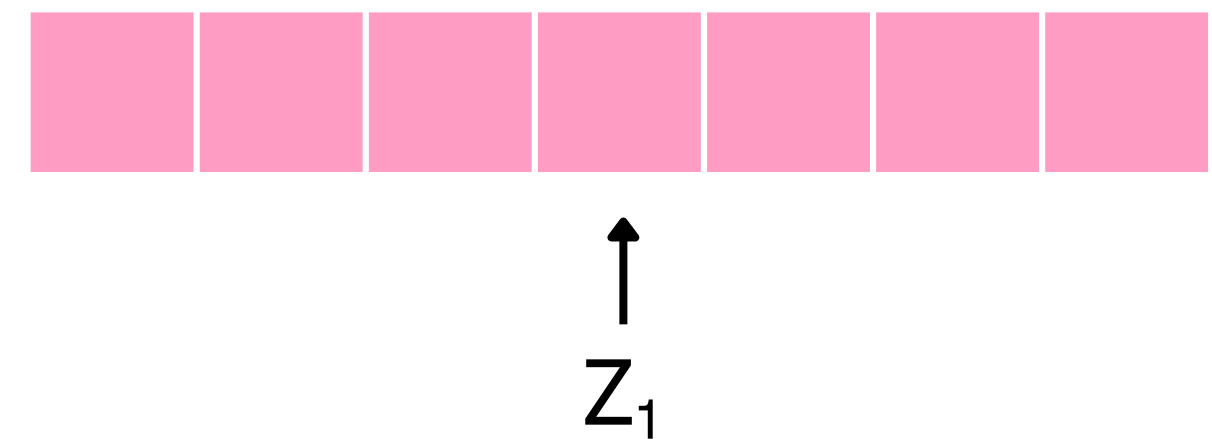
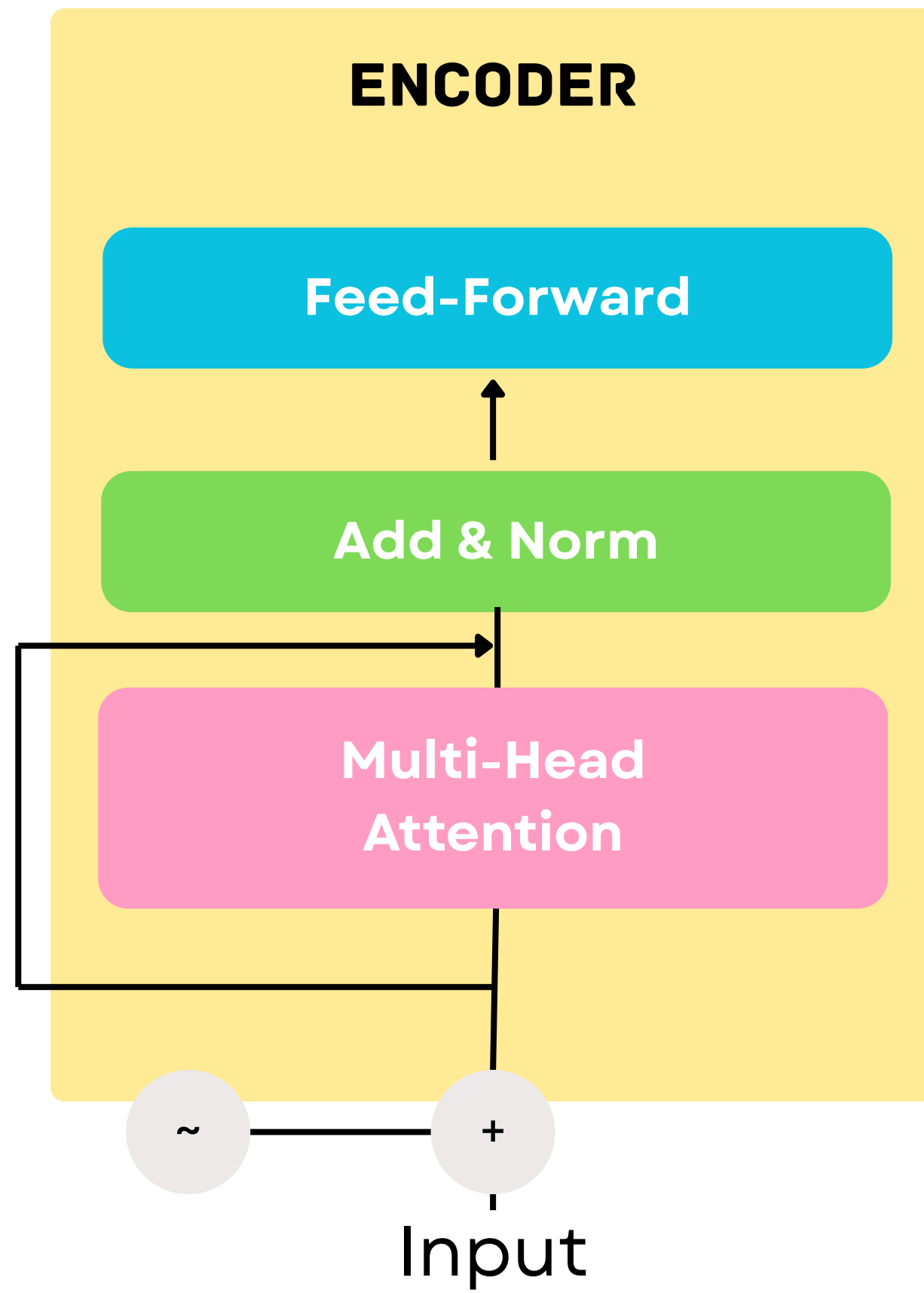


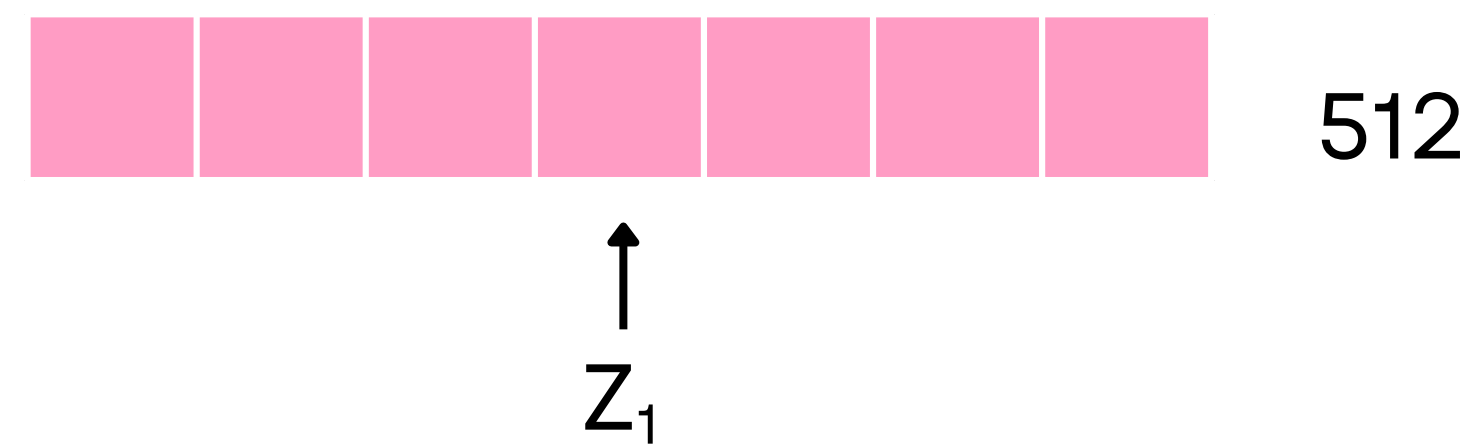
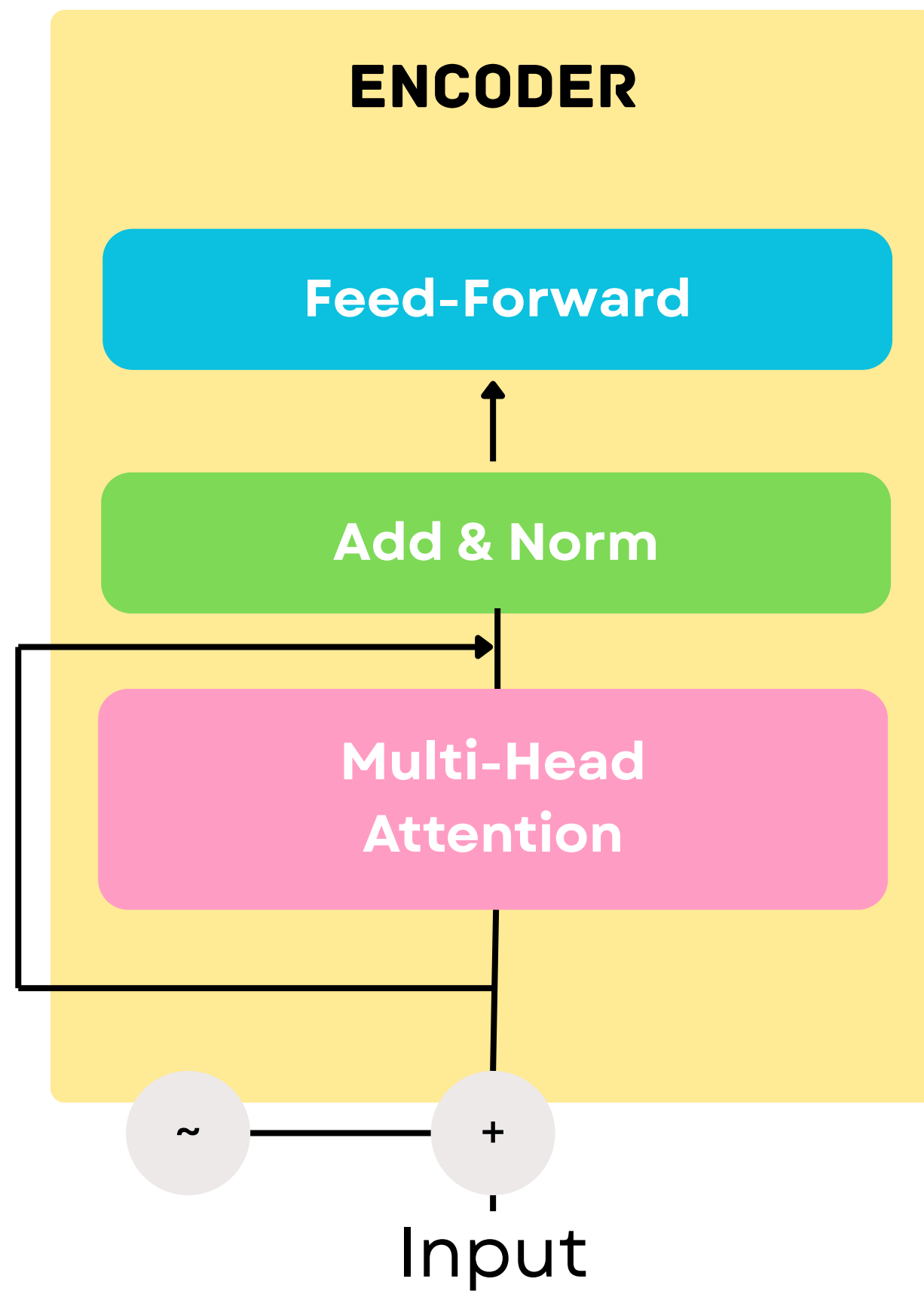


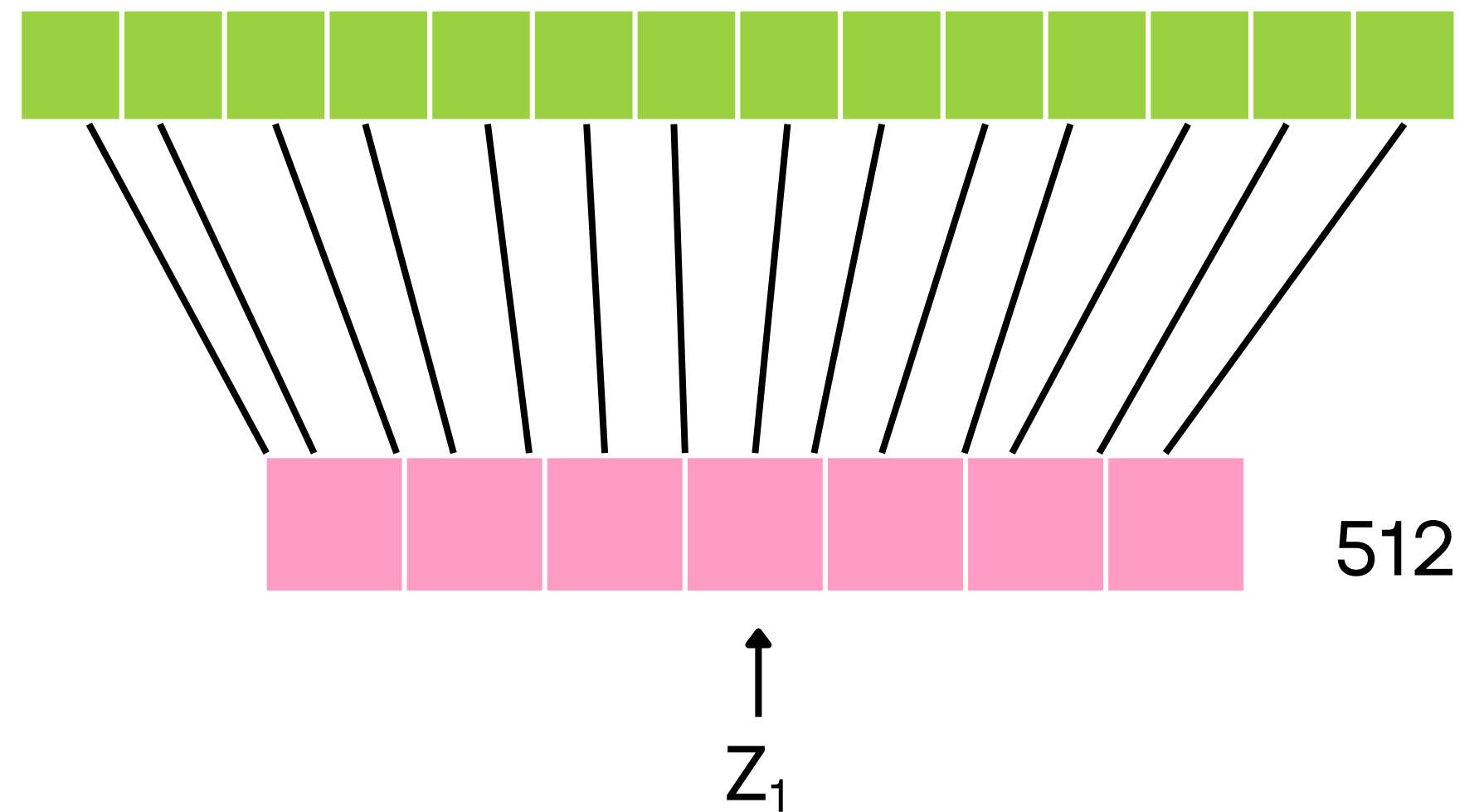
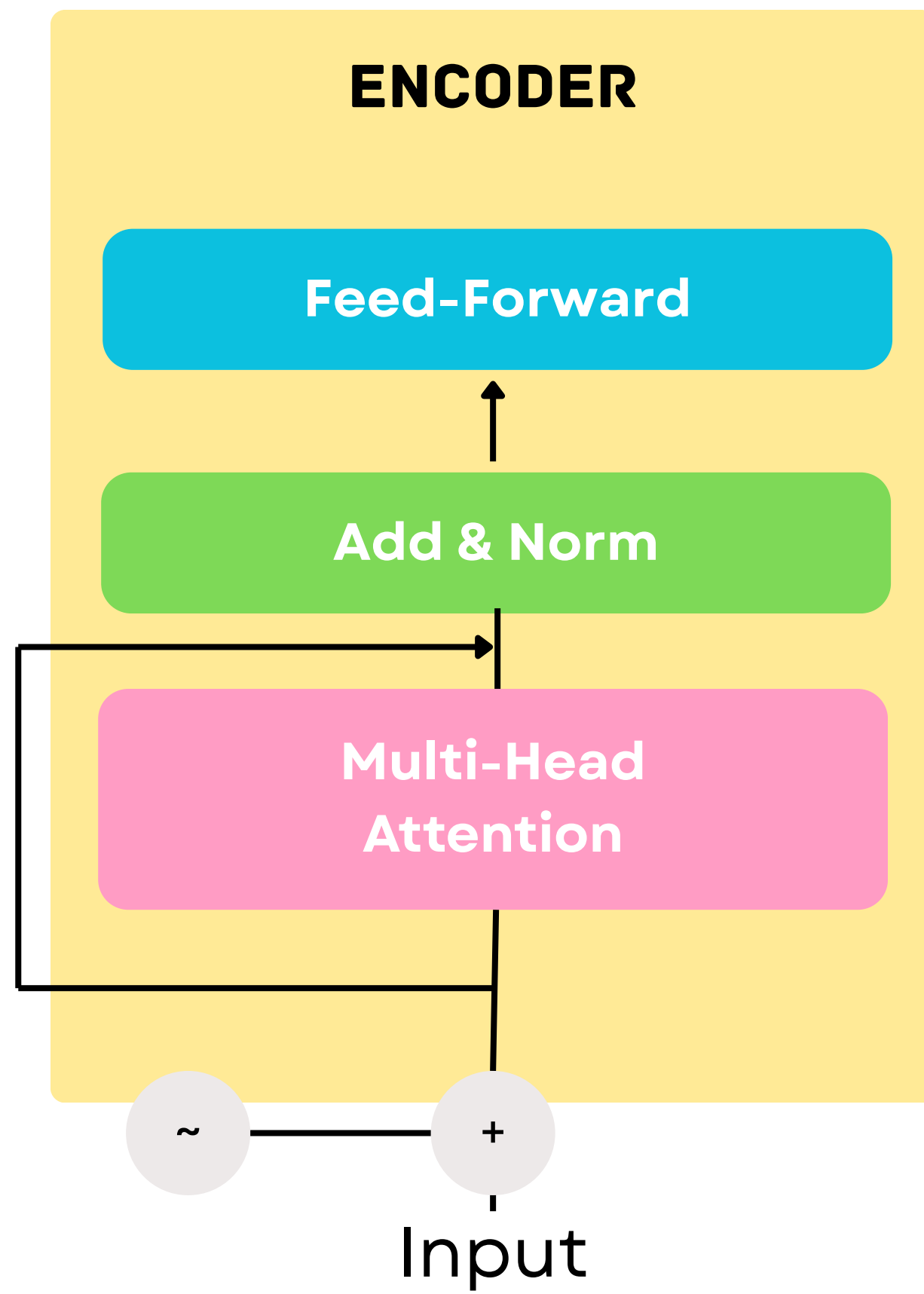


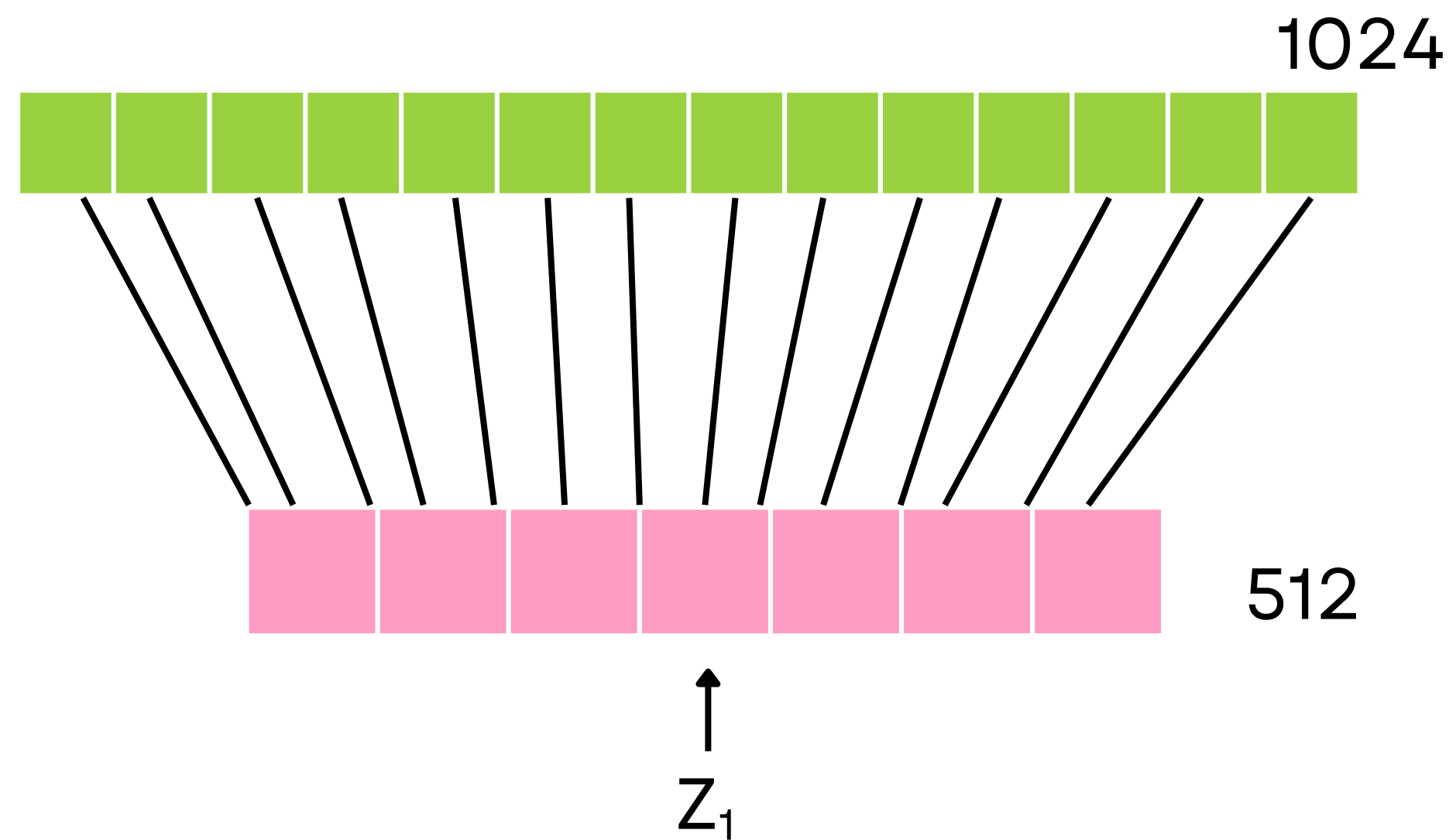
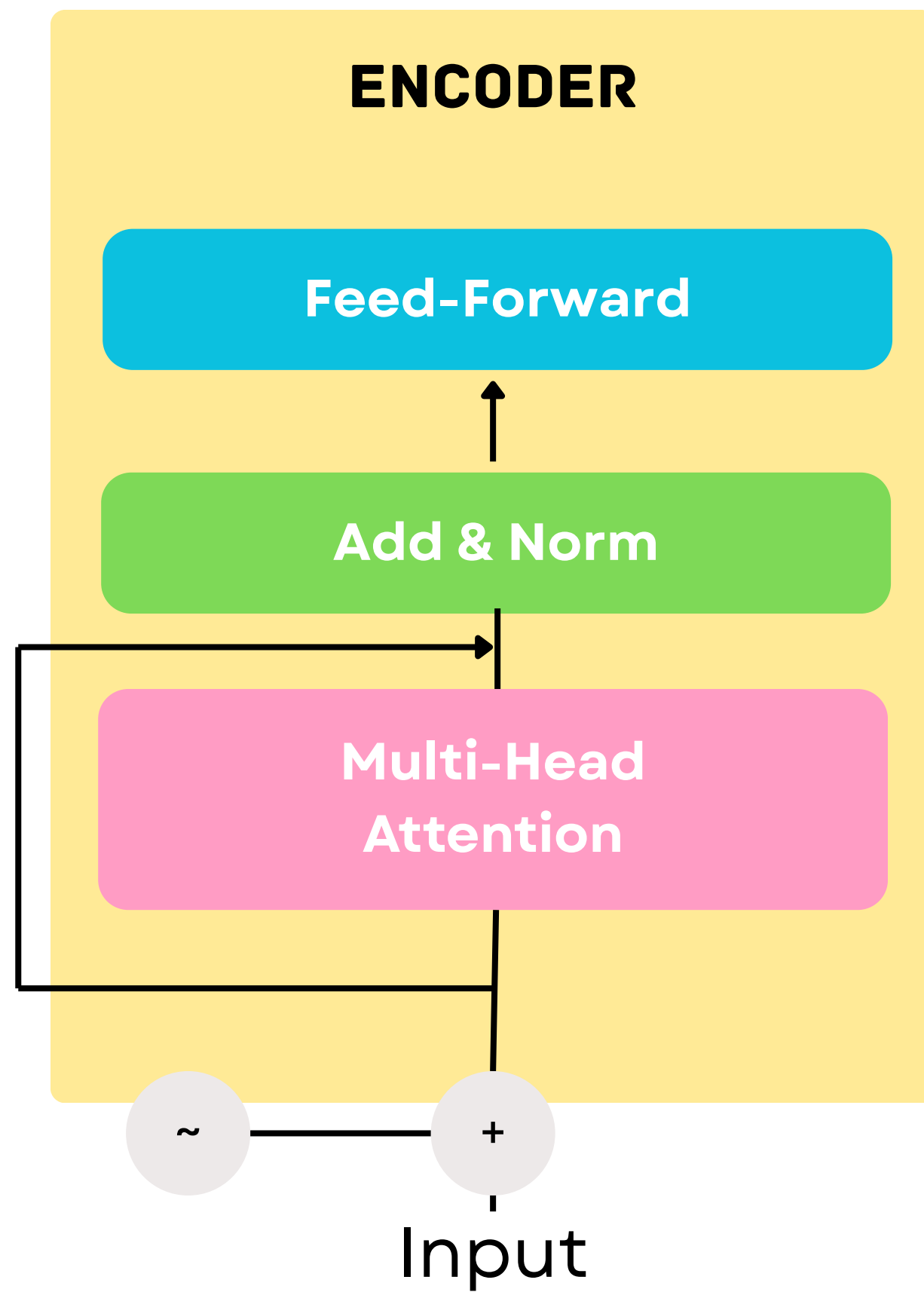
Q: Is there a problem here?

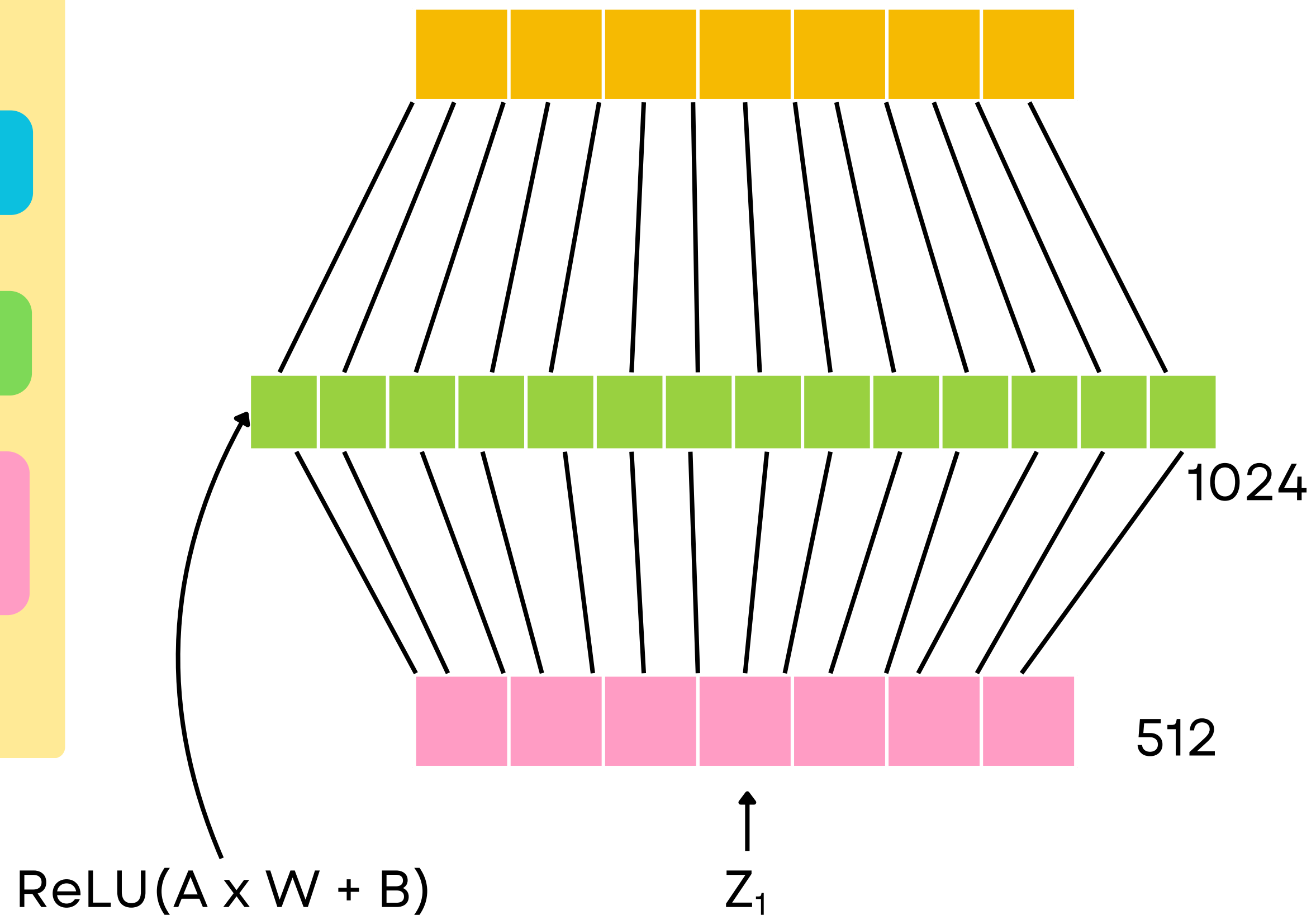
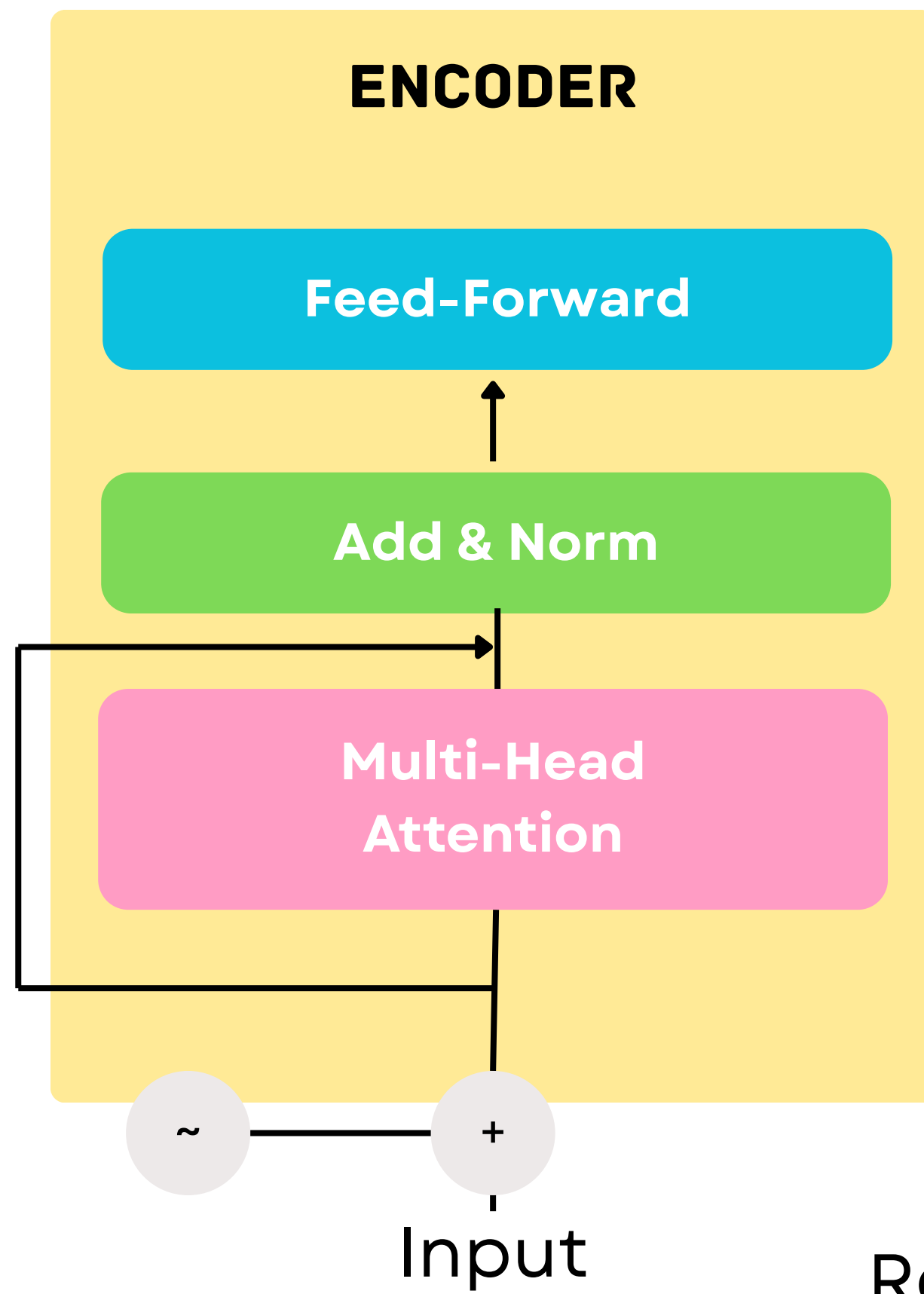


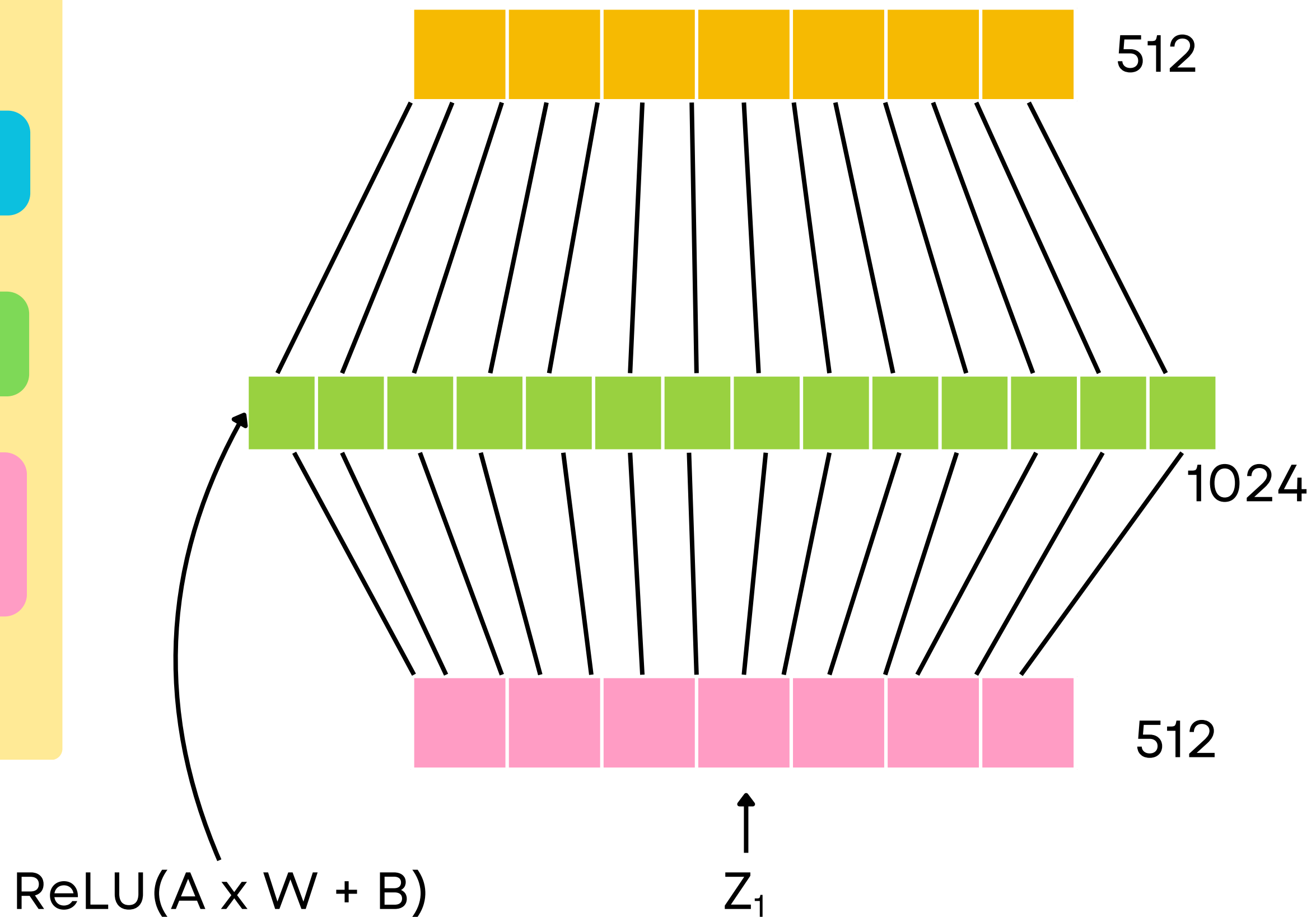
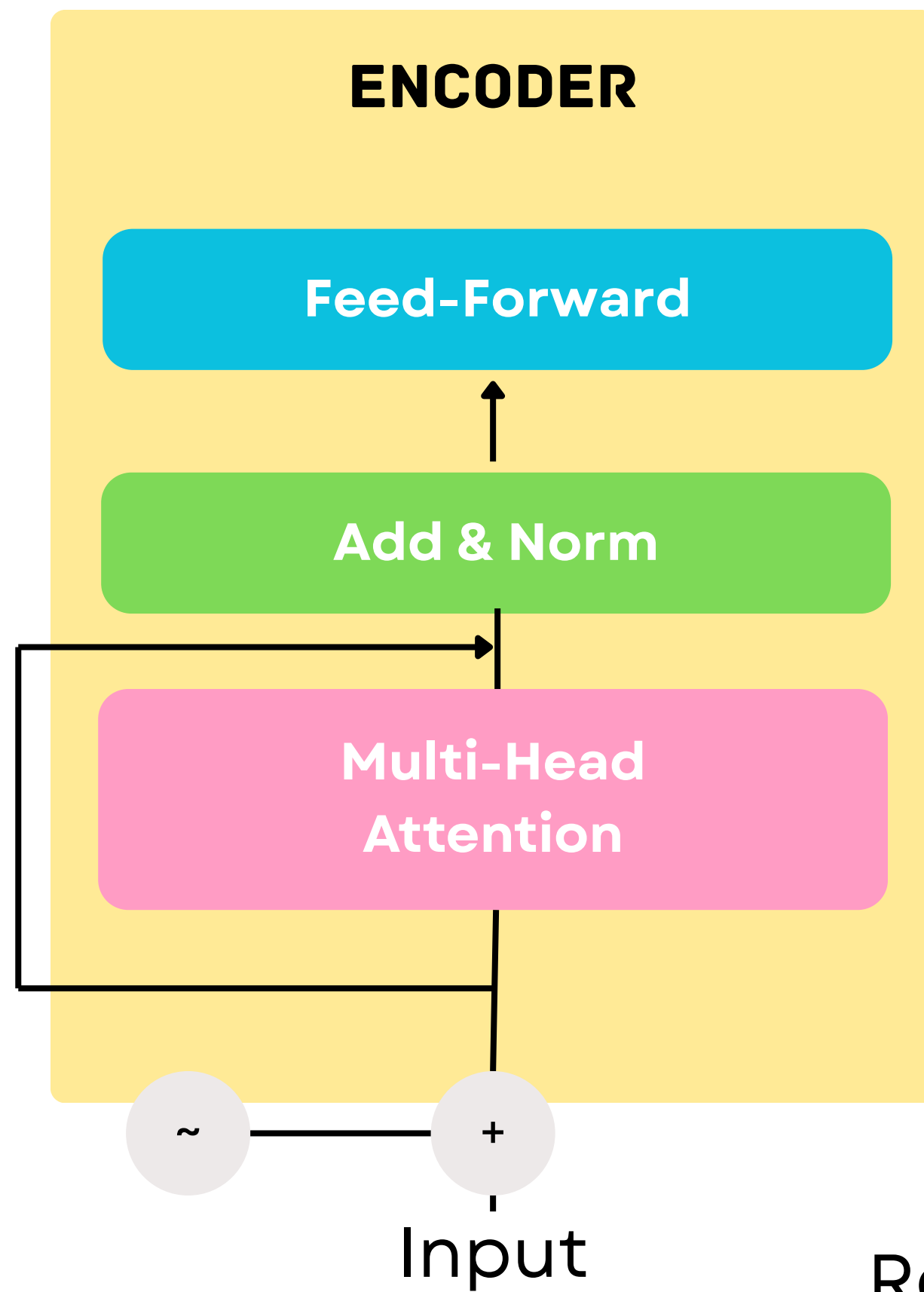


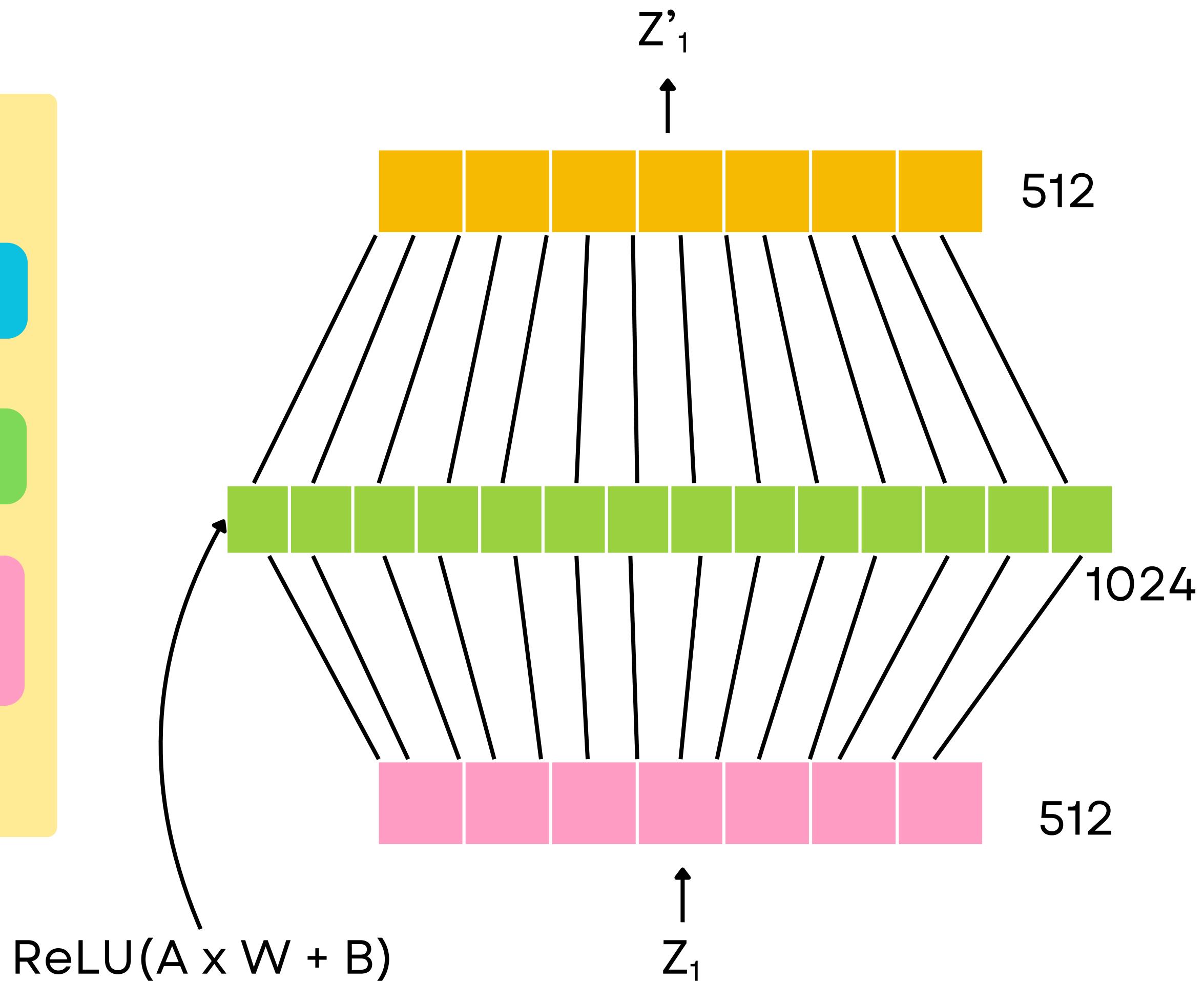
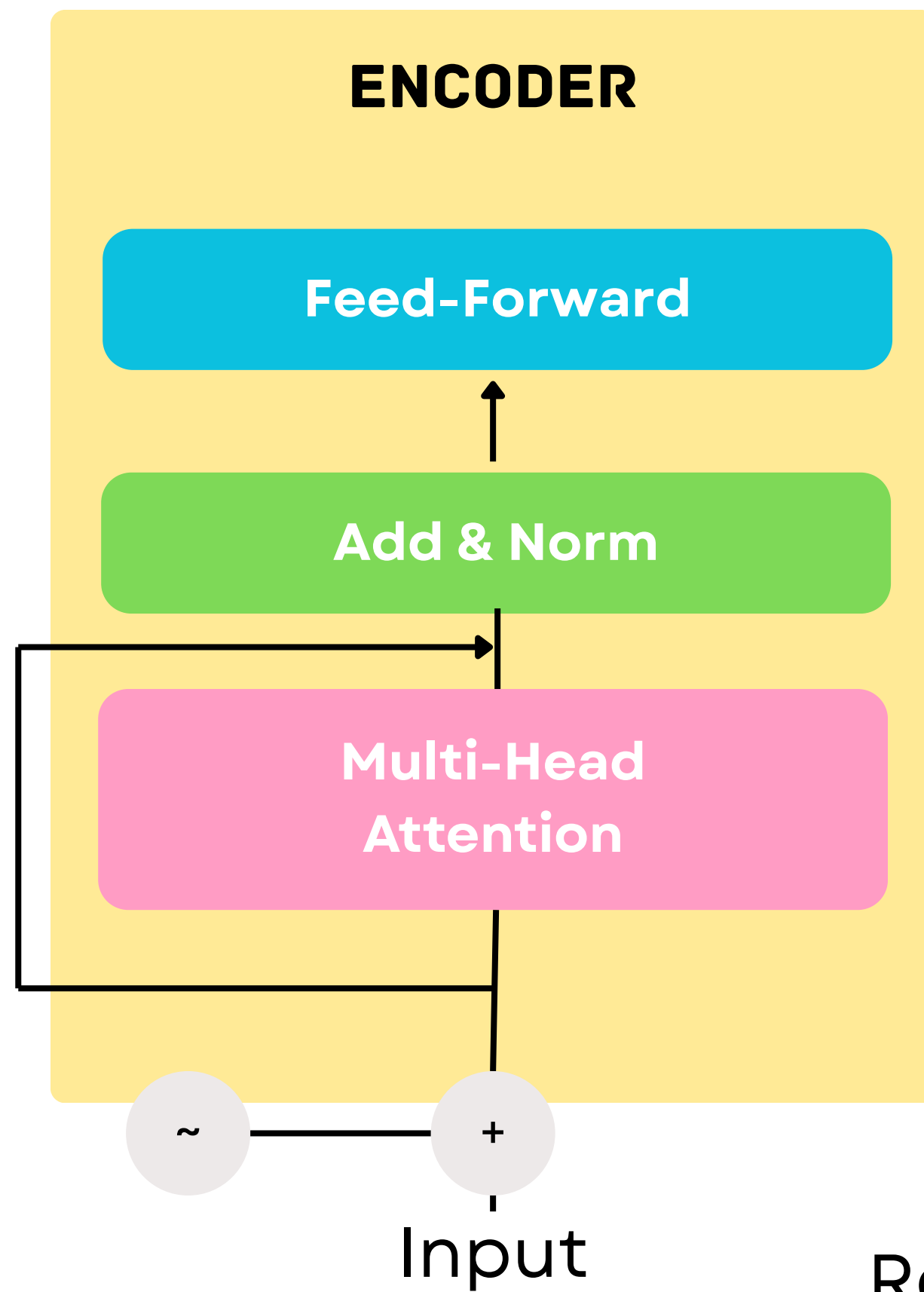


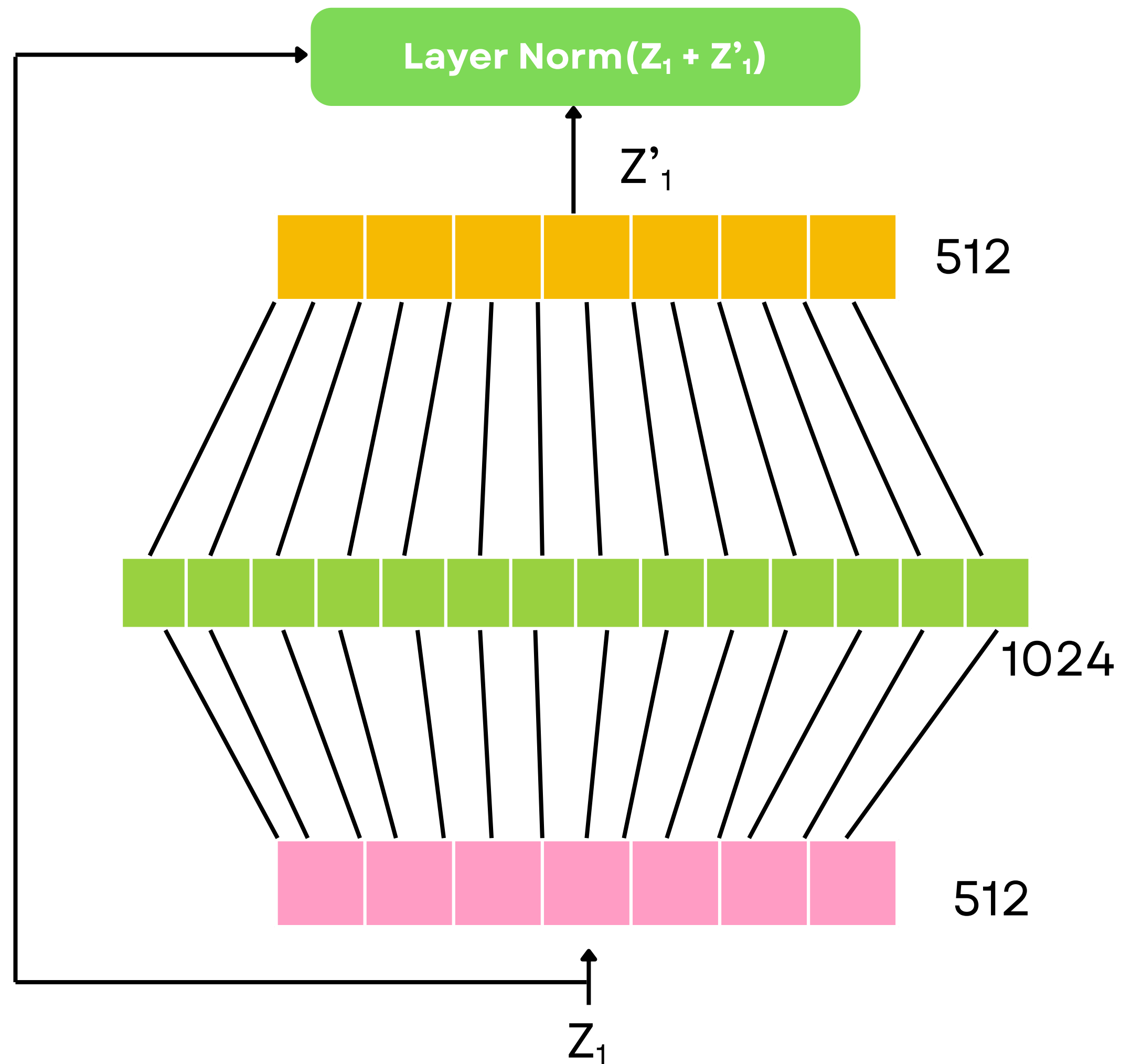
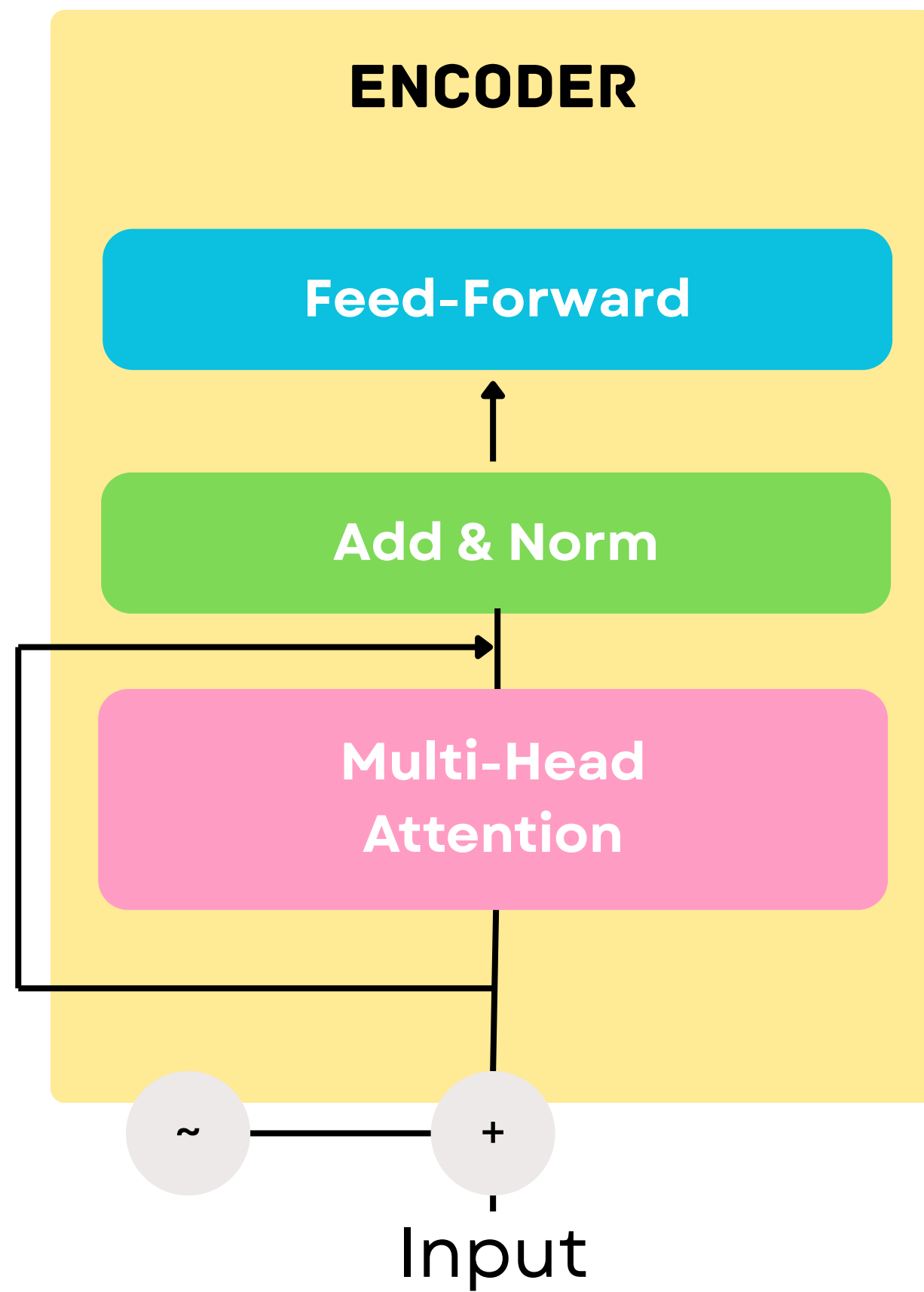


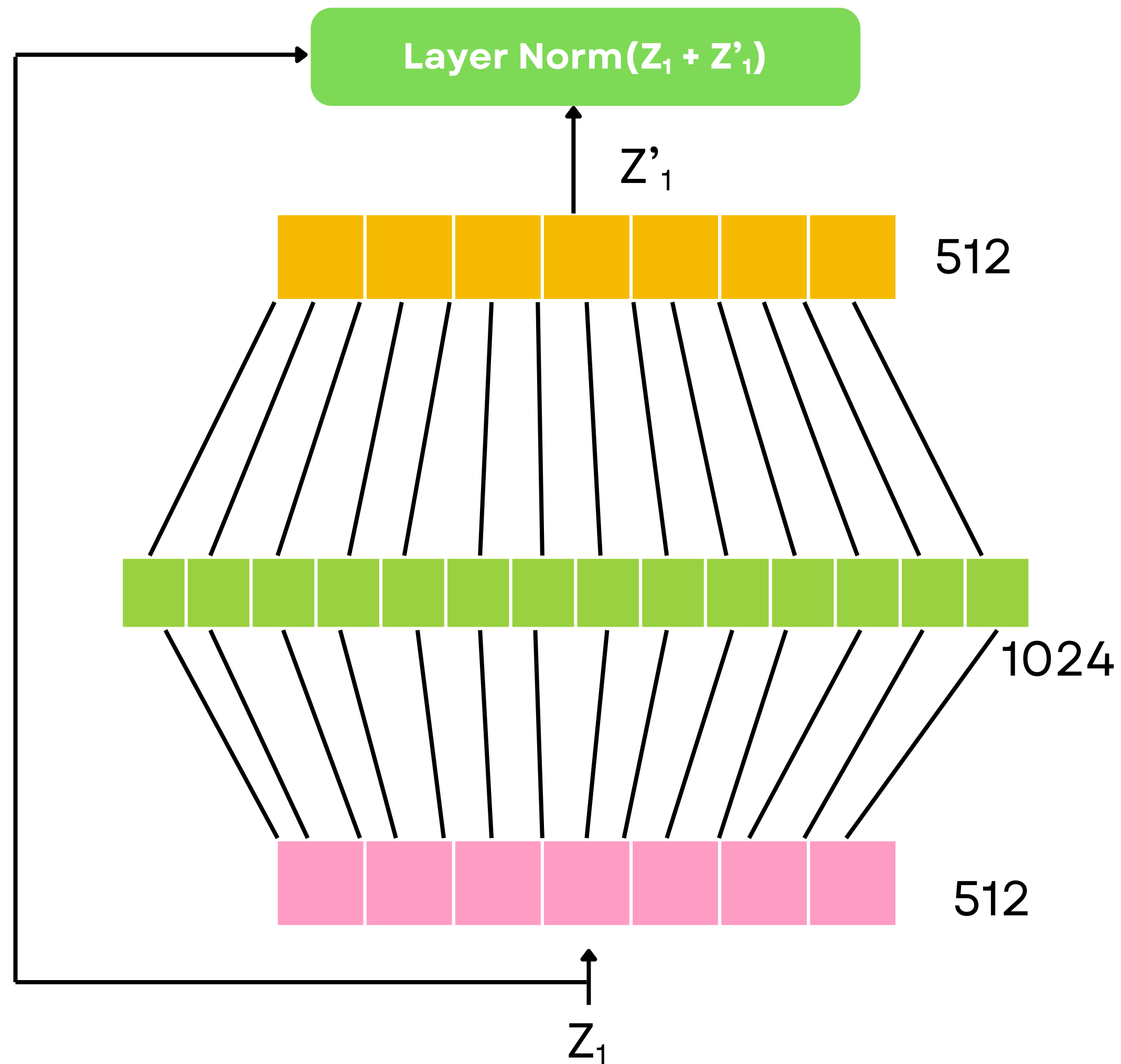
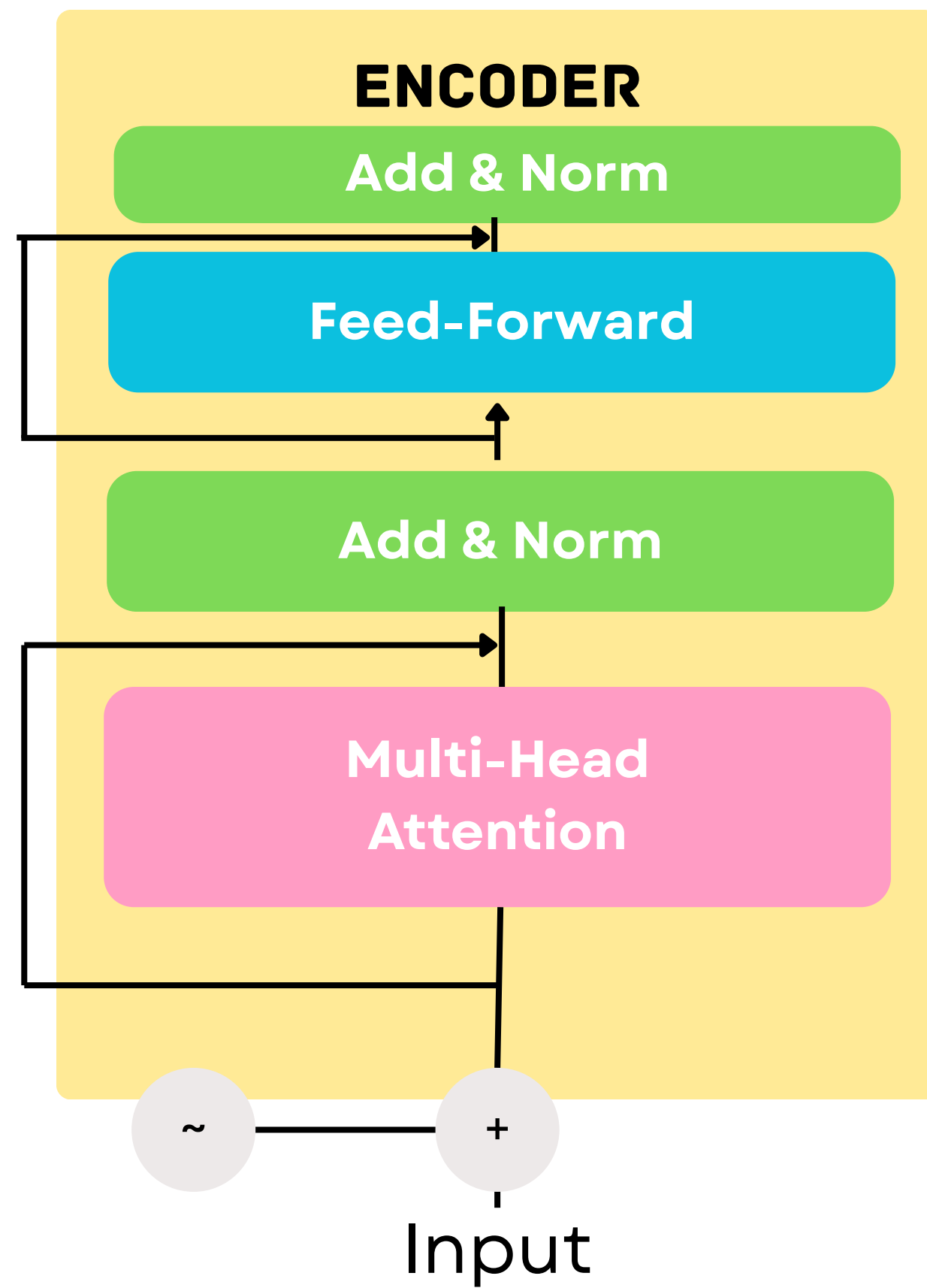












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